



Brightwood® Shutters

Program Reference Guide
2022

2022 Shutter Revision log

Effective Date	Topic	Section	Note
2022/01/01	Frame Hinged Bifold Track 90 Degree Bifold Track T-Post Connector	j1,i	1. Update content and drawings: Panel Configuration (uneven panel width) add content: 1) If the hinge type between panel and frame is 2 3/8" Self Mortise Hinge, the panel width1 (hinged with Self Mortise Hinge) = A, the other panels width (panel width2 / panel width3 / panel width4) = A + 34.5mm; 2) If the hinge type between panel and frame is Invisible Hinge, the panel width1 (hinged with Invisible Hinge) = A, the other panels width (panel width2 / panel width3 / panel width4) = A + 29.5mm; Take the drawings of the hinge type between panel and frame is 2 3/8" Self Mortise Hinge as examples:" 2. 90 Degree Bifold Track & Floating Panels Hanger Location: remove the following drawing: LLL/RRR/LLRRR  3. Corrected available sill plate size: "For Horizontal T Post, Hang Strip (Behind) and shutter with 1.5"/2" Sill, T Post connector is not available and T Post will be installed by screws instead;" should be "For Horizontal T Post, Hang Strip (Behind) and shutter with 1.6"/2" Sill, T Post connector is not available and T Post will be installed by screws instead".

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Features and Benefits

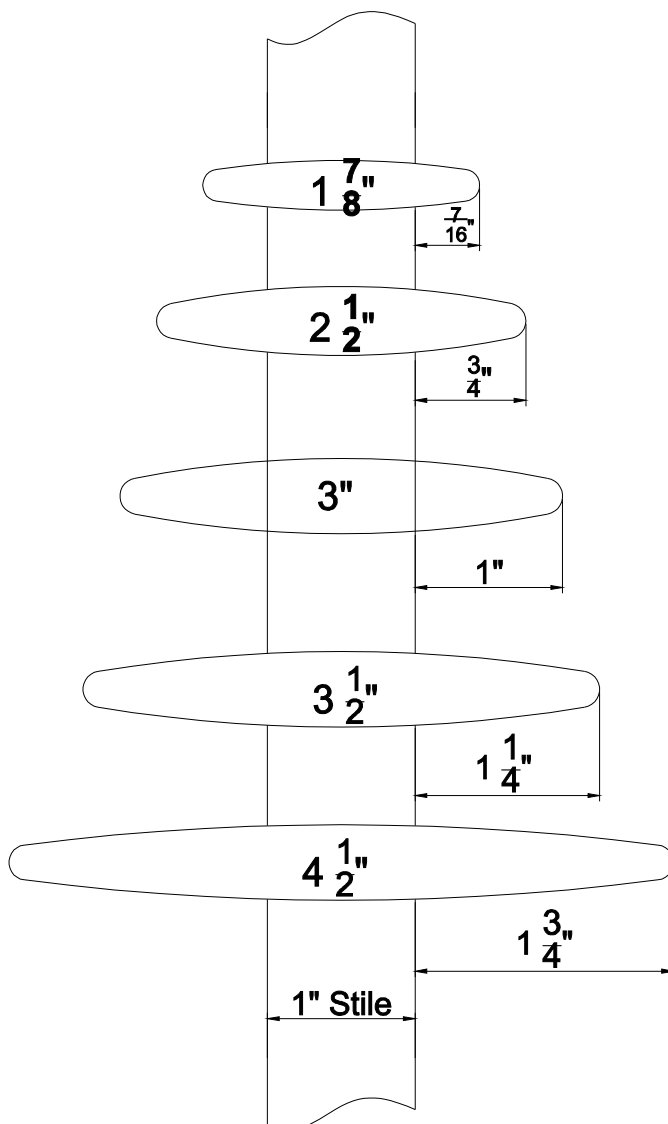
- Enhance view with extremely wide panels. Up to 42" in width for each single panel to create the clean, streamlined look.
- Reinforced engineered stile-- Multiple layers of wood bonded together to reinforce durability to the core of the stile, which results in a robust support system for the shutter panel.
- Employs mortise and tenon joinery that resists shear and racking forces endured by shutters to ensure prolonged quality and durability.
- Quick and easy on-site assembly and installation with the pre-installed magnets and hinges, pre-mitered frames and slotted keyholes.
- Special panel and frame design that is custom cut to accommodate every unique windows.
- Personalize shutters with special options. Choose from fixed louvers, center or offset tilt rod, double hung panels, café shutters, track systems, French door cutout, specialty shapes and many more.
- A guarantee of Norman Shutters exemplary quality with a lifetime limited warranty on materials and workmanship.

Colors & Louver Sizes/Louver Clearance

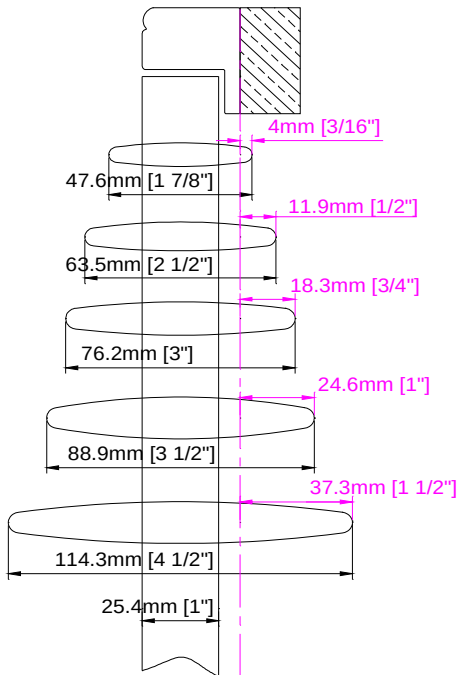
Color ID	Color Name
001	Pure White
002	Extra White
003	Silk White
004	Bright White
006	Pearl
007	Ivory Lace
009	Creamy
012	Crisp Linen
016	Butter
019	String
030	Moondust
032	Sea Mist
042	Hall Gray
046	Ice
048	Chai
049	Stone Gray
051	Brown Gray
053	Clay
063	Decorators White
066	Winchester White 2010
073	Sky Gray
074	Element Gray
075	Storm Gray
077	Dawn
078	Cloud Blue
080	Taupe Gray
836	Classic Black

Louvers	
Elliptical	1 $\frac{7}{8}$ "*
	2 $\frac{1}{2}$ "
	3"
	3 $\frac{1}{2}$ "
	4 $\frac{1}{2}$ "

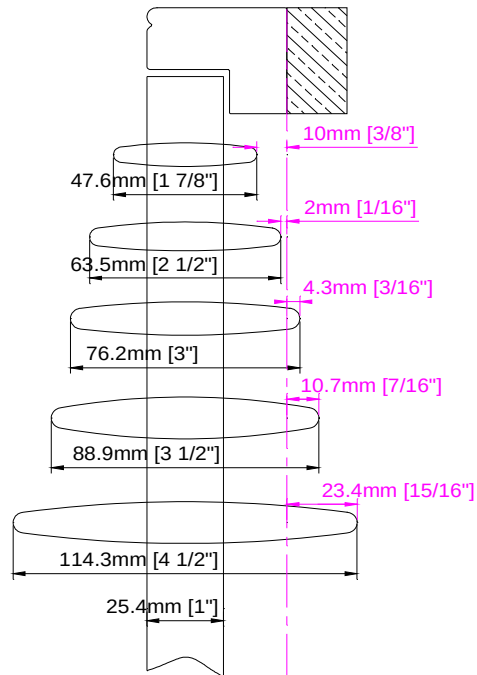
Louver Depth Clearance



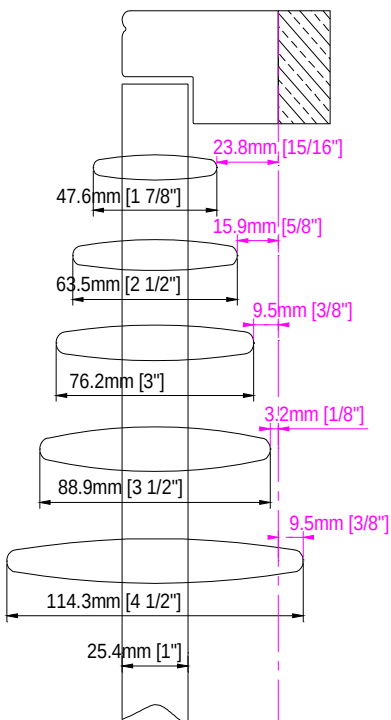
Louver Depth Clearance for Specific Frames



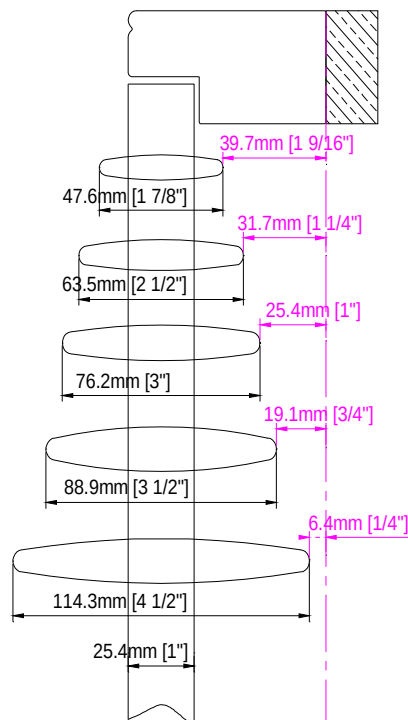
L Frame with 1/4" light block



L Frame

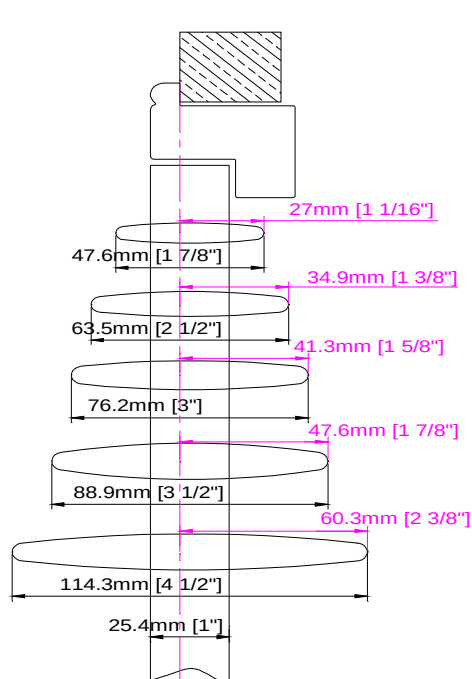


L Frame with 1/2" Buildout

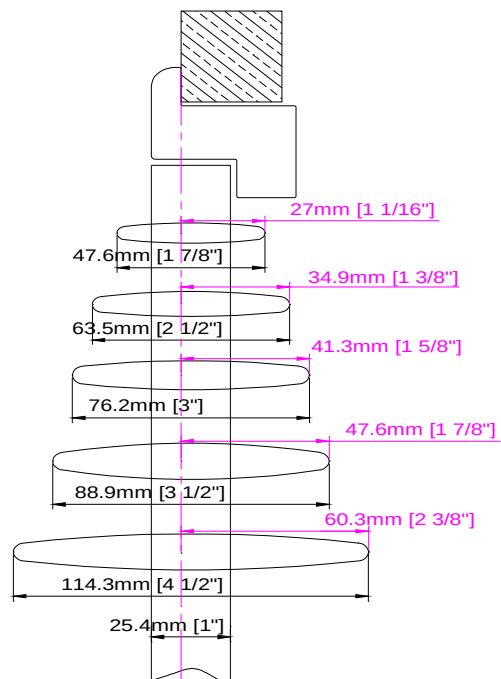


L Frame with 1" Buildout

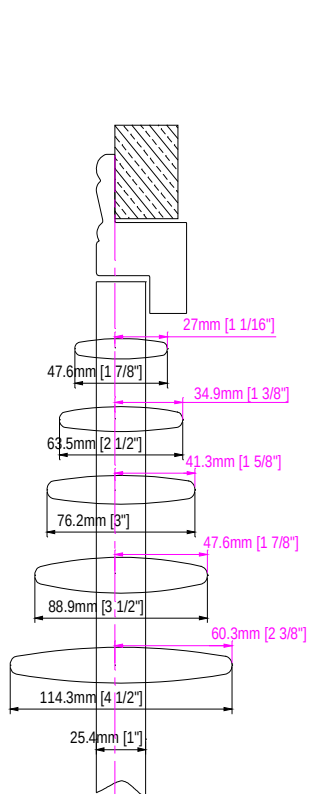
*Some surcharges apply



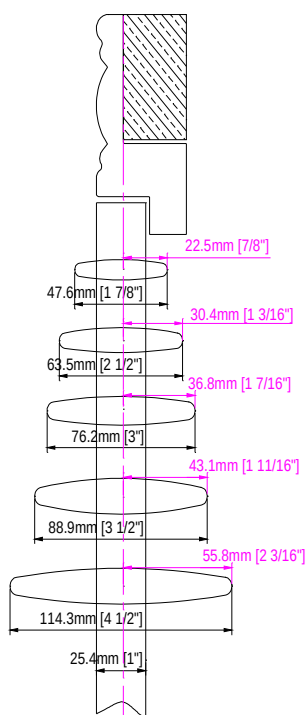
1 1/4" Beaded Z Frame



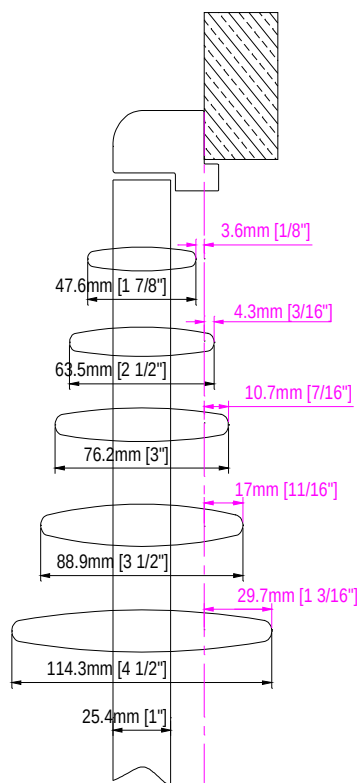
1 1/2" Bullnose Z Frame



2" Bel Air Z Frame

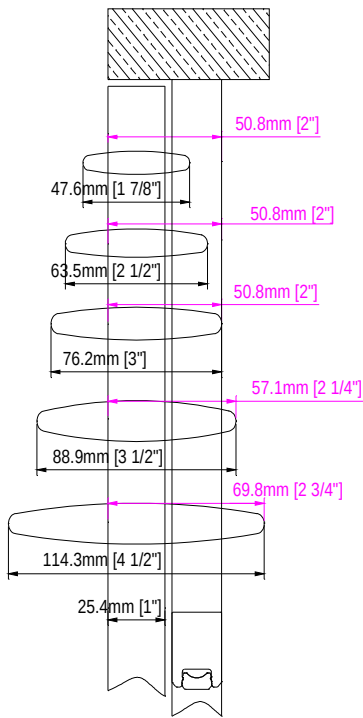


3" Crown Z Frame

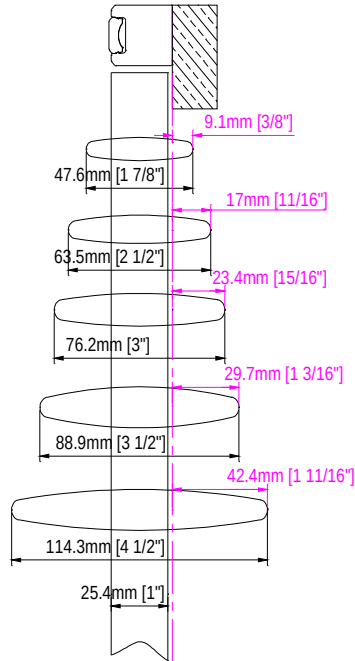


Tilt Out Z Frame

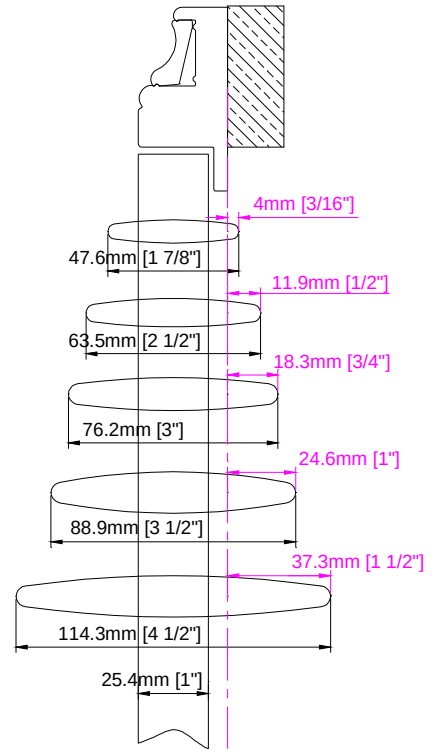
*Some surcharges apply



Hang Strip (Behind the panel)



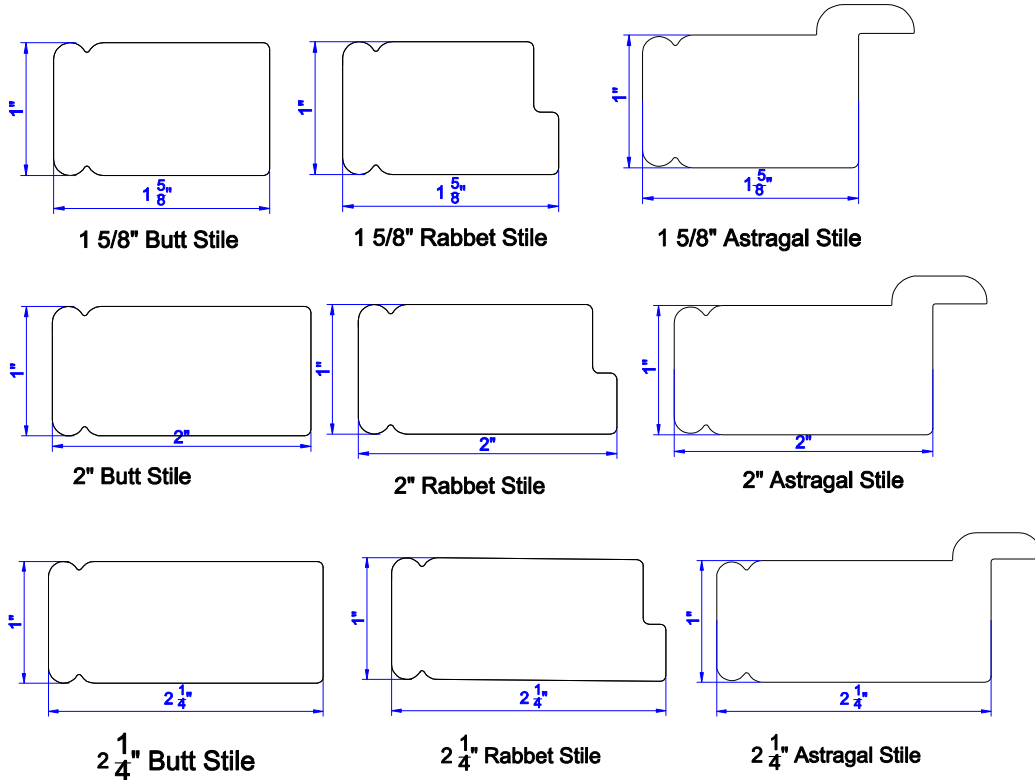
Hang Strip (Beside the panel)



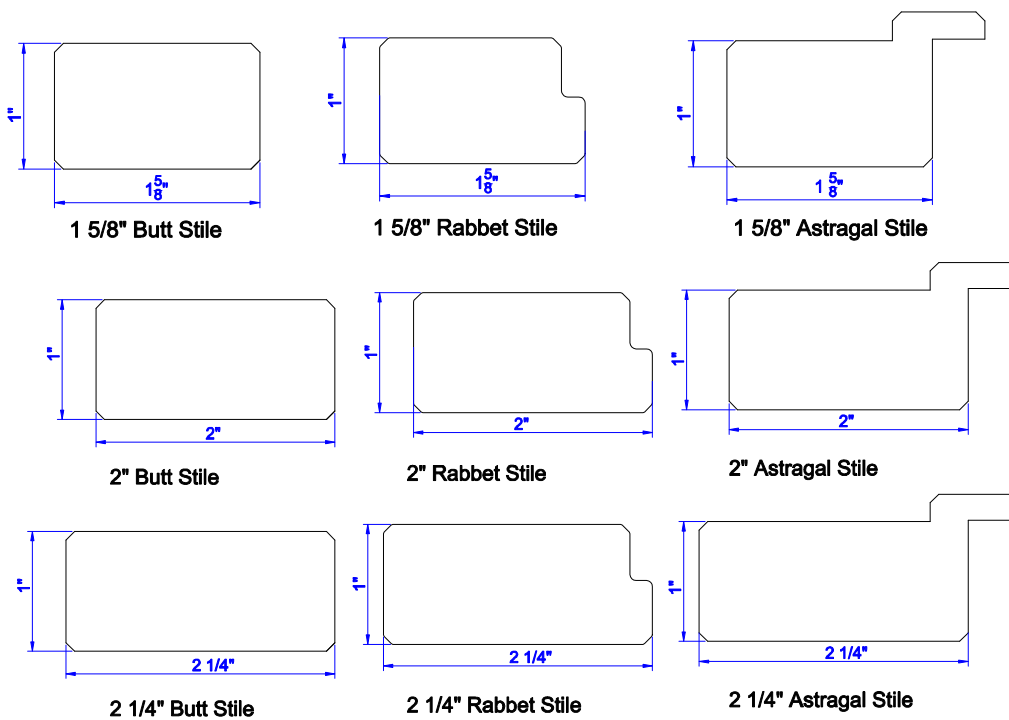
Deco Frame

Stiles

Default: Beaded stile



Optional: Chamfer stile

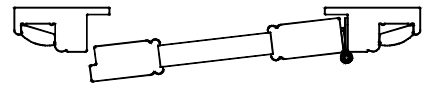


Stiles (continued)

Stile between panel closure

Single panel

- I) Panel width < 9"
 - Rabbeted stile (Deco frame, L frame, Z frame and Hang strip beside panel)
 - 1 ⁵/₈"
 - 2" (default)
 - 2 1/4"
 - Butt stile only (Hang strip behind panel)
 - 1 ⁵/₈"
 - 2" (default)
 - 2 1/4"
- II) Panel width ≥ 9"
 - Butt stile only
 - 1 ⁵/₈" (Optional for panel width ≤ 12")
 - 2" (default)
 - 2 1/4"

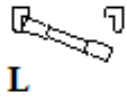


2 panels or more

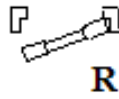
- Rabbet stile
 - 1 ⁵/₈" (Optional for panel width ≤ 12")
 - 2" (default)
- Astragal stile
 - 1 ⁵/₈" (Optional for panel width ≤ 12")
 - 2"
 - 2 1/4"

Panel Configurations

Single Panel



L



R

OR

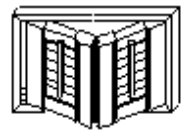
Two Panels



LR

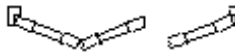
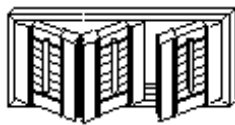


LL

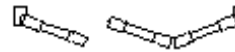


RR

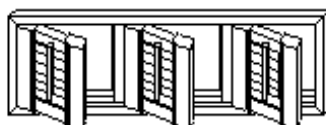
Three Panels



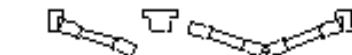
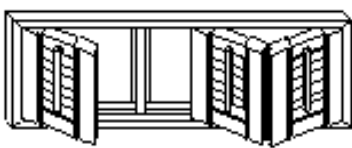
LLR



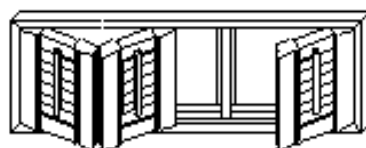
LRR



LTLTL

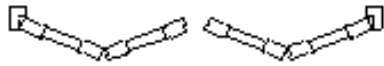


LTRR

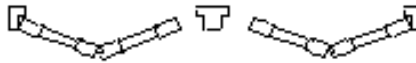
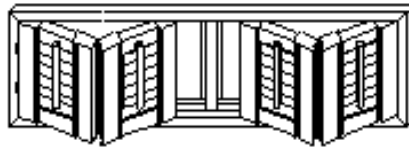


LLTR

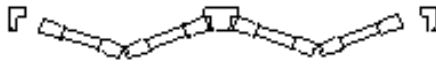
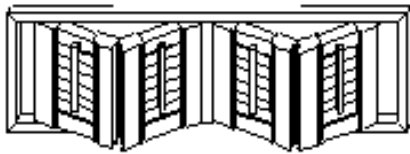
Four Panels



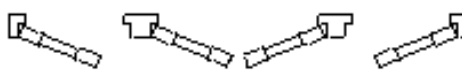
LLRR



LLTRR

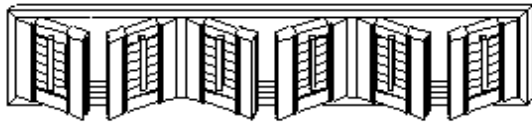


RRTL

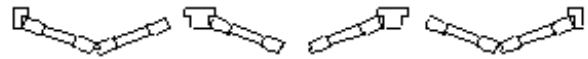


LTLRTR

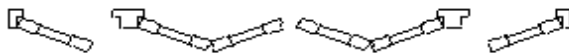
Six Panels



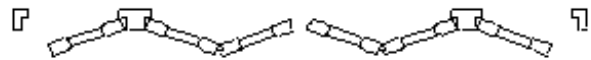
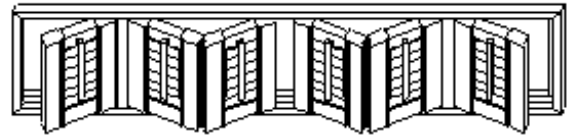
LRTLRTL



LLTLRTR



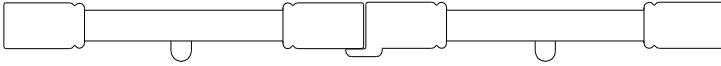
LTLRRT



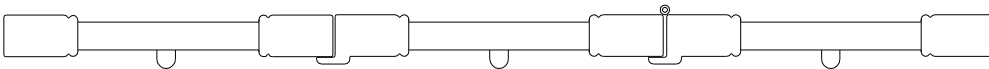
RTLRL

Panel Configurations – Astragal Stile

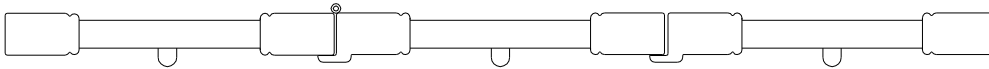
- 2 Panels – LR



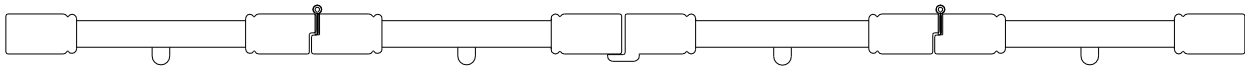
- 3 Panels - LRR



- 3 Panels – LLR

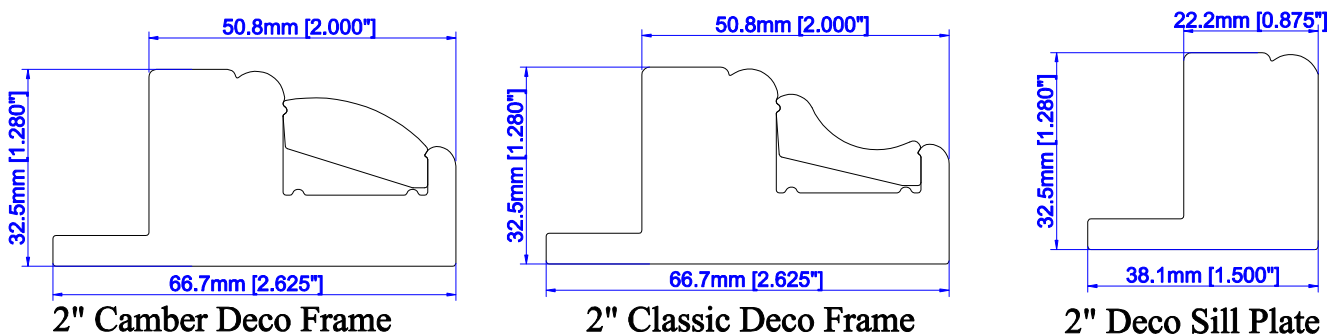
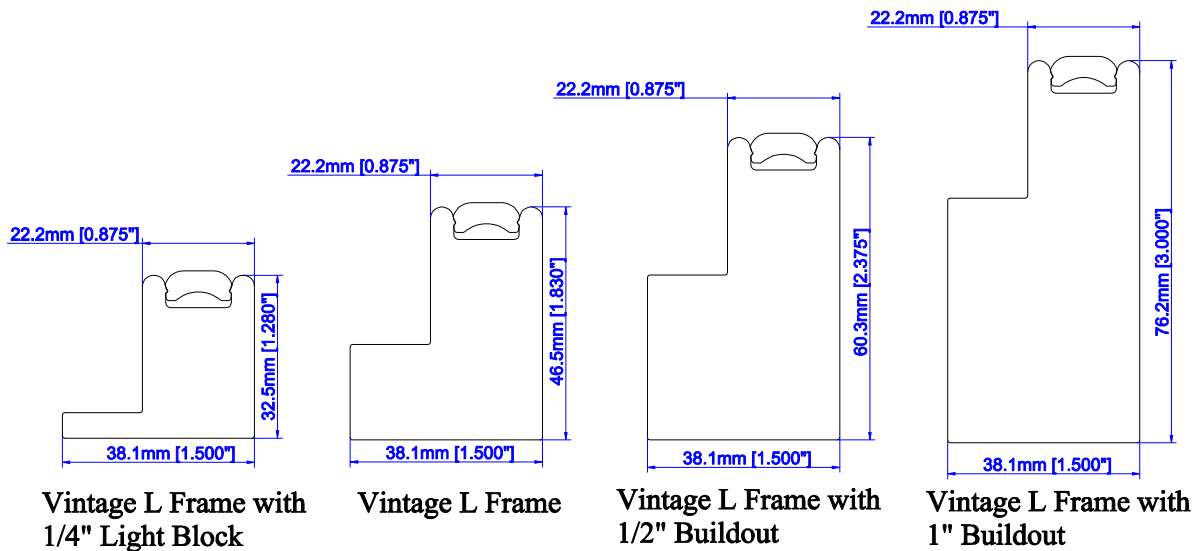
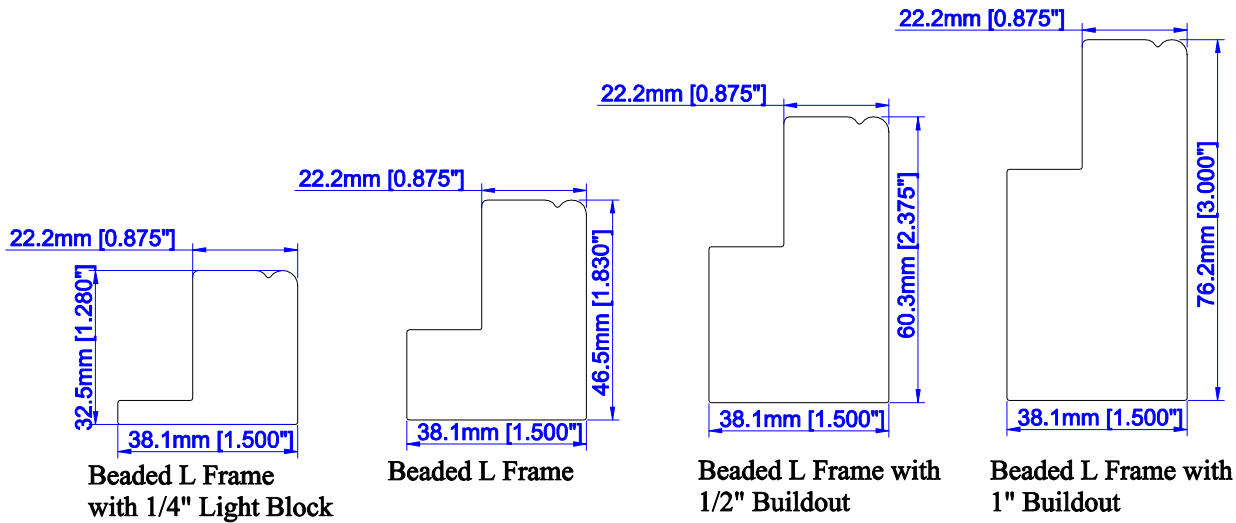


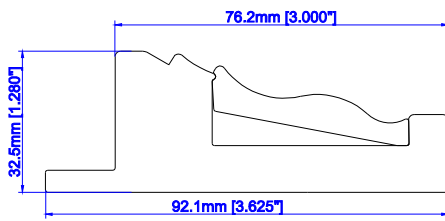
- 4 Panels - LLRR



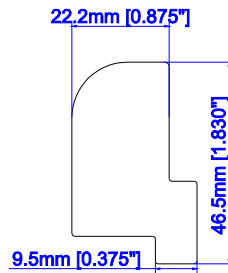
Frame Style

Frame material: Woodlore Plus

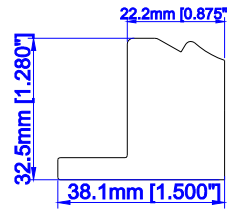




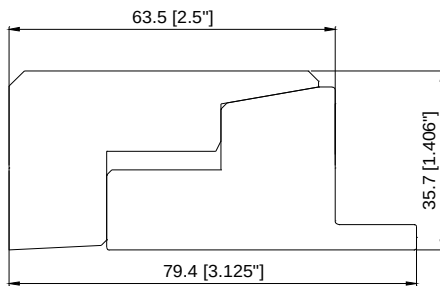
3" Ridge Deco Frame



Tilt Out Z Frame



Ridge Deco Sill Plate



2.5" Mission Deco Frame

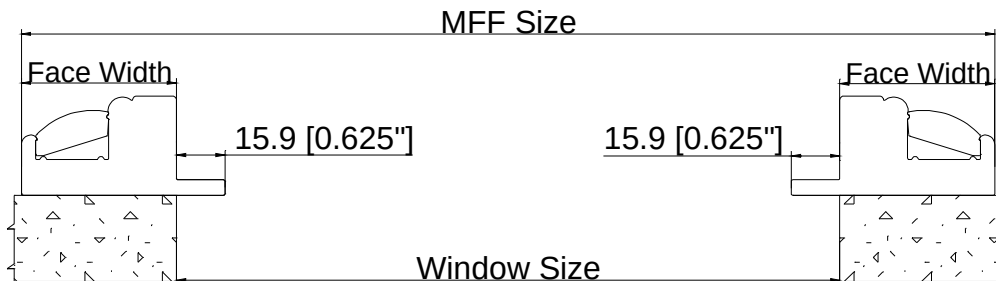
There are 2 options for the Deco frame installation when Window Size is given (Take Camber Deco frame as an example)

1. Default: light block part install into the window opening;

The window size is the exact size of the window opening;

The MFF size is the measurement of the entire shutter, including the frames;

The MFF size = Window Size + Face Width



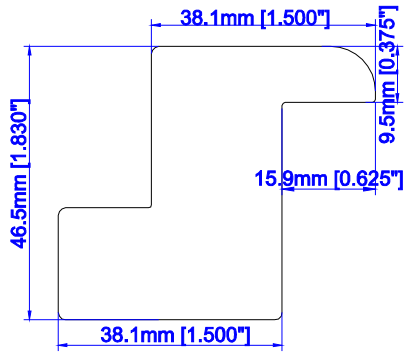
2. Optional: light block part flush with window opening.

The window size is the exact size of the window opening;

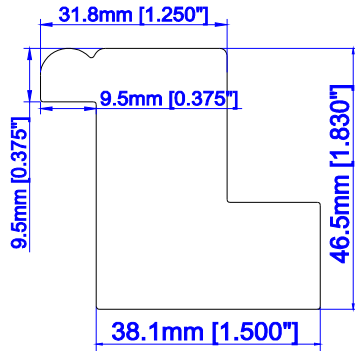
The MFF size is the measurement of the entire shutter, including the frames;

The MFF size = Window Size + Frame Back Width

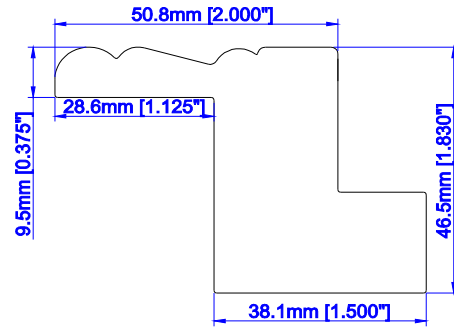




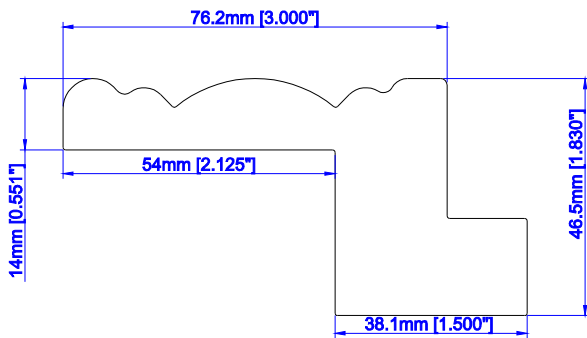
1 1/2" Bullnose Z Frame



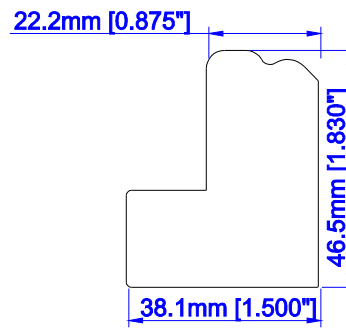
1 1/4" Beaded Z Frame



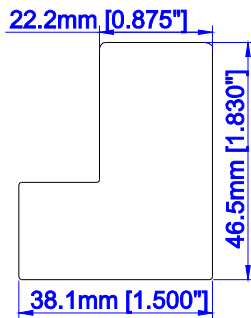
2" Bel Air Z Frame



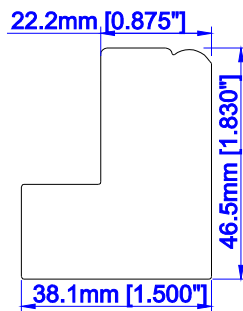
3" Crown Z Frame



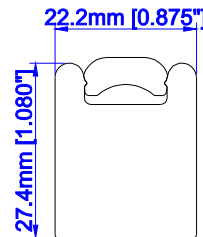
3" Crown Z Sill Plate



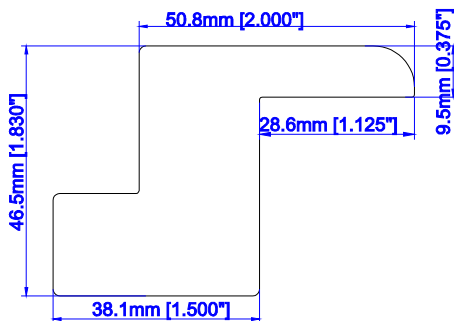
**Bullnose Z
1 1/4" Beaded Z
Sill Plate**



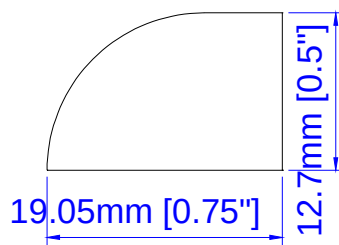
2" Bel Air Z Sill Plate



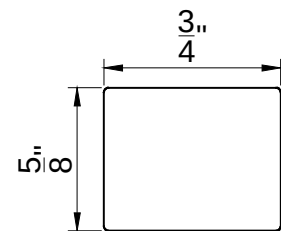
7/8" Vintage Hang Strip



2" Bullnose Z Frame

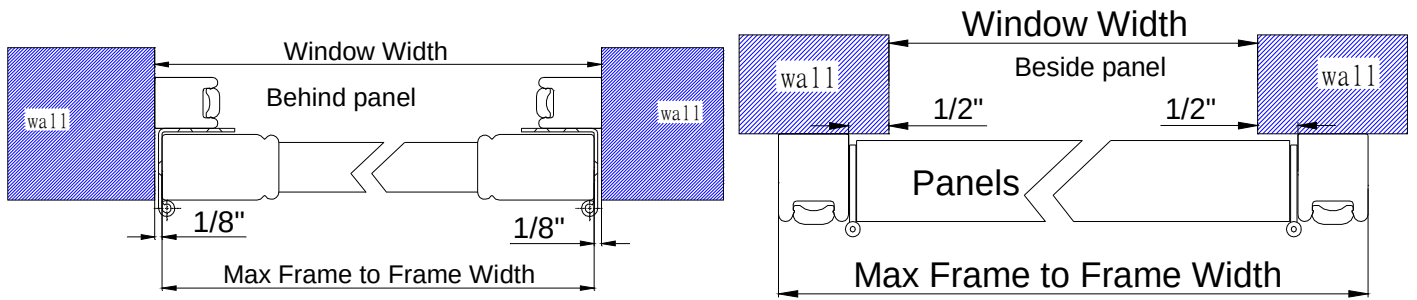


Quarter Round

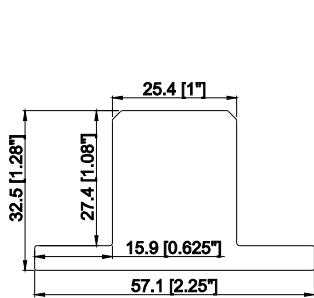


Light Block

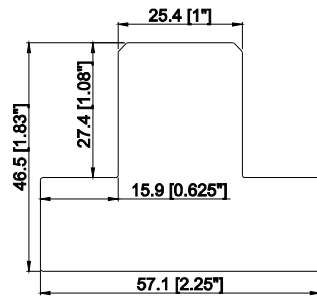
Note: Quarter Round will be made in Wood.



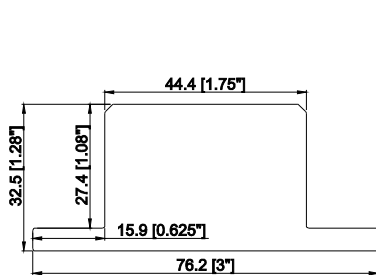
Note: Outside mounted hang strips, behind the panel, is not available for all 4 sides ("all around") or 3 sides inverted.



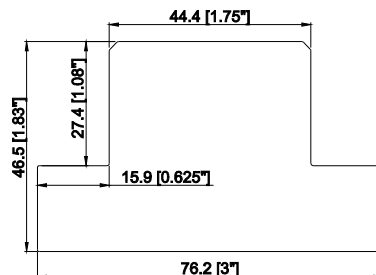
1" Chamfer T Post with 1/4" Light Block (Deco)



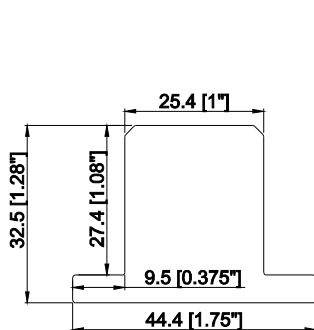
1" Chamfer T Post with 3/4" Light Block (L/Z/HS)



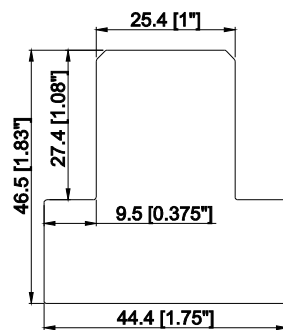
1.75" Chamfer T Post with 1/4" Light Block (Deco)



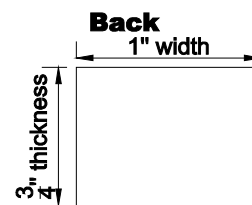
1.75" Chamfer T Post with 3/4" Light Block (L/Z/HS)



1" Chamfer T Post with 1/4" Thick x 3/8" Wide Light Block (Deco)



1" Chamfer T Post with 3/4" Thick x 3/8" Wide Light Block (L/Z/HS)



1" * 3/4" Post (No light block)

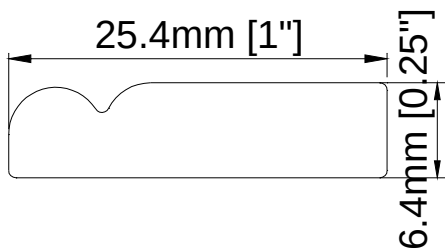
Note: 1" Chamfer T Post with 3/4" Light Block, 1 3/4" Chamfer T Post with 3/4" Light Block, 1" Chamfer T Post with 3/4" Thick x 3/8" Wide Light Block and Custom Chamfer T Post with 3/4" Thick Light Block are not available for all Deco frames and L frames with 1/4" Light Block.

- Custom width* up to 7 1/2" for T Posts (Made of Wood)
 - Size limits (7 1/2")

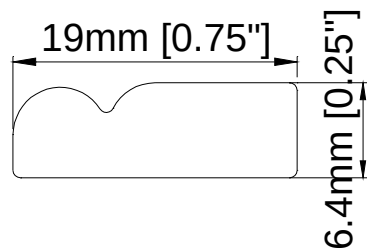
*Some surcharges apply

- Available for composite but it will be made from wood
- That a surcharge applies and customers should refer to their shutter pricing
- Any limitations

Note: The 1"x3/4" Post (No Light Block) is only available for Direct Mount. And no hardware will be supplied to install it on the existing window opening.



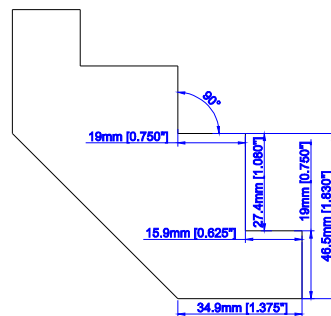
Scribe Trim 1



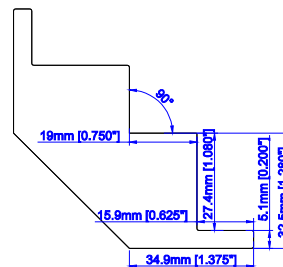
Scribe Trim 2

Note: Scribe Trim 1 & Scribe Trim 2 will be made of wood.

Standard Corner & Bay Posts

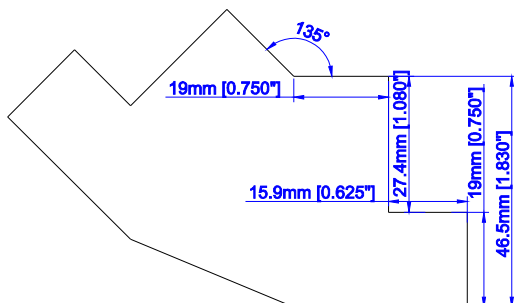


Corner Post with 3/4"
Light Block (L/Z/HS)

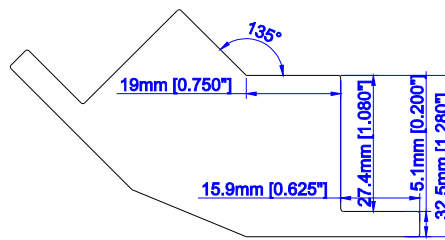


Corner Post with 1/4"
Light Block (Deco)

*Notch out corner post required if ordering Z Frame with solid drywall corner.



Bay Post with 3/4"
Light Block (L/Z/HS)



Bay Post with 1/4"
Light Block (Deco)

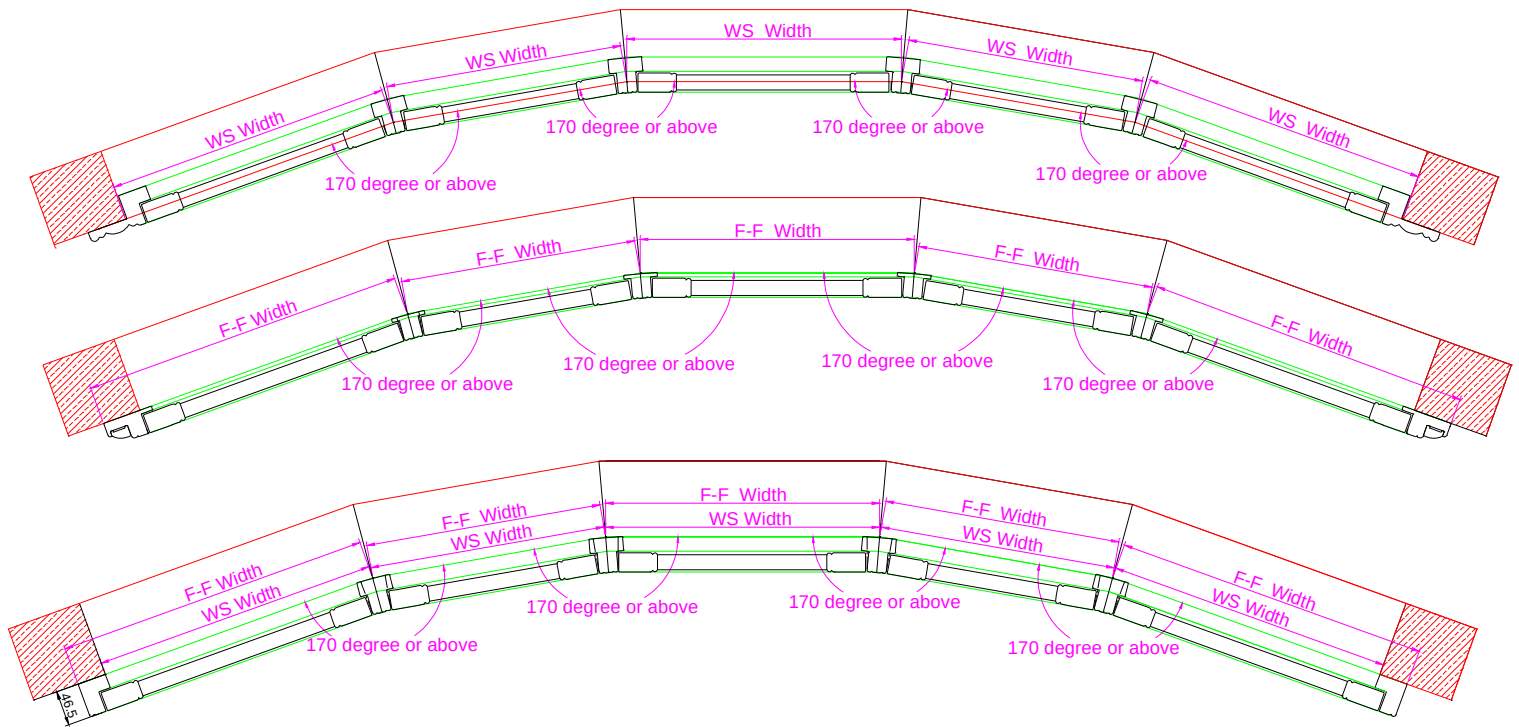
Also available

- Custom angle bay posts *

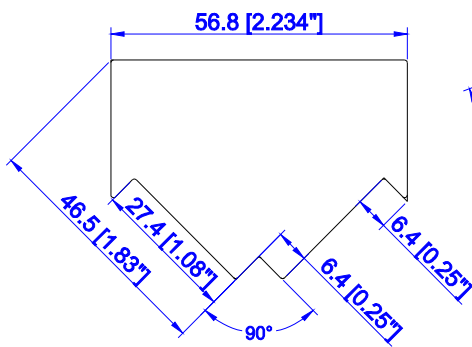
*Some surcharges apply

For bay window at 170 degree or above

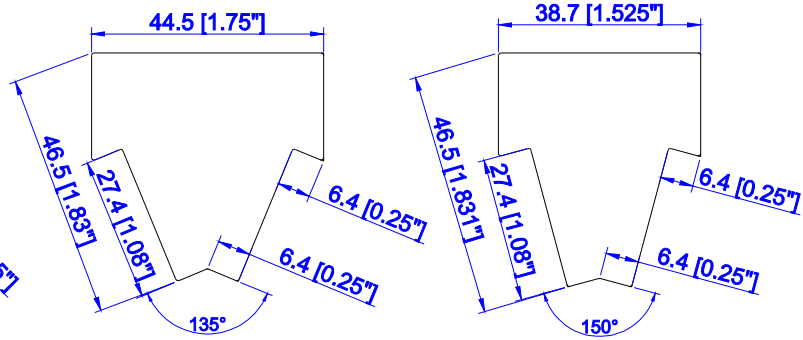
- 1" T Post with 3/8" light block will be used as Bay post;
- For Hang Strip (Behind Panel), bay post will be used.



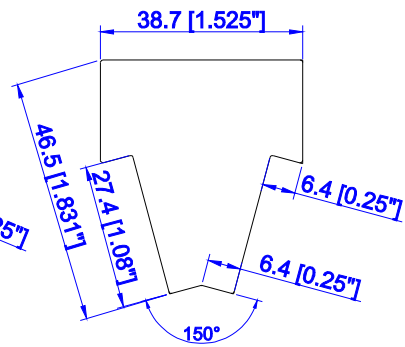
Slim Fit Corner & Bay Posts



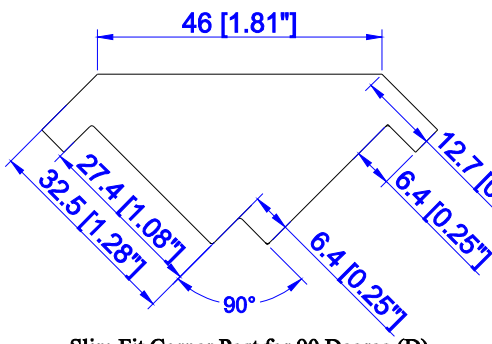
Slim Fit Corner Post for
90 Degree (L/Z/HS)



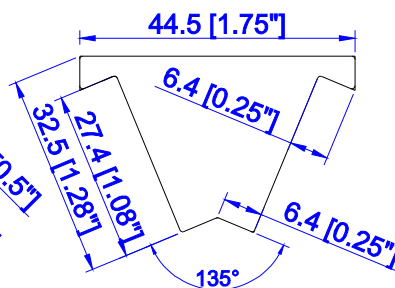
Slim Fit Bay Post for
135 Degree (L/Z/HS)



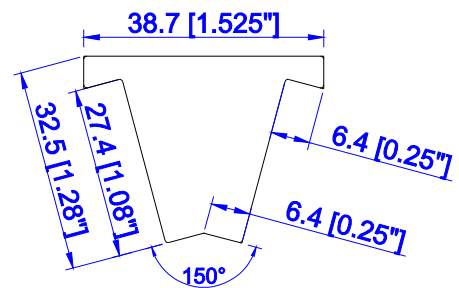
Slim Fit Bay Post for
150 Degree (L/Z/HS)



Slim Fit Corner Post for 90 Degree (D)



Slim Fit Bay Post for 135 Degree (D)



Slim Fit Bay Post for 150 Degree (D)

Note: Due to size and weight limitations, Slim Fit Posts may only be hinged on one side and with a single panel (no bi-folded panels). Please consider this when determining panel configuration.

*Some surcharges apply

- Slim Fit Posts are available in 90, 135 and 150 Degrees only.
- All Bay posts, Corner posts and Slim Fit Posts are made of Wood.

In order to better match corner windows / bay windows:

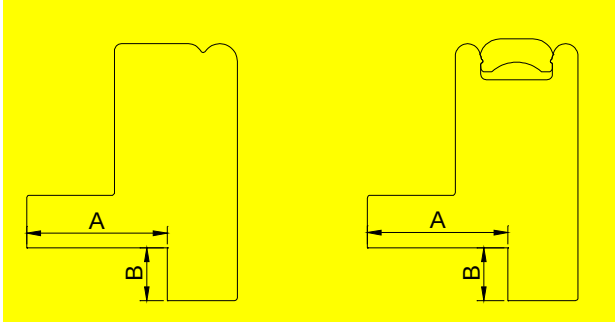
- **For non sample orders, please note that corner windows must have the same:**

- Post Type (Standard or Slim Fit)
- Color
- Hinge Options
- Louver Size
- Frame Type
- Mount Type
- Extension
- Sill Options
- T-Post Options
- Height
- Stile Type and Profile
- Tilt Type
- Divider Rail Options
- Double-Hung Options
- Split Tilt Options
- PerfectTilt Options

- **For non sample orders, please note that bay windows must have the same:**

- Post Type (Standard or Slim Fit)
- Color
- Hinge Options
- Louver Size
- Frame Type
- Mount Type
- Extension
- Sill Options
- T-Post Options
- Height
- Stile Type and Profile
- Tilt Type
- Divider Rail Options
- Double-Hung Options
- Split Tilt Options
- PerfectTilt Options

- Limitation for the Notch Out Size: $A \leq 1"$; $B \leq 1/2"$

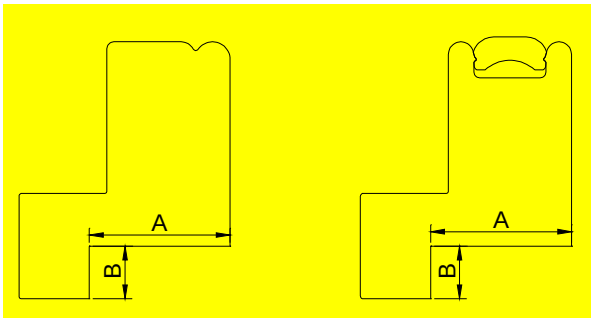


Notch Out

Beaded L Frame (A) *

Notch Out

Vintage L Frame (A) *



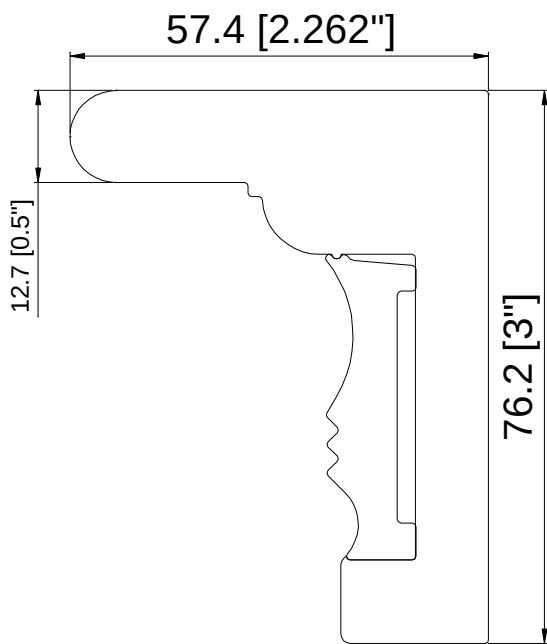
Notch Out

Beaded L Frame (B) *

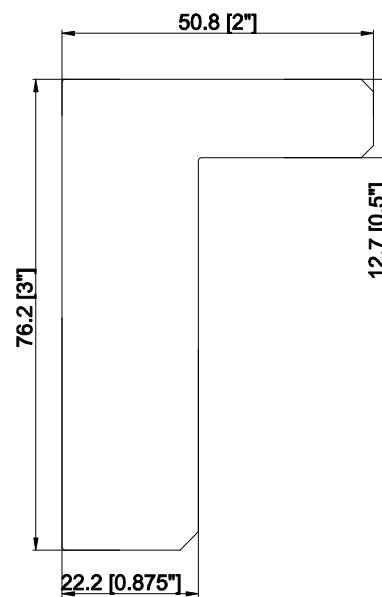
Notch Out

Vintage L Frame (B)

Sill Frames- For use where there is no existing sill

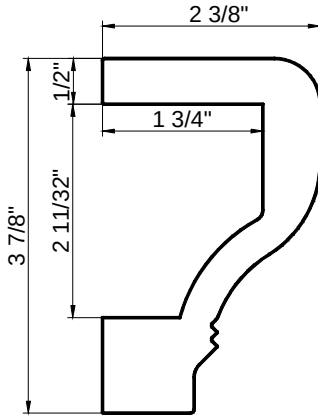


3" Deco Sill Frame

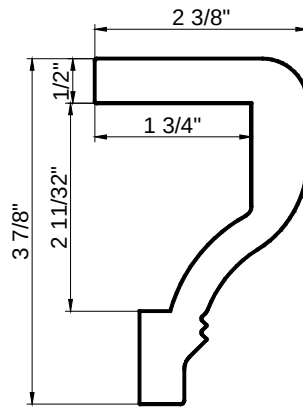


Mission sill Frame

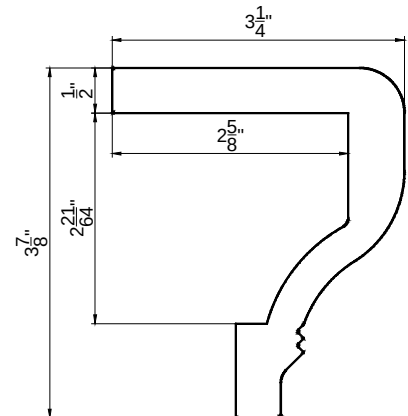
Sill Caps-Used to cover existing sills



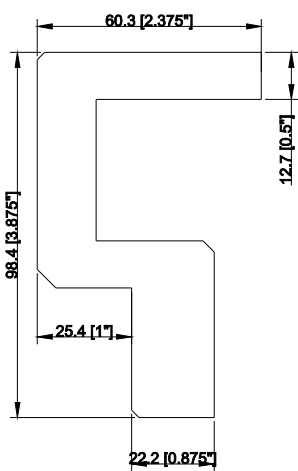
Designer Sill Cap (L)



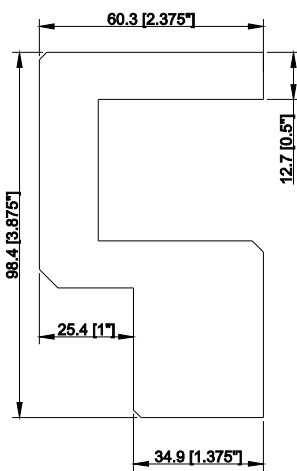
Designer Sill Cap (D/HS)



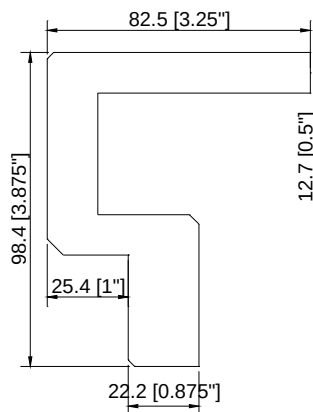
Designer Sill Cap (Z)



Mission Sill Cap (D/HS)



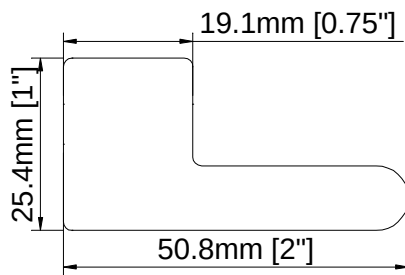
Mission Sill Cap (L)



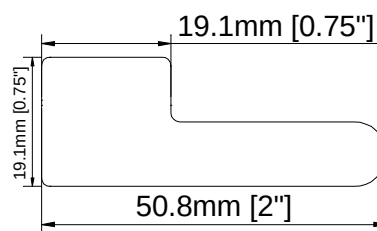
Mission Sill Cap (Z)

Note: For Mission Sill Cap, the extended size at each side needs to be 1" at least.

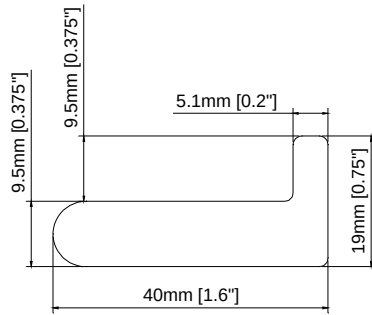
Sill Plates- Used with existing sills



2" Sill Plate with 5/8"
Light Block (L/Z)

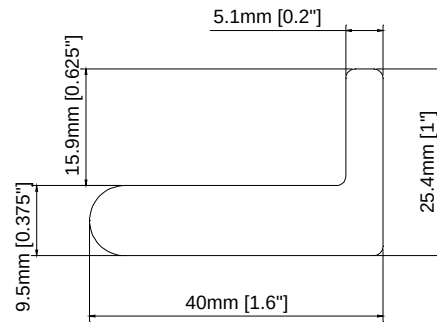


2" Sill Plate with 3/8"
Light Block (L/Z)



1.6" Sill Plate with 3/8" Light Block

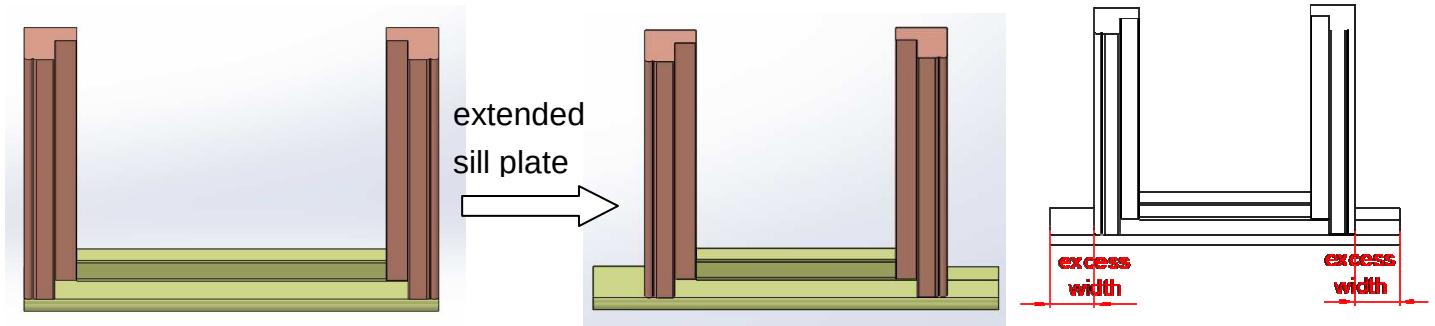
(All Deco & L frames with 1/4" light block)



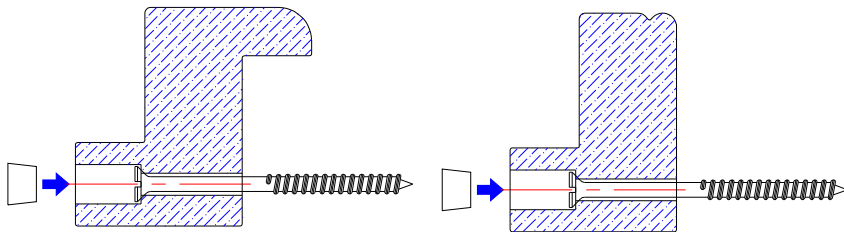
1.6" Sill Plate with 5/8" Light Block

(All Deco & L frames with 1/4" light block)

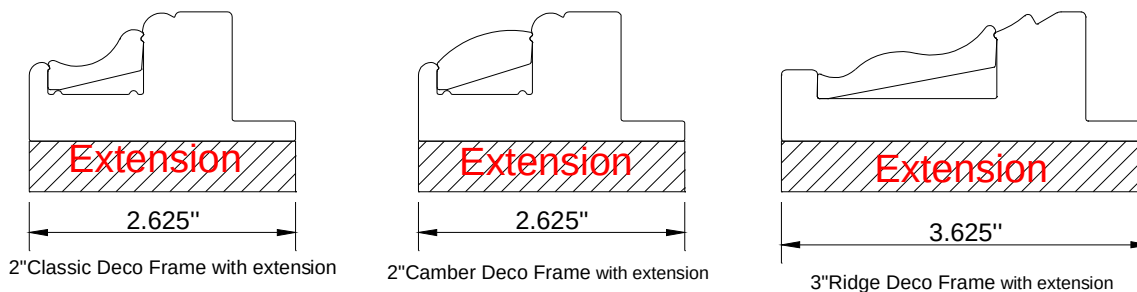
For 1.6" / 2" sill plate with 5/8" (or 3/8") light block, when an extended sill plate is requested, the excess width will be divided evenly on both sides, take the following drawing as an example for reference:



Pre-drill L Frames (Inside Mount Only) or Z frames:



Extension for Deco Frame* (Default to have full extension):



Note: Custom extensions* are available for Deco Frames, L frames & Hang strips (OM, beside the panel) & T posts, when the extension thickness is more than 2", the extension will not be attached to the frames and/or T posts.

Custom extensions* are available for the lips of all Z frames except Tilt out Z frames, when the extension thickness is more than 1", the extension will not be attached to the Z frames.

*Some surcharges apply

● Standard Filler Strips (width x depth):

1" x 3/4"	1" x 2 15/16"	3/4" x 1 1/2"
1" x 1 3/16"	2 7/16" x 7/16"	1" x 1 1/2"
1" x 1 9/16"	3/16" x 1 1/2"	1 3/16" x 1 1/2"
1" x 2"	3/8" x 1 1/2"	1 3/8" x 1 1/2"
1" x 2 3/8"	9/16" x 1 1/2"	1 9/16" x 1 1/2"

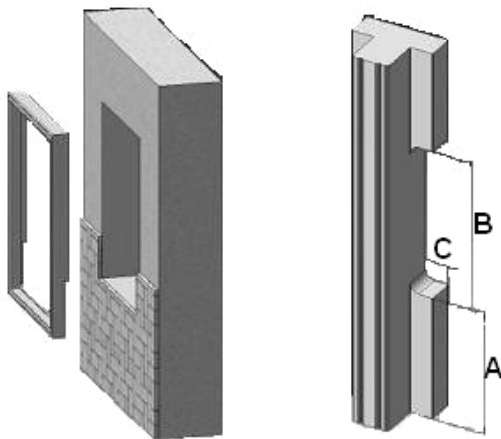
Note:

1. The width or depth of Custom filler strip / Standard filler strip can't be less than 1/8".
2. The width or depth of Custom filler strip / Standard filler strip can't be greater than 7 7/8".
3. The width and depth of Custom filler strip / Standard filler strip can't be greater than 3 15/16" at the same time.

Notch out*:

- o L Frame
- o T Post

Specify dimensions in special instruction box.



Default principle:

Regarding shutter with horizontal T-post type issue: Horizontal T-Post type will always be the Chamfer T post unless otherwise specified.

General Product Specifications

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>
Single Panel Width (Regular)	6"	30" for 1 7/8" louver 36" for 2 1/2" & 3" louver 42" for 3 1/2" & 4 1/2" louvers
Single Panel Width (Bi-fold)	6"	26"
Single Panel Width (Bi-fold Track)	6"	26"
Single Panel Height	10"	132"
Shutter Width (Without T Post)	N/A	4x single panel width (bifold) + frame
Shutter Width (With T Post)	N/A	Unlimited (For orders shipped by sea, frames longer than 142" will be spliced in the middle. For orders shipped by air, frames longer than 120" will be spliced in the middle.)
Shutter Height	N/A	132" + frame

Divider rail is required for panel heights over 78".

Standard divider rail size is 3", Custom Size (3"~7.875") Divider Rail is available with surcharge.

Please note the standard top/bottom rail size should be within 2.75"~6.5". For regular shutters, the step for the rail size is 1/8".

Shutters to be installed where there is no bottom supports are not available.

Shutters without bottom supports may sag and deform, including:

1. Shutters with a bottom gap over 0.1";
2. Shutters with no frame or existing bottom sill to support the panel (or shutter has bottom frame/existing sill, but the panel is frequently left in the open position).

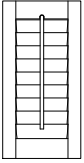
Side by Side / Match Line Number:

When **Side by Side** is selected, it is solely to match top or bottom rails and the number of hinges will also be matched when the panel heights are within +/- 2". When selected, the option to match with another line number or previous work order is solely for matching the top or bottom rails. Neither option will be used to match any other specifications.

Tilt Rods

Standard Tilt Rod

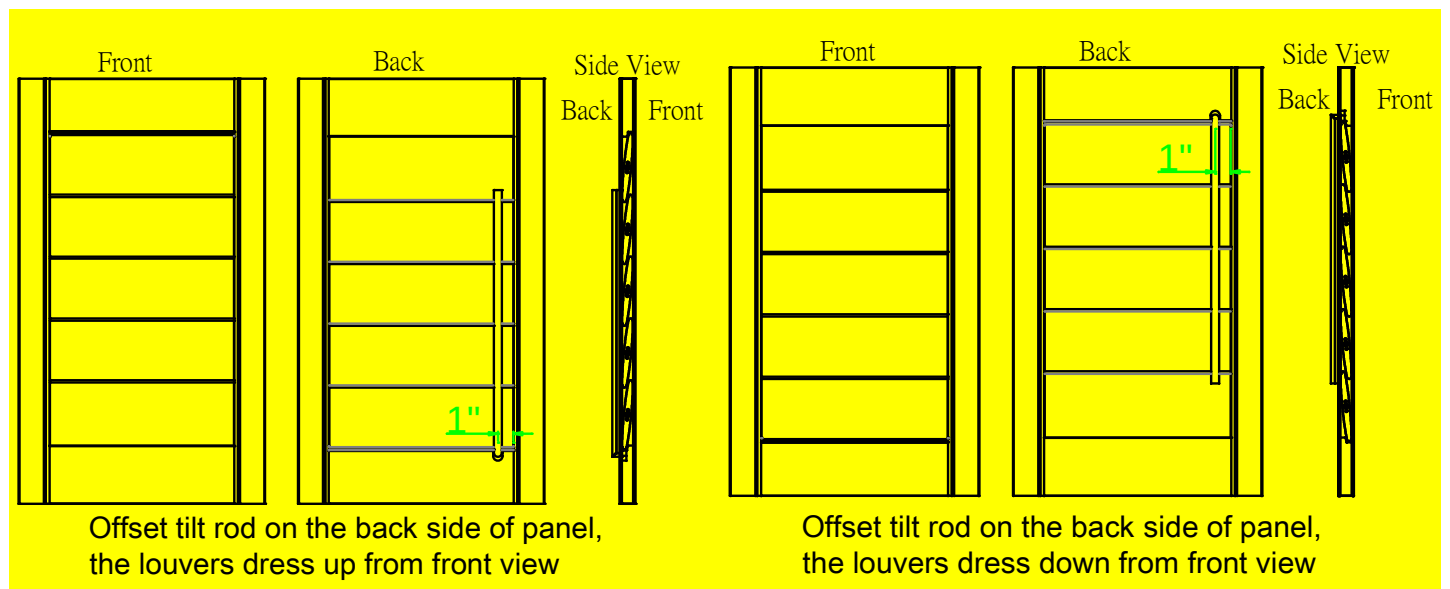
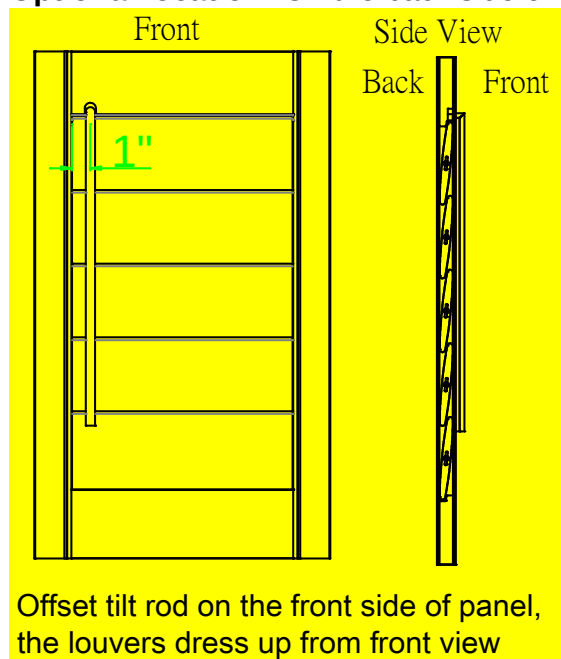
Default Location: In the middle of the panel on front side (default).



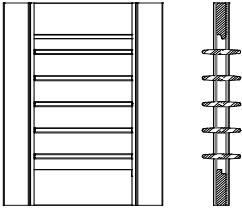
Offset Tilt Rod*

Default Location: On the front side of the panel and at 1" from the louver ends on the hinged side.

Optional location: On the back side of the panel, 1" from the end of the louvers on the hinged side.



InvisibleTilt[™] *



Louver tilting system embedded within panel stile. It is not available for $1\frac{7}{8}$ " louver. The tilting system is split when the number of louvers exceeds the limits as listed below:

Louver Size	Maximum Possible Louvers	Panel Height Range
2½"	24	54 ¾" – 57 ¼"
3"	19	54 ¼" – 57"
3½"	16	54 ¾" – 58"
4½"	12	54 ¾" – 60"

For even numbers of louvers, the number of louvers operated by the top and the bottom tilting system will be the same;

For odd numbers of louvers, the number of louvers operated by the bottom tilting system will be one more than the number of louvers operated by the top tilting system;

To ensure split tilt is included, always specify on the order.

Split Tilt

When selected, the louvers are split evenly by default unless a custom location is requested.

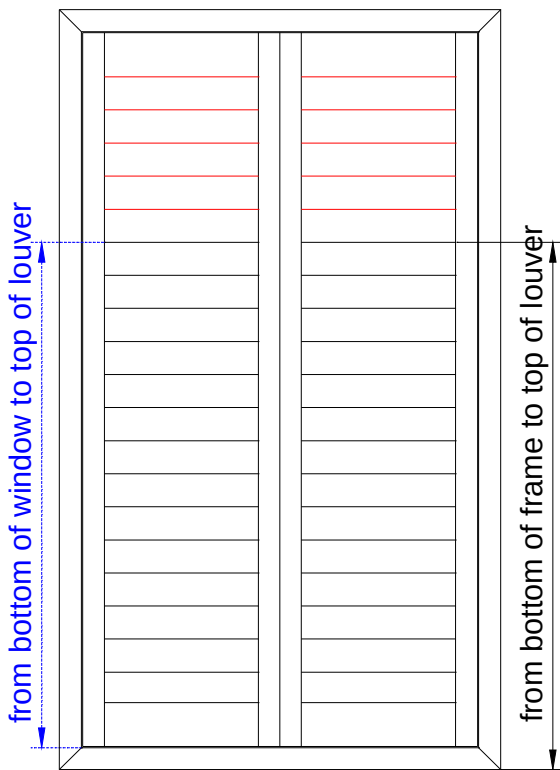
When the customer selects a custom location, the measurement will be as follows (please note that the louvers are in the closed position when measured):

Window size:

- Measure from the bottom of the window to the top of the louver

Max frame-to-frame:

- Measure from the bottom of the frame (or panel if there is no bottom frame, sill or sill cap) to the top of the louver.



Split tilt allowable deviations

Louver Size	Allowable ± Deviation
1 7/8"	3/4"
2 1/2"	1"
3"	1 1/4"
3 1/2"	1 1/2"
4 1/2"	2"

Split Location:

— Per the customer's specification. However, the location of the top louver of the bottom split section may be adjusted to achieve more balanced top and bottom rails when **allow deviations** is selected during order entry. If necessary the location will be moved up from the desired height to ensure any obstruction is covered. The amount of the deviation from the desired height will depend on the size of the louver selected.

If the exact location of the split is critical, customer may select **exact location** in the order entry system when placing the order. This may result in significantly different top and bottom rails. When exact location is selected, a popup will reveal the resulting top and bottom rails measurements from which the customer may either:

1. Accept the listed top and bottom rail sizes
2. Modify the split location (sometimes a slight adjustment in the exact split location may achieve desired top and bottom rail sizes)
3. Change to **allow deviations** so the system will calculate the split location

Note: Top and bottom rail sizes may change if the current line item is being matched with another line item. Carefully review all top and bottom rail sizes and split locations to ensure a desirable look is achieved

Divider Rails

A divider rail is an extra rail in the middle of the panel.

Creates two sections in the panel so louvers can be operated independently.

Strengthens the panel and enhances structural integrity.

Required for panel heights over 78"

Custom Size(3"~7.875") Divider Rail is available with surcharge.

Specifications

Standard size: 3"

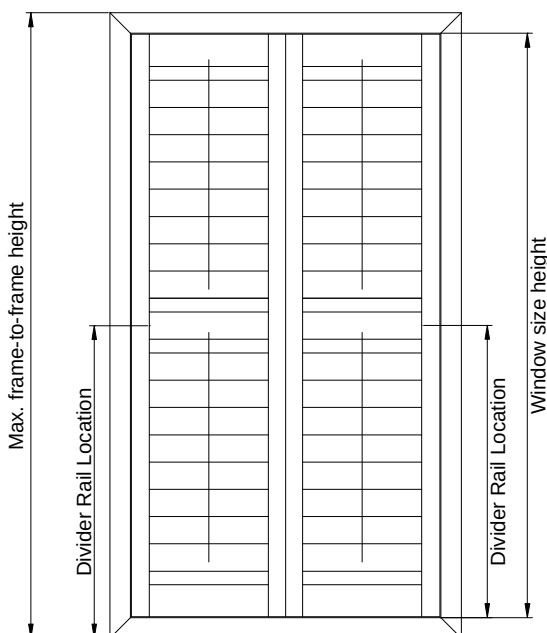
Default location: center of the panel unless specified otherwise

If the Divider Rail Location must be accurate, "Exact Location" must be selected or specified when placing an order.

Measurements:

- Window size:
 - Measure from the bottom of the window to the center of the divider rail.
- Max frame-to-frame:
 - Measure from the bottom of the frame to the center of the divider rail.
- Position:
 - Per customer's specification. However, the divider rail may be adjusted to a lower or higher position, within our allowable deviation, to ensure reinforcement and aesthetics of the shutter.

When placing the order, please clearly specify if the divider rail position needs to line up with that of other shutter divider rails in the same room. Please also refer to the table below for the allowable deviation details.



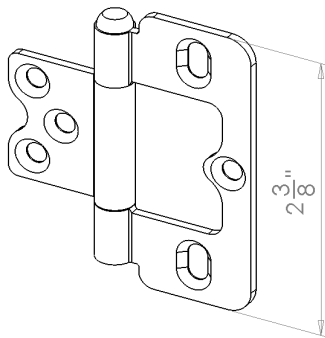
Divider rail allowable deviations

Louver Size	Allowable $\pm$ Deviation
1 $\frac{7}{8}$ "	$\frac{3}{4}$ "
2 $\frac{1}{2}$ "	1"
3"	1 $\frac{1}{4}$ "
3 $\frac{1}{2}$ "	1 $\frac{1}{2}$ "
4 $\frac{1}{2}$ "	2"

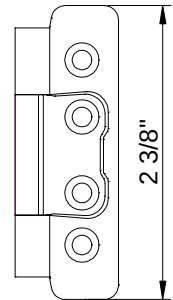
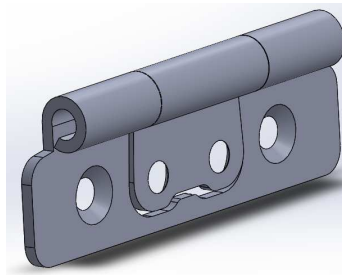
Hardware

Hinges

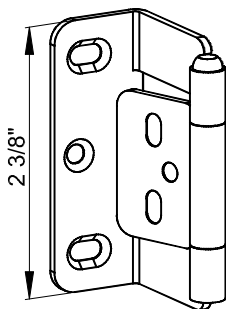
2.375" Self-Mortise Hinge



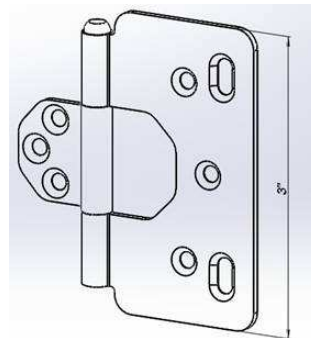
2.375" Bi-fold Hinge



2.375" L Hinge



Direct Mount 3" Hinge



Hinges are available in:

Pure White, Silk White, Bisque, Pearl, Bright Brass, Antique Brass, Black, Nickel plated, Brushed Nickel, Stainless Steel*, Crisp Linen, String, Sea Mist, Stone Gray, Brown Gray, Taupe Gray.

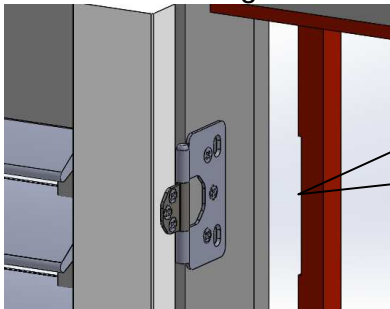
While hinge colors of Crisp Linen, String, Sea Mist, Stone Gray, Brown Gray, Taupe Gray are not available for Direct mount 3" Hinges.

False hinges are not available.

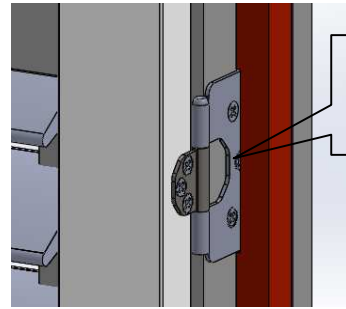
- Direct Mount 3" Hinge:
 1. The hinge position of the direct mount 3" hinge will be different from that of the 2.375" ones. For shutters used both the 2.375" and the direct mount 3" hinge, the hinge position of the 2.375" ones will match that of the direct mount 3" hinge;
 2. The hinge is only to install between the panel and the existing window. For other situation, 2.375" hinges will be used;

*Some surcharges apply

3. To avoid conflict, there will be cut-out for the light block of the hinged side. Please refer to the drawing below.

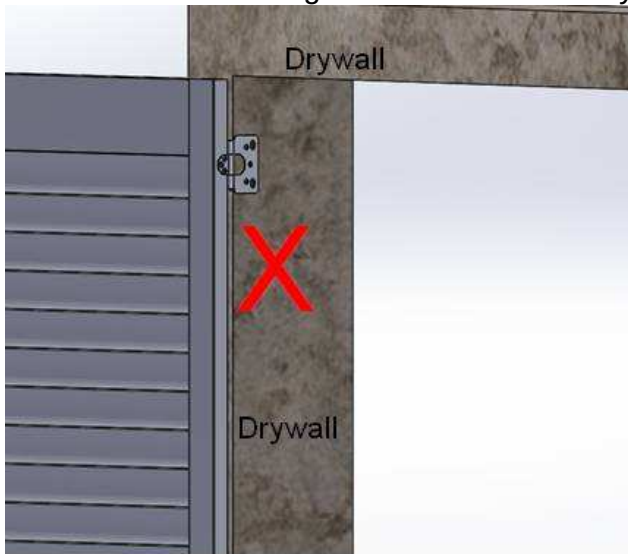


Cut-Out on
the light
block



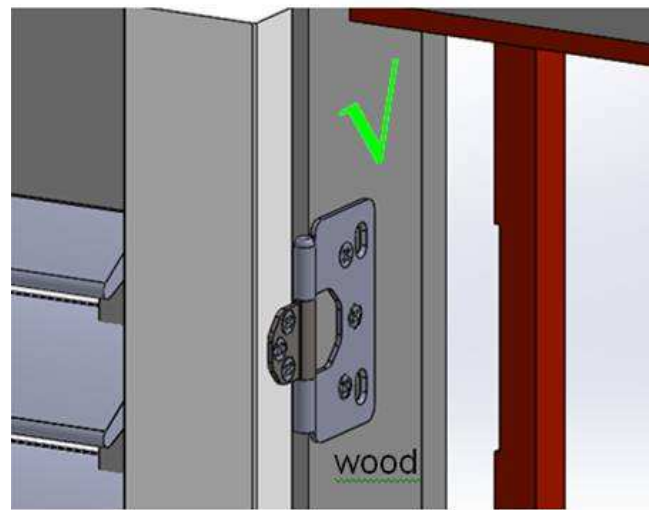
Cut-Out when the
hinged is installed.

4. Direct Mount 3" Hinge is not suitable for drywall installations.



Direct Mount 3" hinge is not suitable for drywall installations.

It's not recommended.



It's acceptable.

Invisible Hinge*:



1. Invisible hinge is not available for multifold panels.
2. Invisible hinge can only be between the panel & frame/T post, connections from panel to panel will use a regular hinge.
3. Color Option: Only available for Pearl Chrome Plated.

*Some surcharges apply

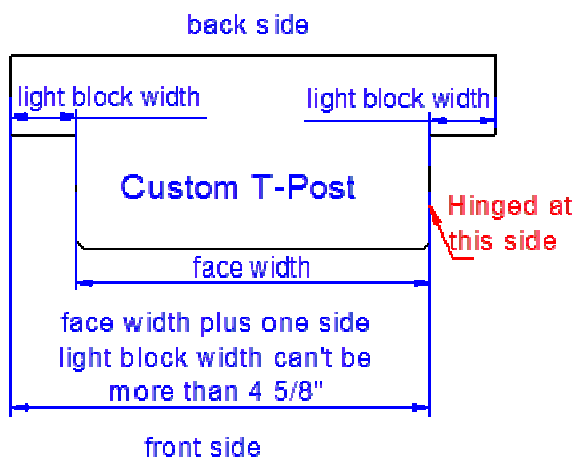
Frame Options	Hinge	Notes
2" Camber Deco Frame	Y	/
2" Classic Deco Frame	Y	/
2.5" Mission Deco Frame	N	/
3" Ridge Deco Frame	N	/
Vintage L Frame W / 1/4" Light Block	N	/
Vintage L	N	/
Vintage L Frame w/ 1/2" Buildout	N	/
Vintage L Frame w/ 1" Buildout	N	/
Beaded L w 1/4"light block	Y	/
Beaded L	Y	/
Beaded L Frame w/ 1/2" Buildout	Y	/
Beaded L Frame w/ 1" Buildout	Y	/
Tilt out Z Frame	N	/
1.25" Beaded Z Frame	Y	/
1.5" Bullnose Z Frame	Y	/
2" Bullnose Z Frame	Y	/
2" Bel Air Z Frame	Y	/
3" Crown Z Frame	N	/
7/8" Vintage Hang Strip	N	/
1" Chamfer T-Post w/ 1/4" Light Block(for Deco)	Y	only available with hinge on one side, for example, LTL is available, but RTL is not.
1" Chamfer T-Post w/3/4" Light Block(for L/Z/HS)	Y	
1" Chamfer T-Post w/ 1/4" thick 3/8" wide Light Block(for Deco)	Y	
1" Chamfer T-Post w/3/4" thick 3/8" wide Light Block(for L/Z/HS)	Y	
1 3/4" Chamfer T-Post w/1/4" Light Block(for Deco)	Y	available with hinge on BOTH side, for example, RTL.
1 3/4" Chamfer T-Post w/3/4" Light Block(for L/Z/HS)	Y	
Corner Post w/1/4" Light Block(for Deco)	Y	/

Corner Post w/3/4" Light Block(for L/Z/HS)	Y	/
Bay Post w/1/4" Light Block(for Deco)	Y	/
Bay Post w/3/4" Light Block(for L/Z/HS)	Y	/
Invisible Hinge Qty		
Panel Height (H)	Hinge Qty	
H<=48"	2	
48"<H<=78"	3	
78"<H<=96"	4	
96"<H<=120"	5	
H>120"	6	

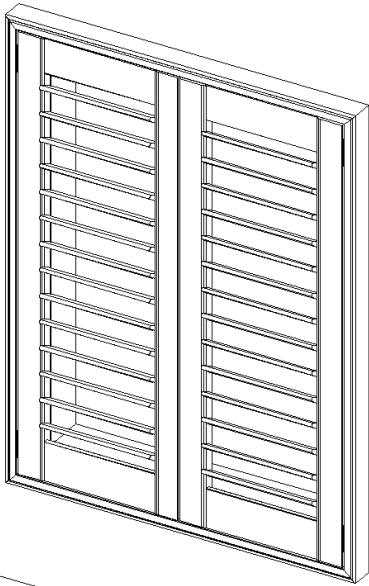
Preinstallation:

Panels and frames will be packaged individually, and the hinge is preinstalled onto the frame/post, with the screws for the other half part on the panels in the hardware box.

Note: If Custom T-posts need to install Invisible Hinge, the size of Custom T-post face width plus one side light block width can't be more than 4 5/8"; Please see drawing for reference.

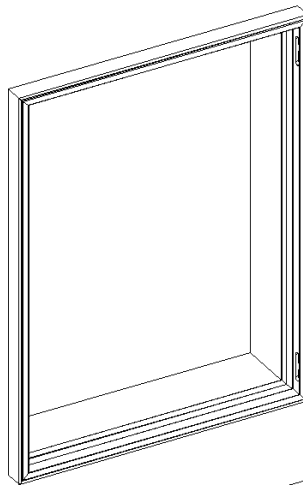
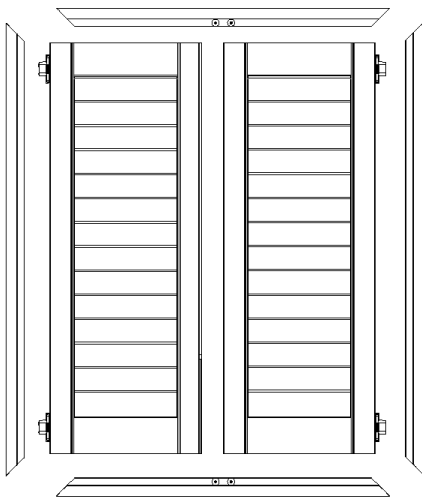


Installation Instructions for the Invisible hinges
Take L Frame as an example:

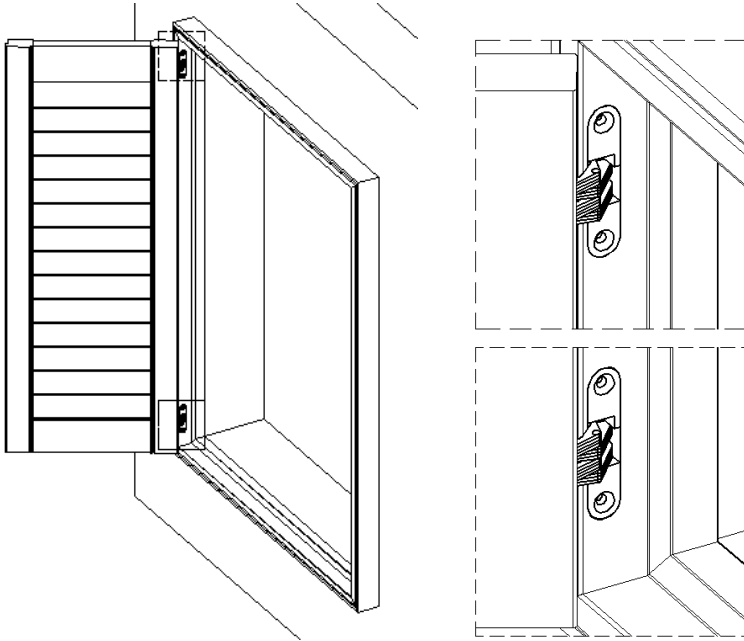


- 1). Remove the corresponding panels and frames from their packaging and match the panels with the correct frames.

Install the frames on the corresponding windows and make sure they are square (diagonals are equal)



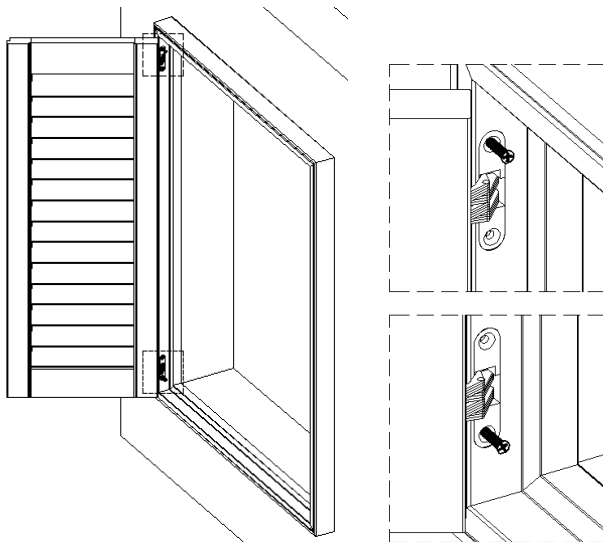
- 2). Please insert **ALL of** the Invisible hinges on the panel into the hinge holes on the corresponding frames prior to trying to secure any single hinge. This may require the assistance of a second person to hold the panel to prevent the hinges from sliding out of their slots while the screws are secured (Do not let the panels hang from just one hinge).



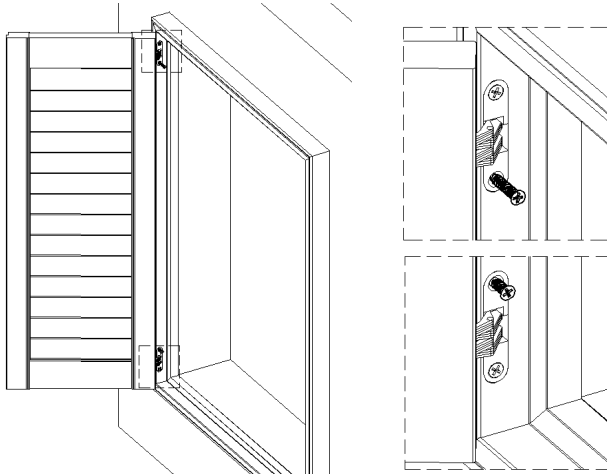
3). To properly secure the panels, first please tighten the upper screw of the uppermost top hinge and the lower screw of the lowest hinge. Close the panel to check whether the shutter closure is normal and the gap is uniform.

Note:

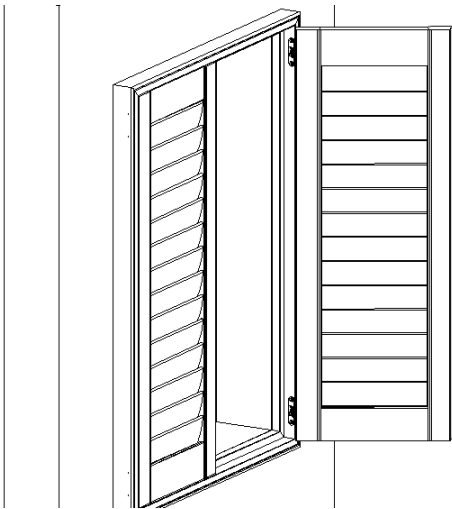
- a) When locking the screw, again it is important to prevent the invisible hinges from sliding out of their slots.
- b) Only the screws provided in the hardware box may be used and they cannot be replaced by other screws.



4). When the closure gap is uniform, secure the remaining screws.



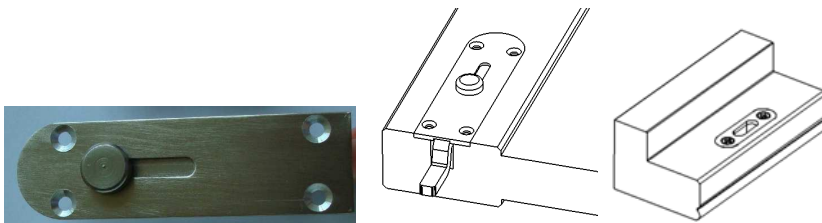
5). Install the remaining panels in the same manner.



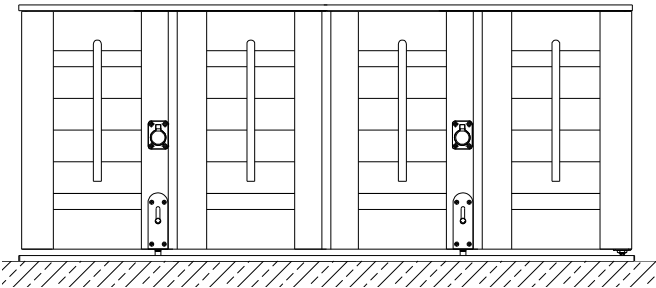
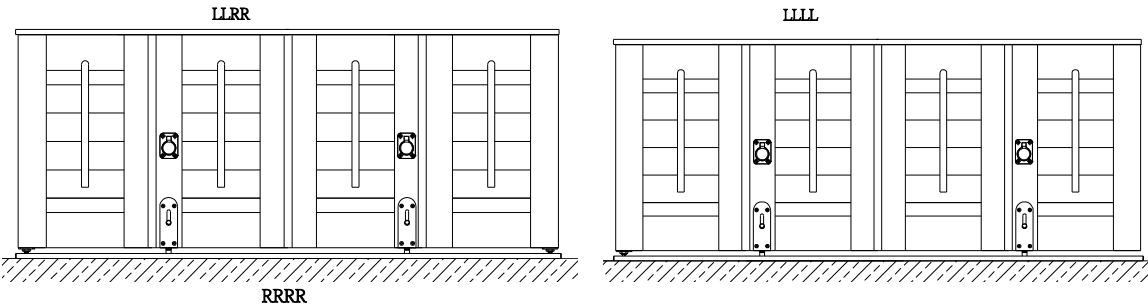
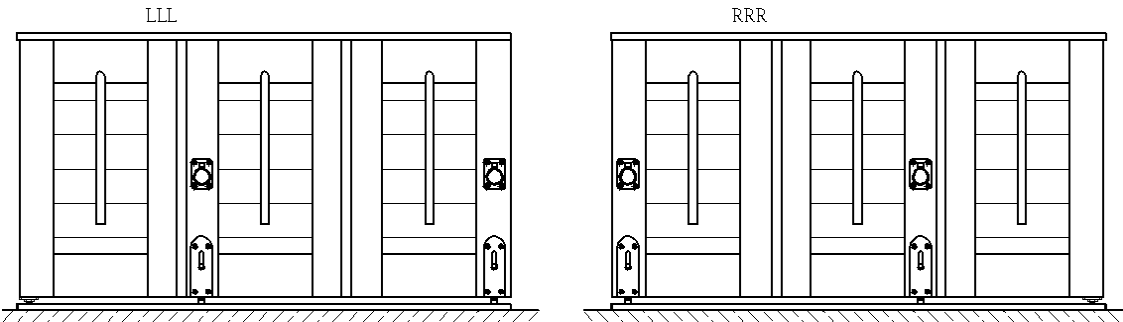
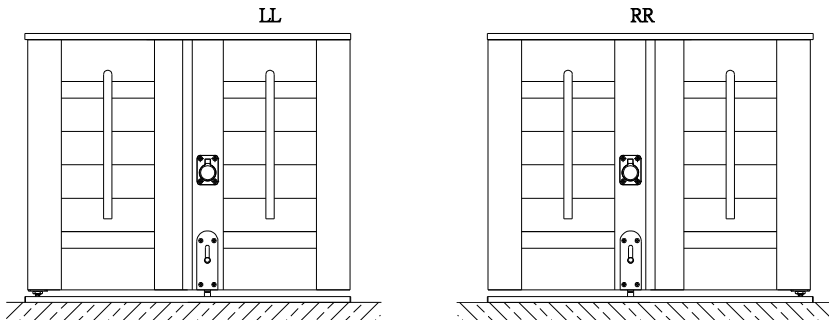
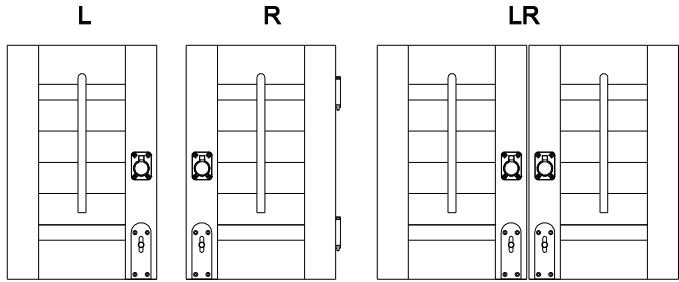
Panel Lock: (not available for 1.625" Stile)

Available Material: All

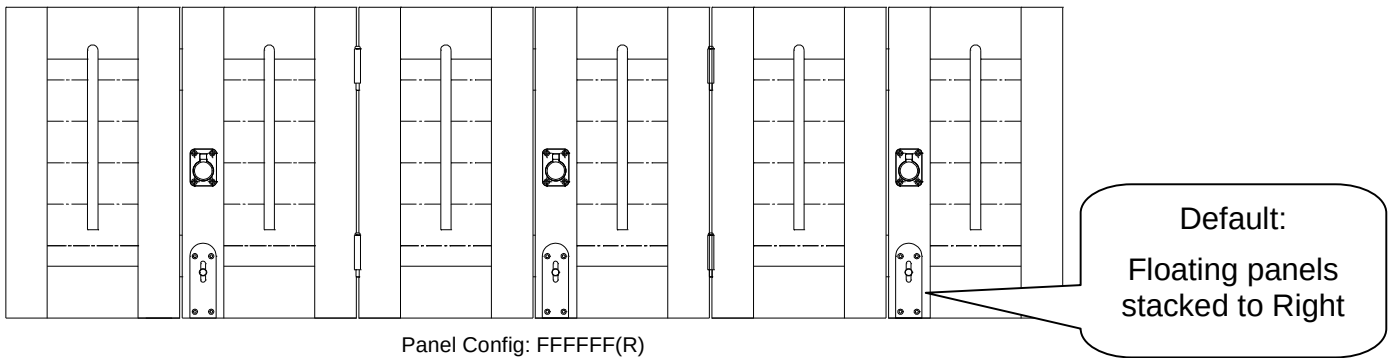
Color: Brush Nickel Plated, Pure White, Pearl.



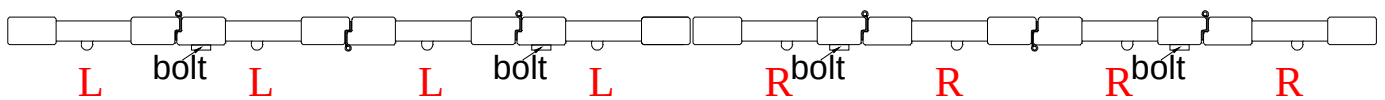
Position: (Only available on Bottom for Track Shutters with Bottom Track; For Regular, available on Top/Bottom/both Top and Bottom)



*Some surcharges apply

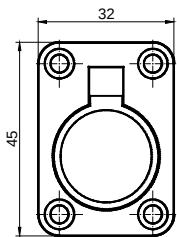


The Panel Lock location for both track system shutters and regular shutters would be the same. The general principle is that the Panel Lock position will avoid the stile which will install the spring-loaded guide at bottom (same stile as the Wheel carrier).



Ring pull: for easy opening and closing. (Not available for 1.625" Stile)

Available in all shutter programs.



Ring pull location principle for Floating Panels:

1. When floating panel config is in pairs, it will be split into halves, the left will be for LL and the right will be for RR. So the ring pull location for FF/FF, FF/FF/FF/FF will be same as that of panel config as LL/RR, LL/LL/RR/RR.
2. When floating panel config is FF, the ring pull location will be same as that of panel config LL.
3. When floating panel config is FFFF/FF, the ring pull location will be same as that of panel config LLLL/RR.
4. When floating panel config is singular, it will be split into 3 sections, the left and mid sections are for LL and the right section is for RR. So the ring pull location for FF/FF/FF, FF/FF/FF/FF/FF will be same as that of LL/LL/RR, LL/LL/LL/RR/RR.

Please note the ring pull color will match with the hinge color.

When shutter order with Invisible Hinge, the ring pull color will match with the normal hinge color.

Please see the matching chart below:

Hinge color	Pure White	Silk White	Bisque	Pearl	Black	Nickel Plated	Brushed Nickel	Antique Brass	Bright Brass	Stainless Steel
Ring Pull color	White	White	Bisque	Pearl	Black	Brushed Nickel	Brushed Nickel	Antique Brass	Bright Brass	Brushed Nickel

Hinge color	Crisp Linen	String	Sea Mist	Stone Gray	Brown Gray	Taupe Gray
Ring Pull color	Bisque	Bisque	Brushed Nickel	Brushed Nickel	Brushed Nickel	Brushed Nickel

- The ring pull color is selectable.

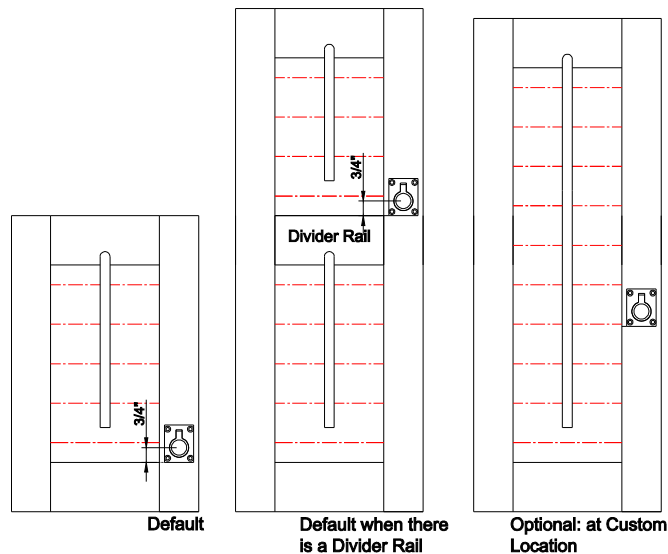
Color ID	Color Name	2.375" Hinge color (Default)	Direct Mount 3" Hinge color (Default)	T post connector color (Default)	Ring pull color (Default)	Panel Lock (Default)	Invisible hinge color	New Top & Bottom Tracks color (Default)	Hoffman Key Color (Default)
001	Pure White	Pure White	Pure White	Pure White	White	Pure White	Pearl Chrome Plated	Bright White	White
002	Extra White	Silk White	Silk White	Silk White	White	Pure White		Bright White	White
003	Silk White	Silk White	Silk White	Silk White	White	Pure White		Bright White	White
004	Bright White	Silk White	Silk White	Silk White	White	Pure White		Bright White	White
006	Pearl	Pearl	Pearl	Pearl	Pearl	Pearl		Bright White	White
007	Ivory Lace	Bisque	Bisque	Bisque	Bisque	Pearl		Metallic Gray	White
009	Creamy	Pearl	Pearl	Pearl	Pearl	Pearl		Bright White	White
012	Crisp Linen	Crisp Linen	Bisque	Bisque	Bisque	Pearl		Metallic Gray	White
016	Butter	Crisp Linen	Bisque	Bisque	Bisque	Pearl		Metallic Gray	White
019	String	String	Bisque	Bisque	Bisque	Pearl		Bronze	White
030	Moondust	Sea Mist	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated		Metallic Gray	Transparent
032	Sea Mist	Sea Mist	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated		Metallic Gray	White
042	Hall Gray	Sea Mist	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated		Metallic Gray	White
046	Ice	Silk White	Silk White	Silk White	White	Pure White		Bright White	White

*Some surcharges apply

048	Chai	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Bronze	Transparent
049	Stone Gray	Stone Gray	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Bronze	Transparent
051	Brown Gray	Brown Gray	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Bronze	Transparent
053	Clay	Nickel Plated	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Bronze	Transparent
063	Decorators White	Silk White	Silk White	Silk White	White	Pure White	Bright White	White
066	Winchester White 2010	Pearl	Pearl	Pearl	Pearl	Pearl	Metallic Gray	White
073	Sky Gray	Silk White	Silk White	Silk White	White	Pure White	Bright White	White
074	Element Gray	Sea Mist	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Metallic Gray	White
075	Storm Gray	Sea Mist	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Metallic Gray	Transparent
077	Dawn	Pearl	Pearl	Pearl	Pearl	Pearl	Metallic Gray	White
078	Cloud Blue	Silk White	Silk White	Silk White	White	Pure White	Metallic Gray	White
080	Taupe Gray	Taupe Gray	Nickel Plated	Nickel Plated	Brushed Nickel	Brush Nickel Plated	Metallic Gray	White
836	Classic Black	Black	Black	Black	Black	Brush Nickel Plated	Black	Black

- When the bottom track colors are bright white and metallic gray, the default spring loaded guide color will be white; When the bottom track colors are bronze and black, the default spring loaded guide color will be coffee.
- Please note that the color options for pin/tension gears are White and Coffee. For custom color orders please select either White or Coffee when placing an order.
- For custom color orders, the supplied Hoffman keys will match the selected pin/tension gear color (White or Coffee).
- For custom color orders, please select the desired T-post connector color from the available options when placing an order.
- For custom color orders with Invisible Hinge, but without Bifold hinge, please select the desired magnet catch (round magnet catch or sill magnet catch or 8.5x25mm magnet catch) color from the available options when placing an order.

Position: For different panel configuration, please refer to the location of Panel Lock to show on which stile the pull will be fixed.

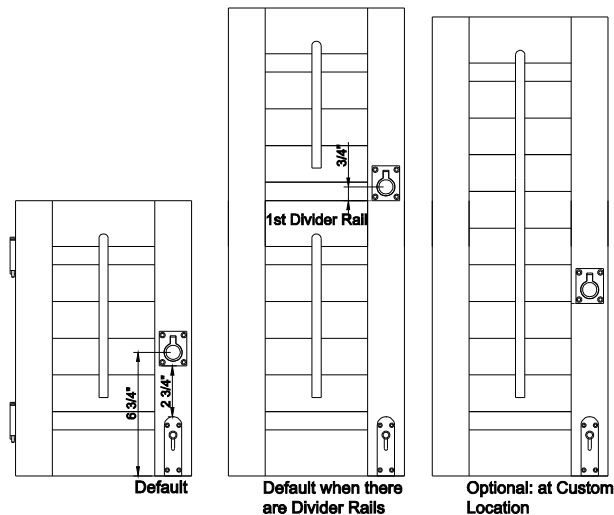


Position:

- A. Above Bottom Rail (Default)
- B. Above 1st Divider Rail (Default when there are Divider Rails)
- C. Custom Location

If both Panel Lock & ring pull are specified, the distance between them is 2 3/4", and the ring pull location is 6 3/4" from the bottom of panel.

When there is no mid rail, ring pull will be just on the top of Panel Lock (2 3/4" above Panel Lock) as default location; if there are mid rails, bottom of ring pull lines up with top of the 1st mid rail as default location.



Position:

- A. 2 3/4" Above Panel Lock (Default when both Ring Pull and Panel Lock are specified)
- B. Above 1st Divider Rail (Default when there are Divider Rails)
- C. Custom Location

Measurements:

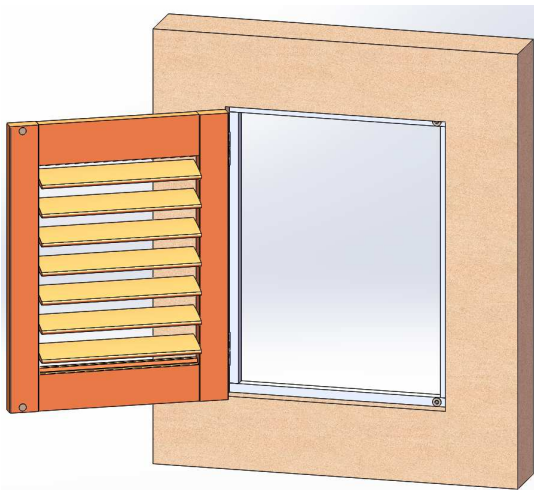
- Window size:
 - Measure from the bottom of the window to the center of the ring pull.

*Some surcharges apply

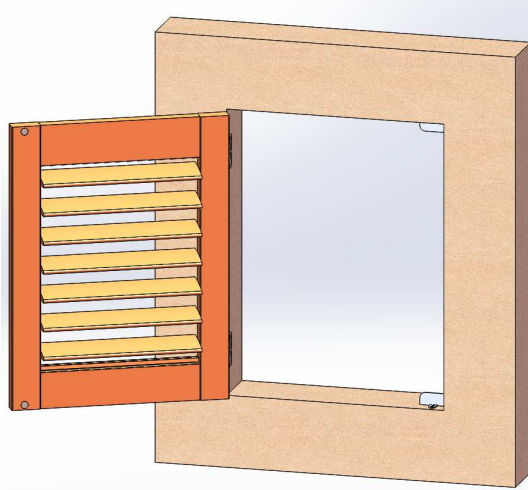
- Max frame-to-frame:
 - Measure from the bottom of the frame to the center of the ring pull.

Mount Options for Direct Mount:

Light block



L magnet catch



12.5mm Magnet with screw hole (recessed into panels)



Round Magnet Catch (installed on light blocks)

L Magnet Catch (sill magnet catch): used for catching the magnets when ordering panels without light blocks.



Round Magnet Catch and L Magnet Catch are available in:
Pure White, Silk White, Bisque, Pearl, Bright Brass, Antique Brass, Black, Nickel plated, Crisp Linen,
String, Sea Mist, Stone Gray, Brown Gray, Taupe Gray.

*Some surcharges apply

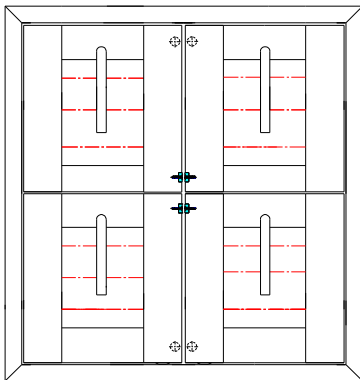
Side Magnet: Available for Butt, rabbet and Astragal stile



Side Magnets will be applied in following situation:

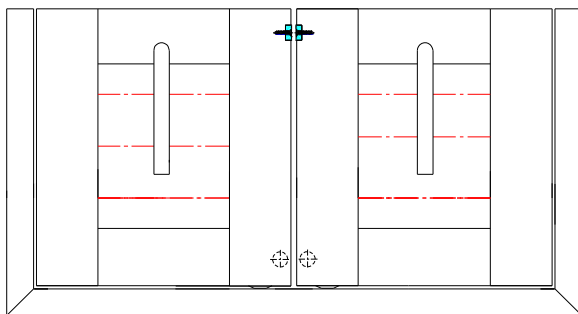
1. Double Hung shutter but without horizontal T post:

Side magnets will be applied to the meeting stiles of the L and R panel. (Default)



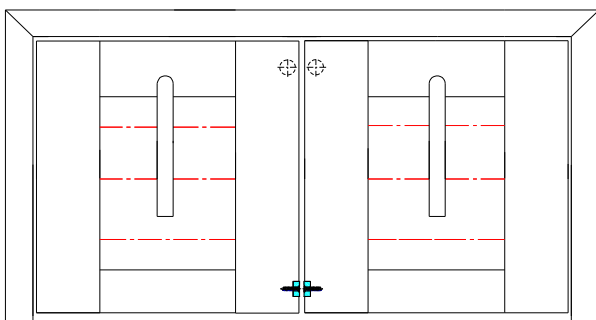
2. Café Shutter

Side magnets will be applied to the meeting stiles of the L and R panel. (Default)



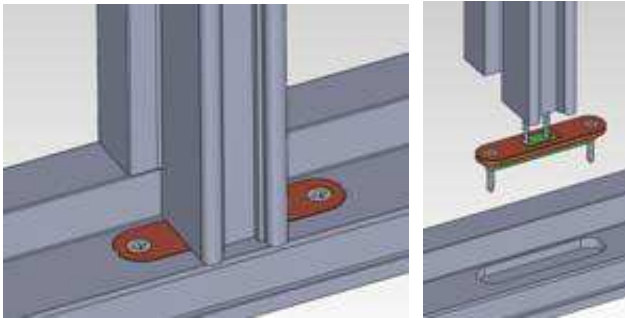
3. No frame or light block or L-shape striker plate on either top or bottom, or if larger than standard top or bottom gap is requested.

Side magnets will be applied to the meeting stiles of the L and R panel. (Optional)



4. Note: if panels only with side magnets and there is no standard magnet on the back of panels, panels may open for external factors.

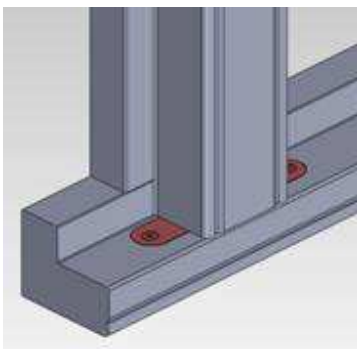
T-Post Connector



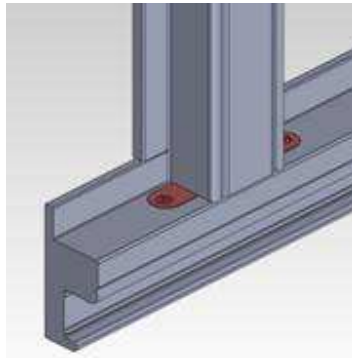
- Available colors: Pure White; Silk White; Pearl; Bisque; Black; Nickel-Plated.
- Available for all types of T Posts (custom-width T posts excluded).

T Post Connector can be used with the following Frames.

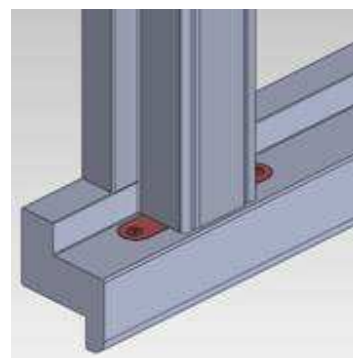
L Frame:



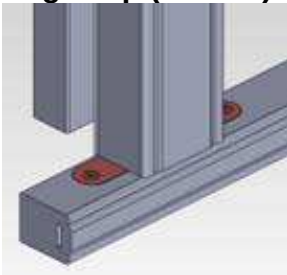
Deco Frame:



Z Frame:



Hang Strip (Beside)

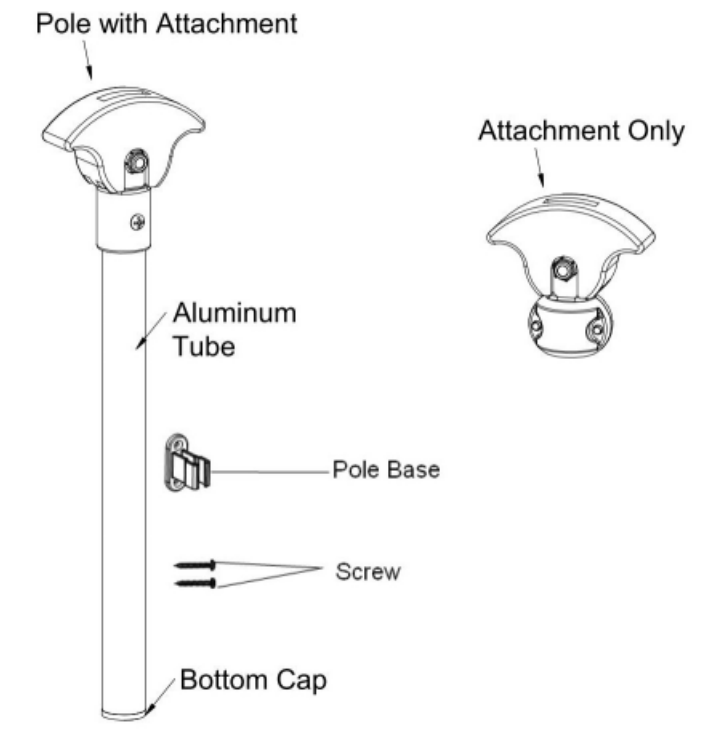


Note:

- For Horizontal T Post, Hang Strip (Behind) and shutter with 1.6"/2" Sill, T Post connector is not available and T Post will be installed by screws instead;
- If there is a panel lock which may interfere with the T-Post connector, all the T-Posts for the entire shutter will be installed with screws and the frames to be pre-drilled, even in cases where the panel lock may only interfere with one T-Post;
- The screw color is nickel-plated regardless of the color of the T Post connector;
- T post connectors can only be installed on a straight piece, and screws will be used instead for T Post installed on a curved or raked piece.

Optional Poles:

1. Terminology



Component

NO.	Name	Note
1	Pole with Attachment	Aluminum Tube, Pole Base and Bottom Cap included *Surcharge applies.
2	Attachment only	• Compatible with extension poles found in most home improvement stores. • Screw thread (3/4") and thread pitch (1/5") *Surcharge applies.

2. Available Pole Length

Two lengths available:

- 36"
- 60"

*Some surcharges apply

Bi-fold 180[™]

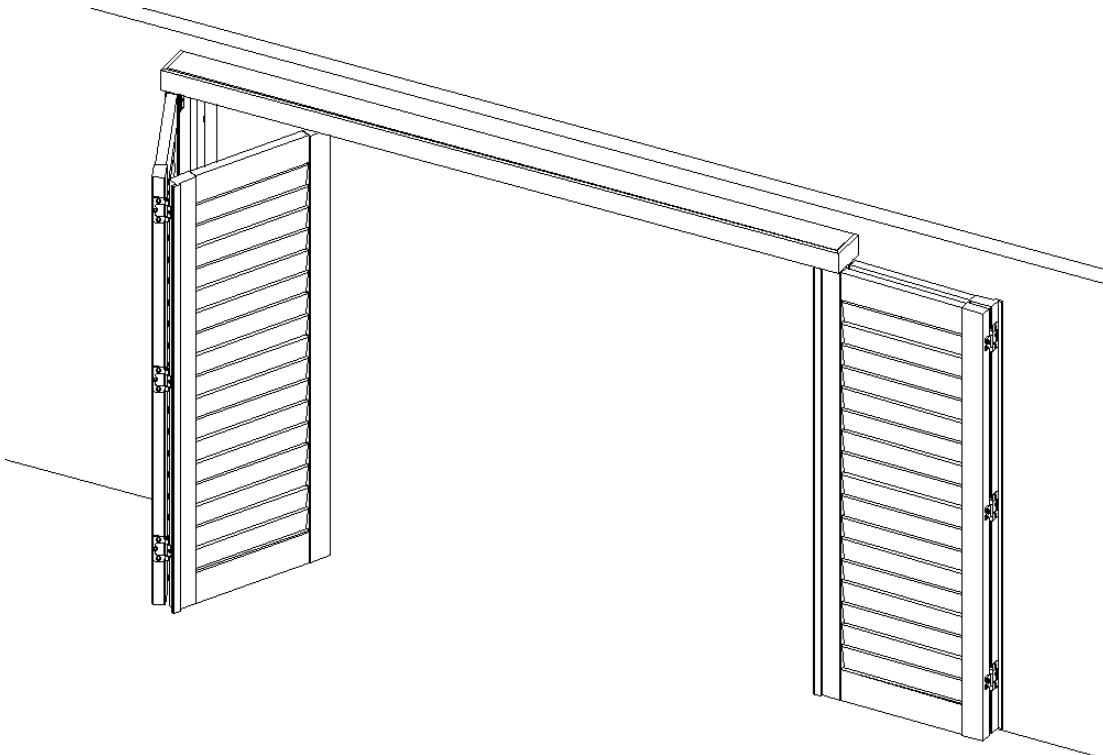
Guidelines

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>
Single-Panel Width (Bi-fold Track)	6"	26"
Single-Panel Width (LLL/RRR)	6"	20"
Single-Panel Height	10"	132"

Outside Mount

Max frame-to-frame width: window width + 3 1/2"

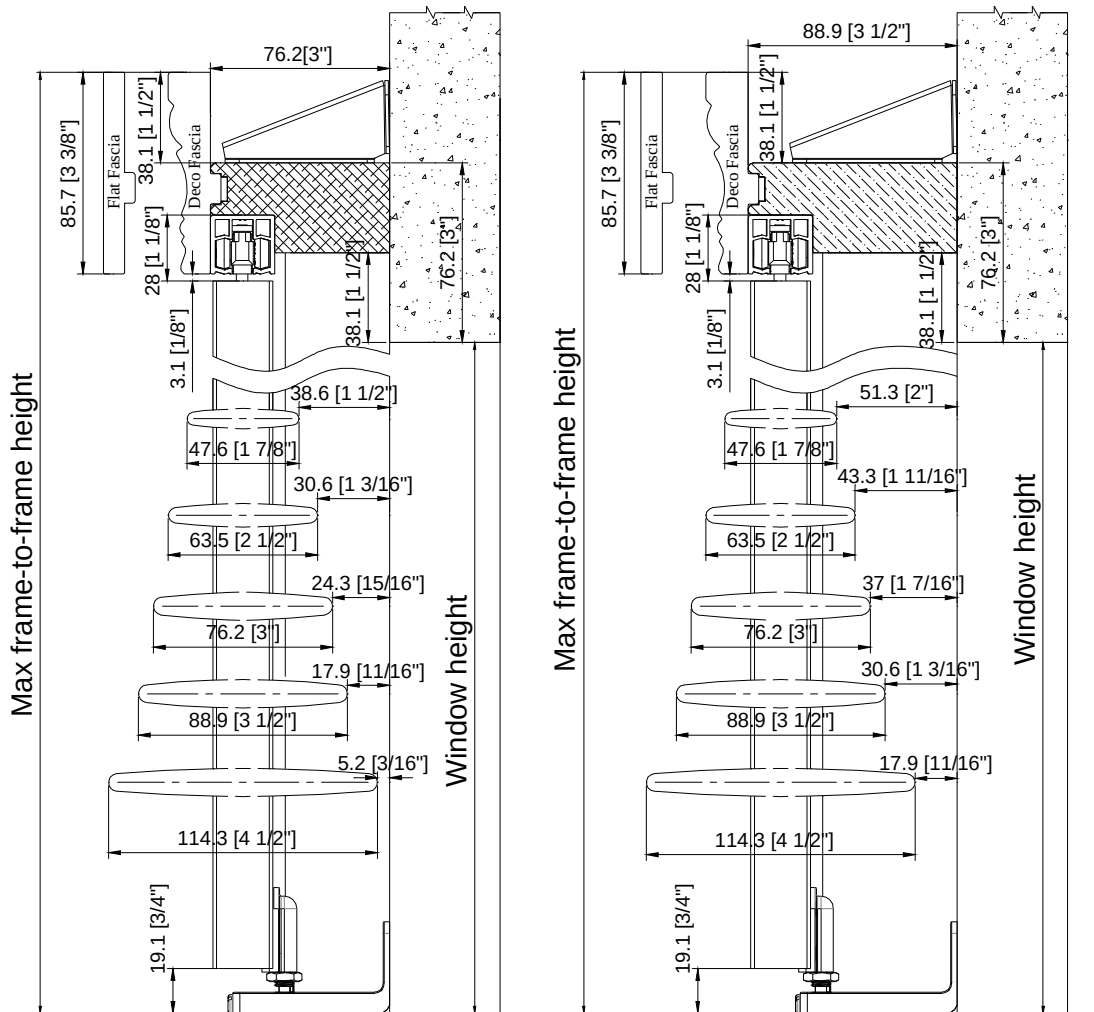
Max frame-to-frame height: window height + 4 1/2"



- Track is pre-installed on the header; standard 1 track only.

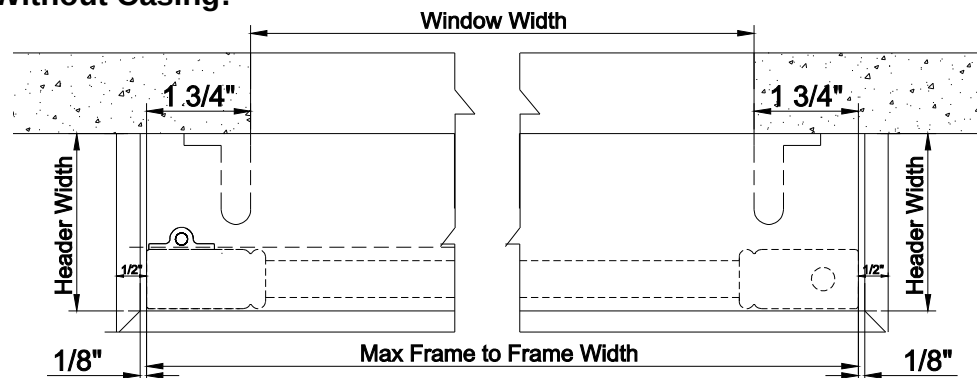
Mounting Options: Outside Mount only

Side View



Top View

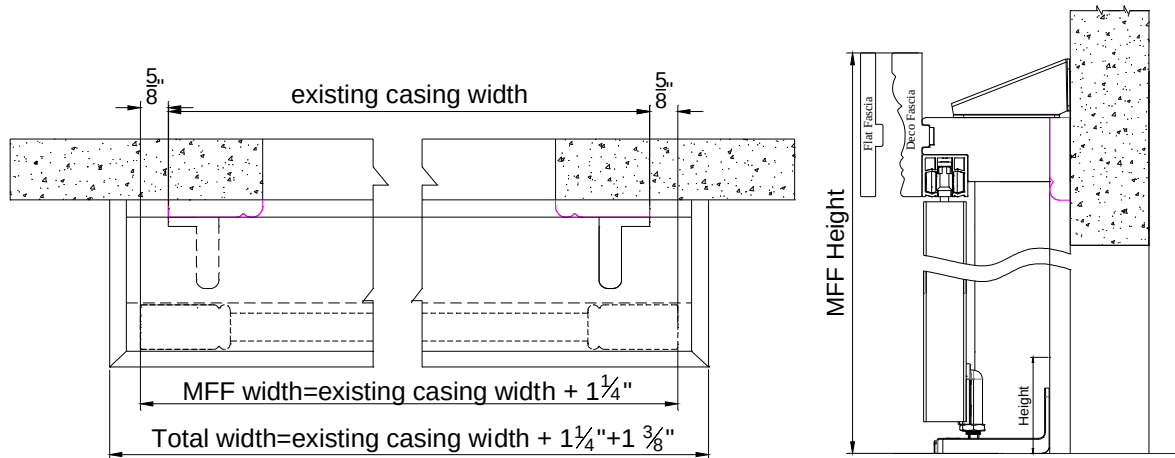
Without Casing:



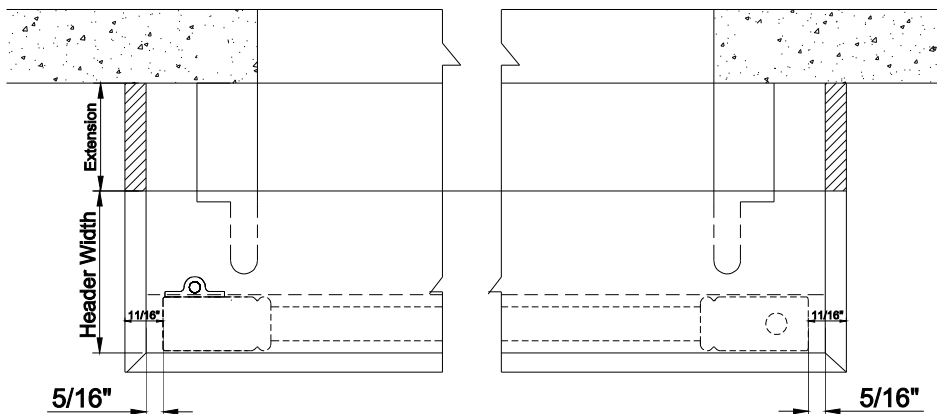
Note:

- ◇ Fascia return will protrude the panel about 1/2" for each side.
- ◇ The header will protrude the panel about 1/8" for each side.
- ◇ Header, light blocks, and bottom pivot bracket(s) must be installed on a flat surface.

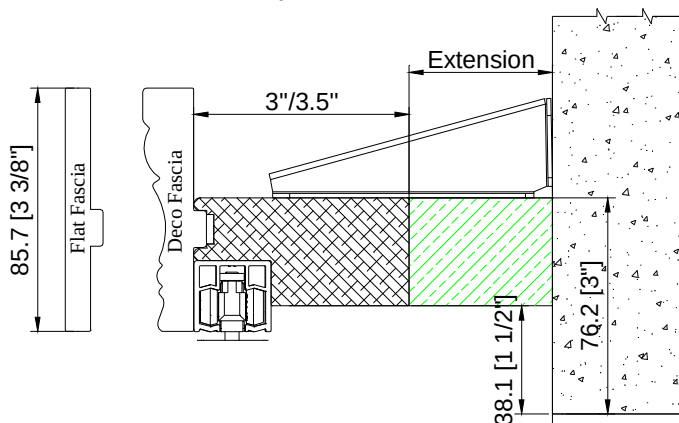
With Existing Casing:



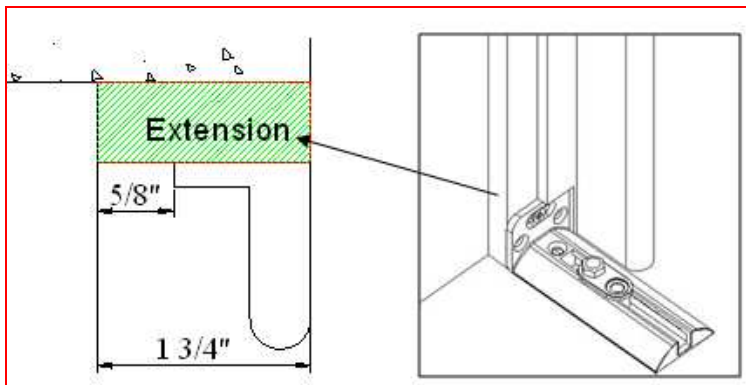
- ✧ Header Extension: Optional
- ✧ Custom extension up to 2" for Header
- ✧ When extension is needed, the header will protrude the panel 5/16" for each side and the fascia return will protrude the panel 11/16" for each side.



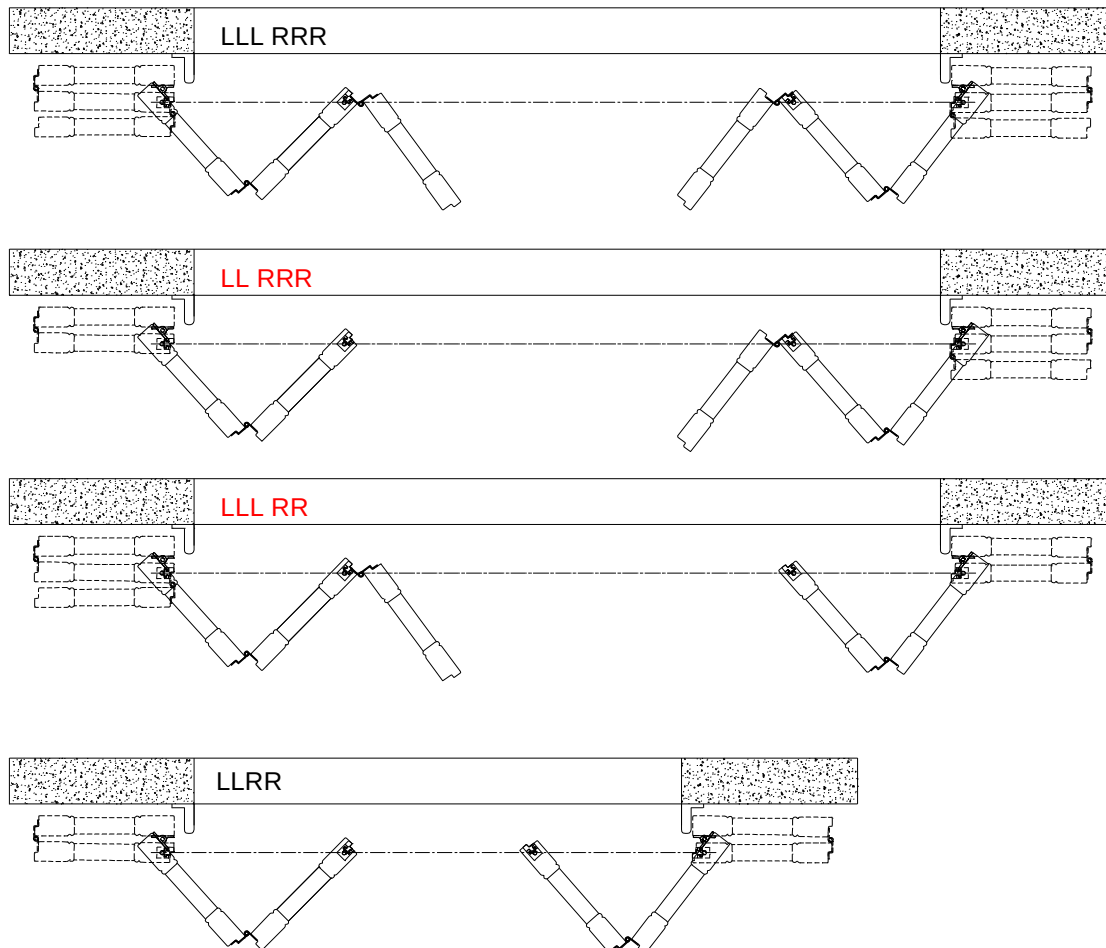
The extension will be pre-attached.

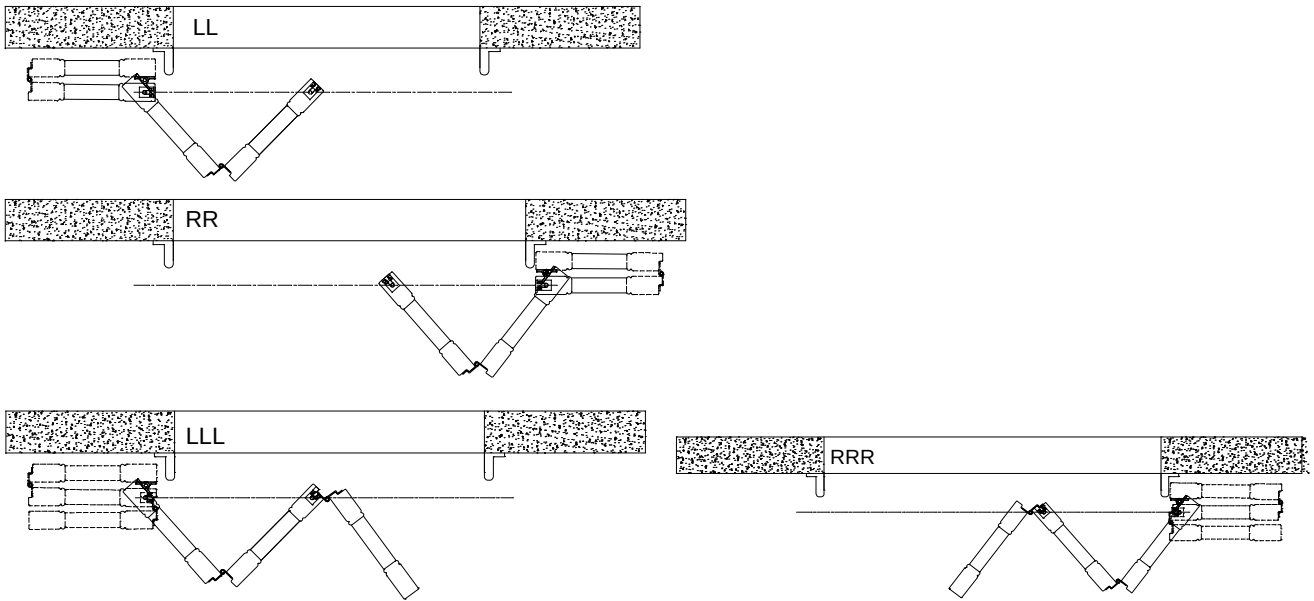


For 180 Bi-fold, if Bottom Pivot L Bracket is required, the extension for the L shape light block will be 1 3/4" in width. Please see picture below:



Panel Configuration : Only 2", 2 1/4" rabbet and butt stiles are available. Center stile is not available.





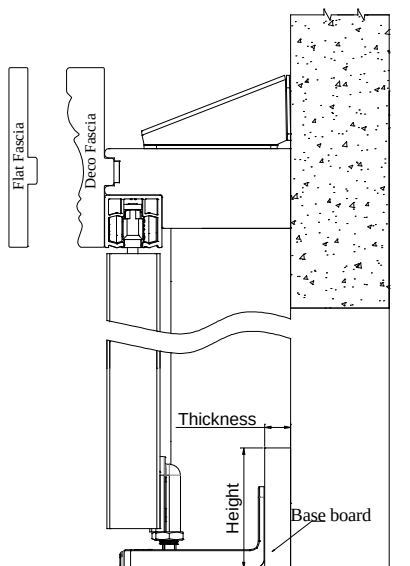
Notes:

If there is no baseboard to the wall, 3" header can be used.

If there is baseboard to the wall, then it should be as below:

1. If base board thickness is less or equal to 5/8", 3 1/2" header can be used.
2. If base board thickness is over 5/8", it is suggest using the 3" header with build out. And the build out thickness=base board thickness-1/8".

Please see the drawing below:



180° Bi-fold Track Shutters Framing Materials

Type	Header Dimension	Top Fascia Dimension	L Shape Light Block Strip Dimension
Header & Top Fascia & Light Block Strip	3"	3 3/8"	1 9/16"
	3 1/2"		2 1/16"

180° Bi-fold Track Shutters Hardware Box

Parts
Wrench
Hinge Pin
Triangle Bracket (based on the length of header)
2" screws for Triangle Bracket
¾ " screws for Triangle Bracket
Bottom Pivot Bracket
1 ½ " screws for bottom pivot bracket
2 ½ " screws for Frame with predrilled screw holes

Bi-fold 90[™]

Guidelines and Specs:

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>
Single-Panel Width (Bi-fold Track)	6"	26"
Single-Panel Width (Multifold track up to 4 panels)	6"	20"
Single-Panel Height	10"	132"

Outside Mount

Max frame-to-frame width: window width + 3 1/2"

Max frame-to-frame height: window height + 4 1/2"

Semi-Inside Mount

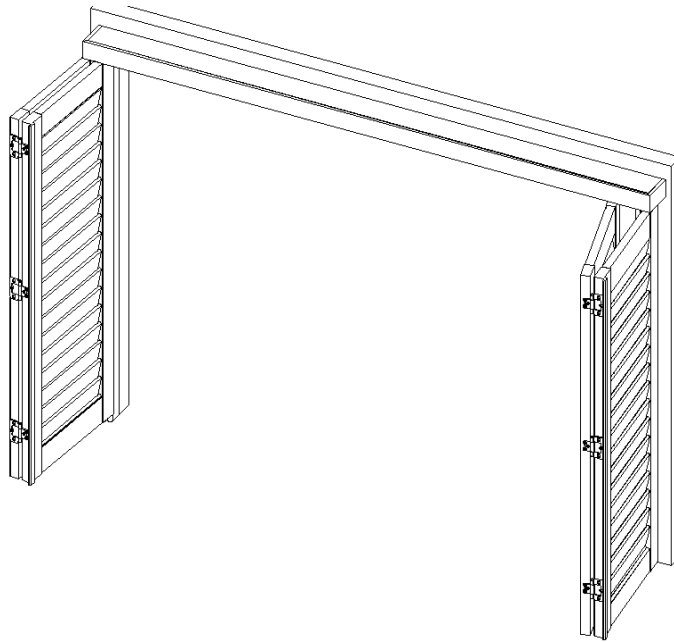
Max frame-to-frame width: window width - 1/8"

Max frame-to-frame height: window height + 1 3/8"

Inside Mount

Max frame-to-frame width: window width - 1/8"

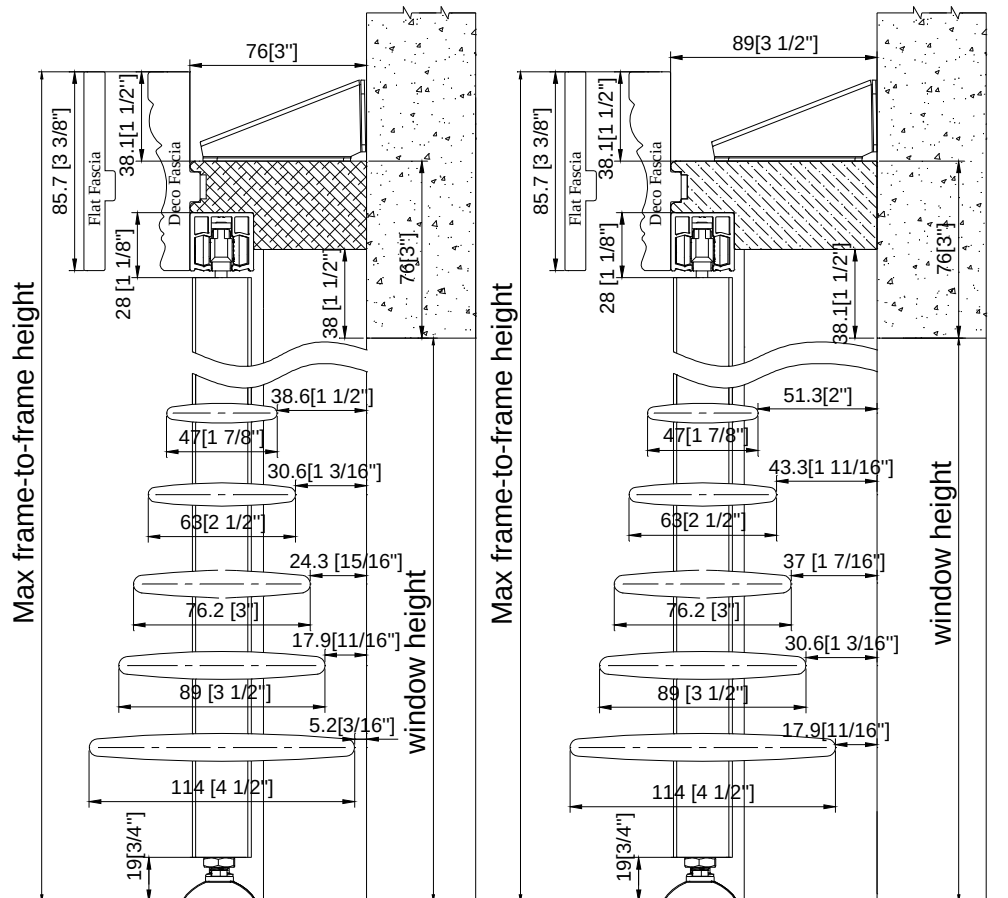
Max frame-to-frame height: window height - 1/8"



Track is pre-installed on the header; standard 1 track only.

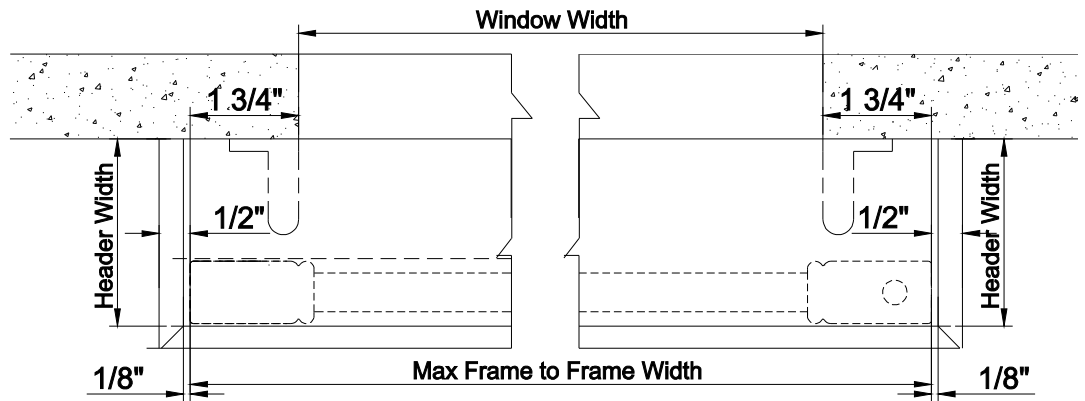
Mounting Options: Outside Mount

Side View



Top View

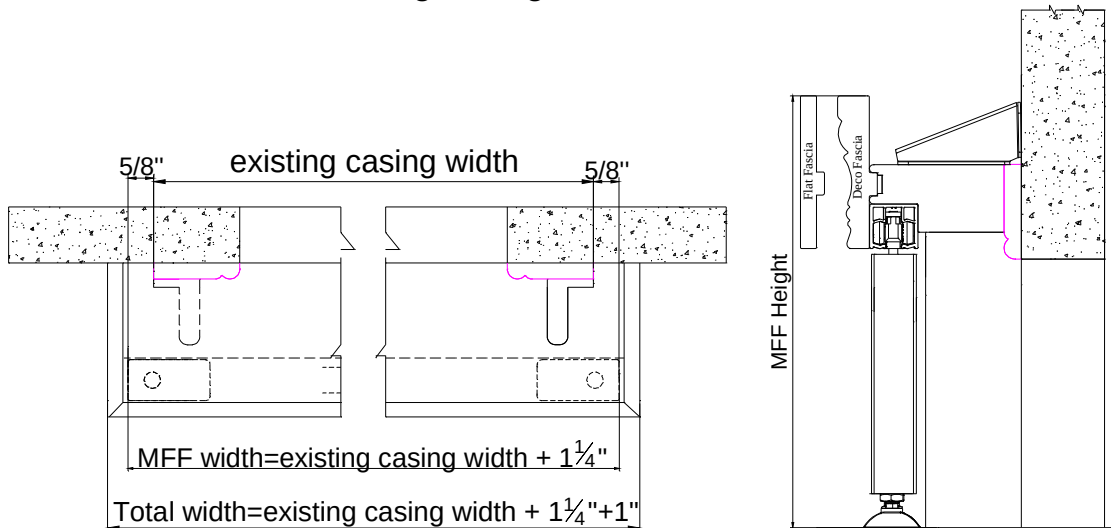
Without Casing:



Note:

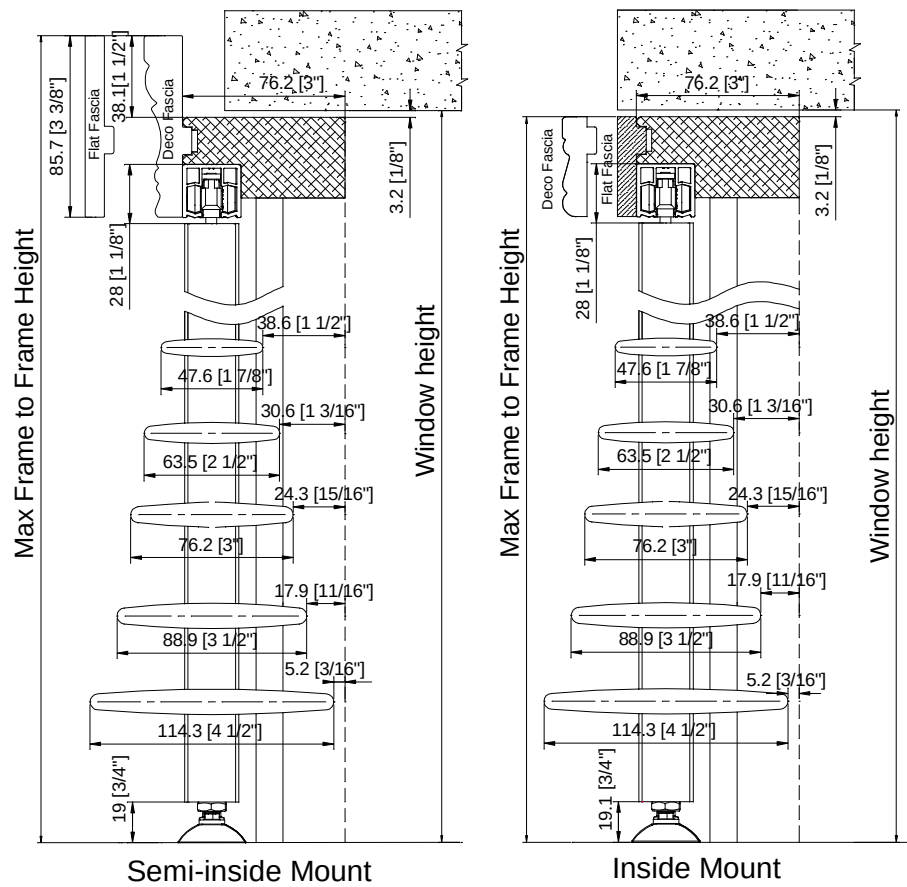
- ◇ Fascia return will protrude the panel about 1/2" for each side.
- ◇ The header will protrude the panel about 1/8" for each side.
- ◇ Header, light blocks, and bottom pivot bracket(s) must be installed on a flat surface.

With Existing Casing:

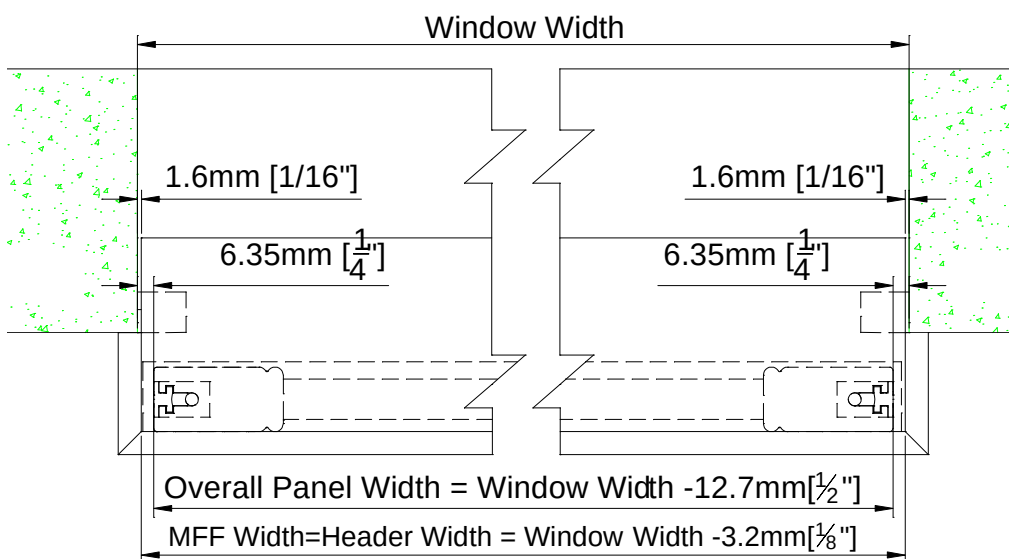


Semi-Inside/Inside Mount

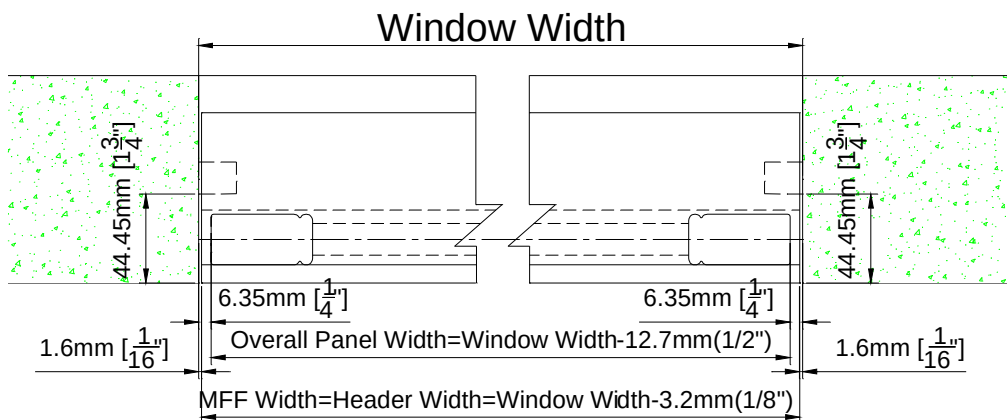
Side View



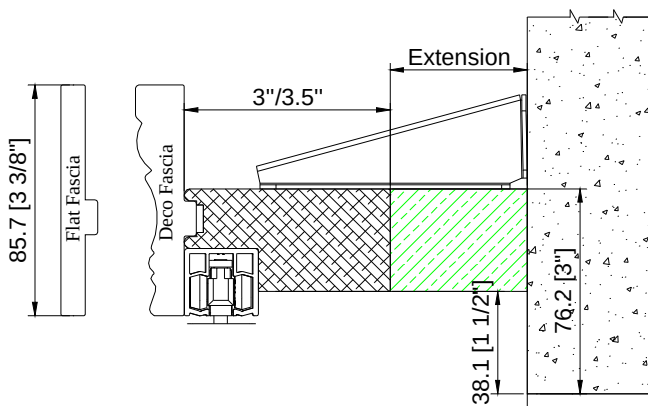
Semi-Inside Mount Top View



Inside Mount Top View

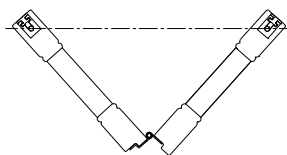


- ✧ Header Extension: Optional.
- ✧ Custom extension up to 2" for Header
- ✧ The extension will be pre-attached.

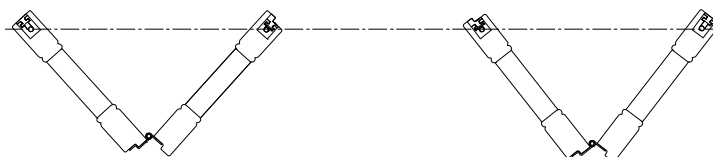


Panel Configuration : Only 2", 2 1/4" rabbet and butt stiles are available. Center stile is not available.

LL or RR



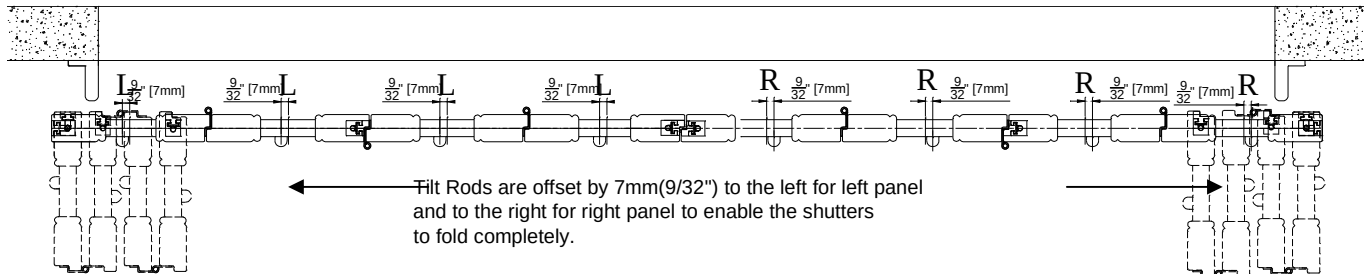
LLRR



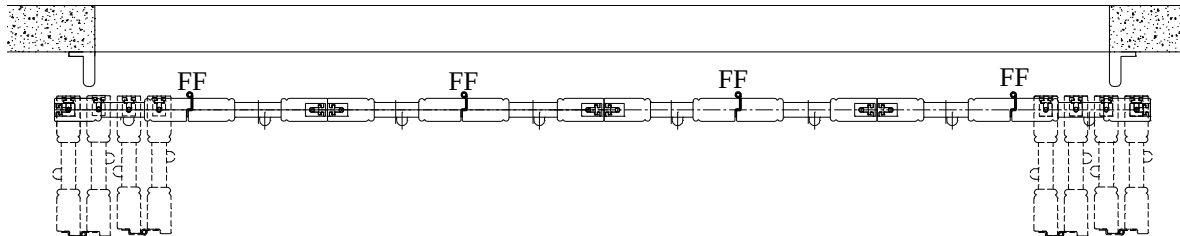
90° Multi-Fold Track Shutters (Wood only)

Maximum single panel width for multi-fold is 20"

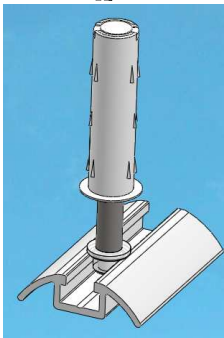
Panel Configuration: LLLL, RRRR, or LLLLRRRR



90° Floating Panels



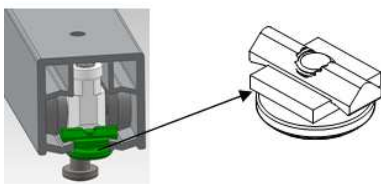
- o Two or four panels can be hinged together, fold and slide to either direction
- o Application comes with wheel carrier, spring loaded guide and floor track
- o Can be used on large window



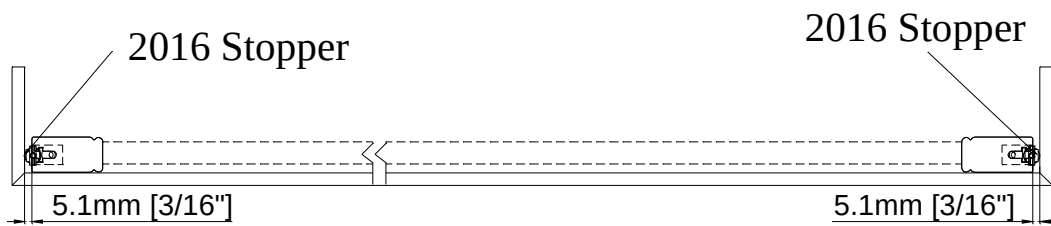
Note: For 90° Floating Panels outside mount/semi-inside mount without side boards, the 2016 Stopper will be supplied by default. The gap between the fascia return / end of top track and the panel is 5mm each side.

The 2016 stopper can also be used in the center or any position of the top track as an option.

2016 Stopper

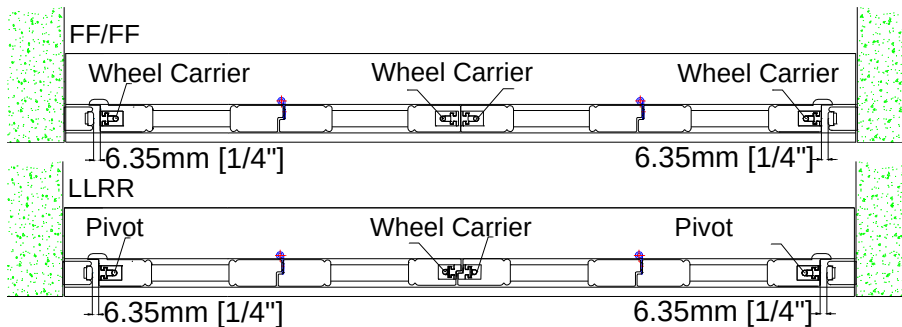
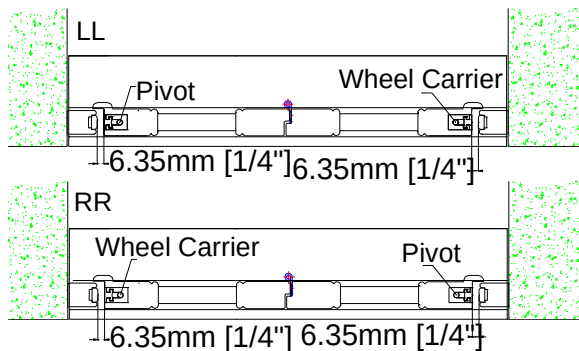


Please see drawing below to show the gap.

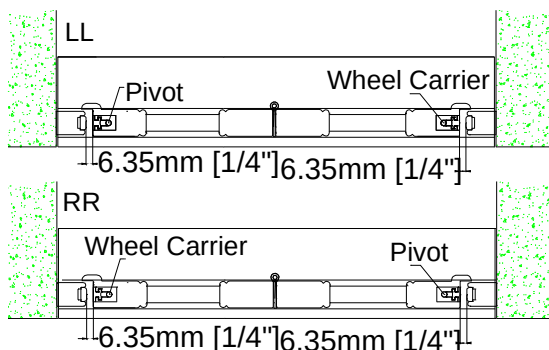


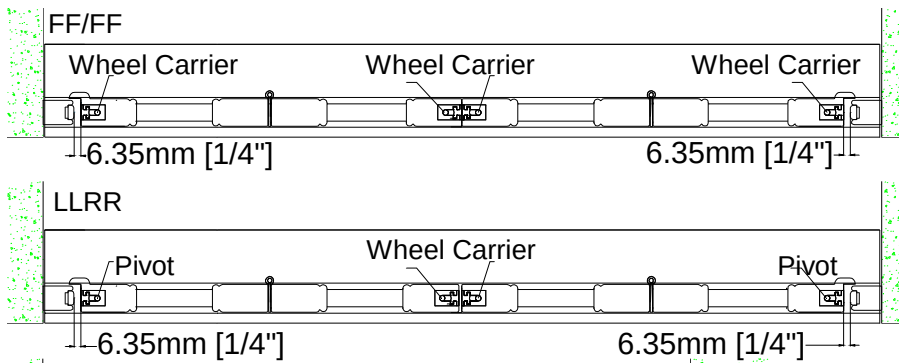
2. When rabbet stile required, the meeting stile of unhinged stile will be butt stile by default.

Optional (Hang Strip for Inside Mount Only)



Note: When Butt/rabbet Stile is required, the stile next to the hang strip will be astragal stile.





90 Degree Bi Fold Track Shutters Framing Materials

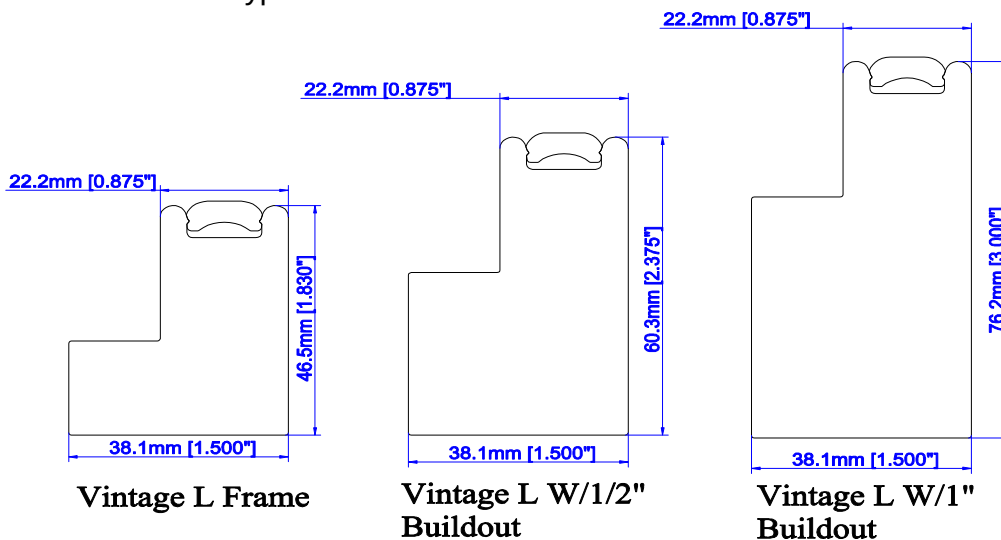
Type		Header Dimension	Top Fascia Dimension	Light Block Strip Dimension	Hang Strip Dimension
Header & Top Fascia & Light Block Strip	OM	3"	3 3/8"	1 9/16"	
		3 1/2"		2 1/16"	
	Semi-IM	3"	1 7/8"	5/8"x3/4"	1.08"x0.875"
	IM	3"		5/8"x3/4"	

Note: It will be L shape light block strip for outside mount.

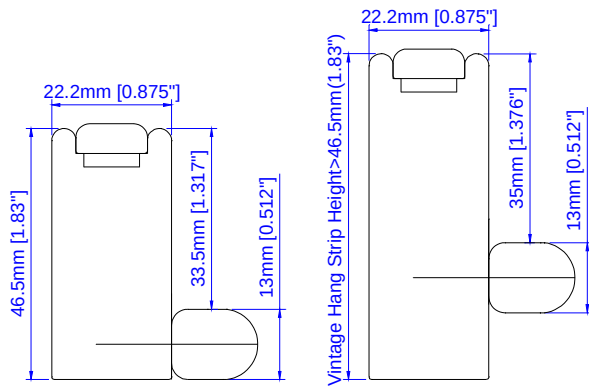
Frame Hinged Bifold

Available Stile: Rabbet Stile Only

Available Frame type



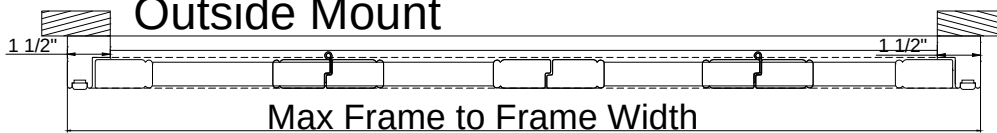
Vintage Hang Strip with 13mm light block: For single direction shutter with even panel qty, the frame on the unhinged side will be Vintage Hang Strip with 13mm [0.512"] light block, the height of the Vintage Hang Strip will match with that of the hinged side frame. The Vintage Hang Strip will be made in wood with surcharge.



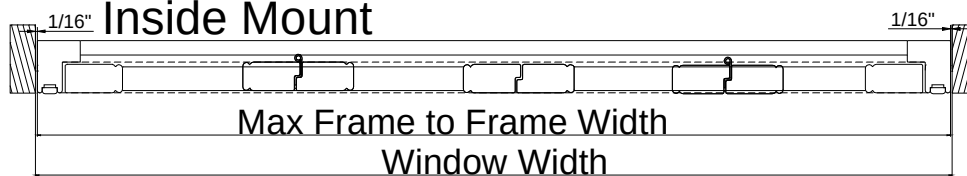
Top View

Hinged at Left or Right with wheel assembly at top;

Outside Mount



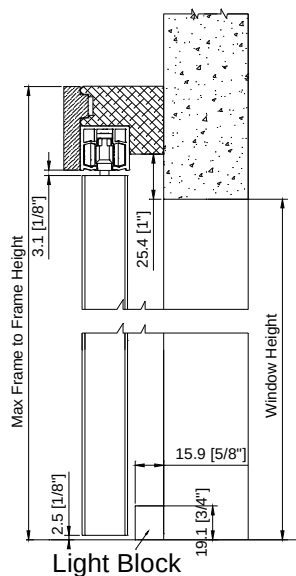
Inside Mount



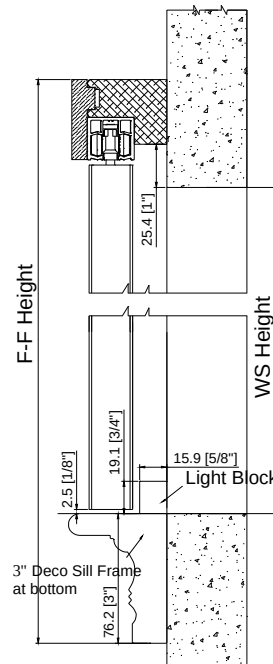
Side View

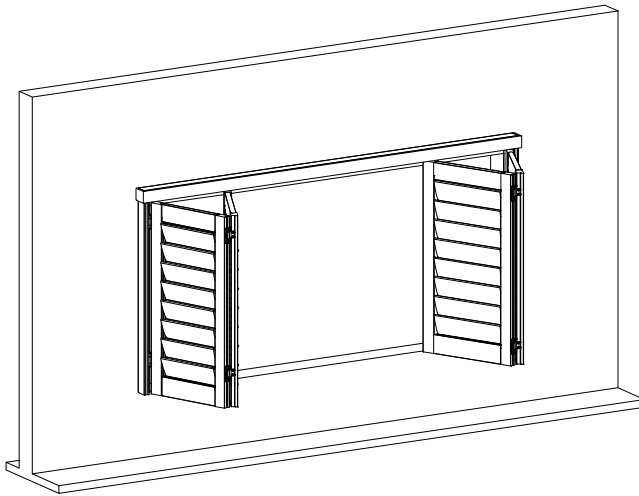
Outside Mount

3-Sides + bottom light block



3-Sides + 3\" Deco Sill Frame

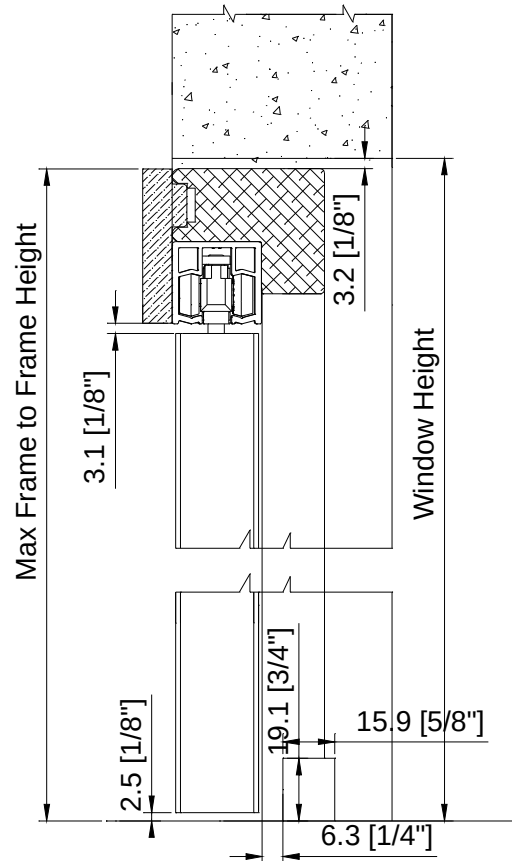
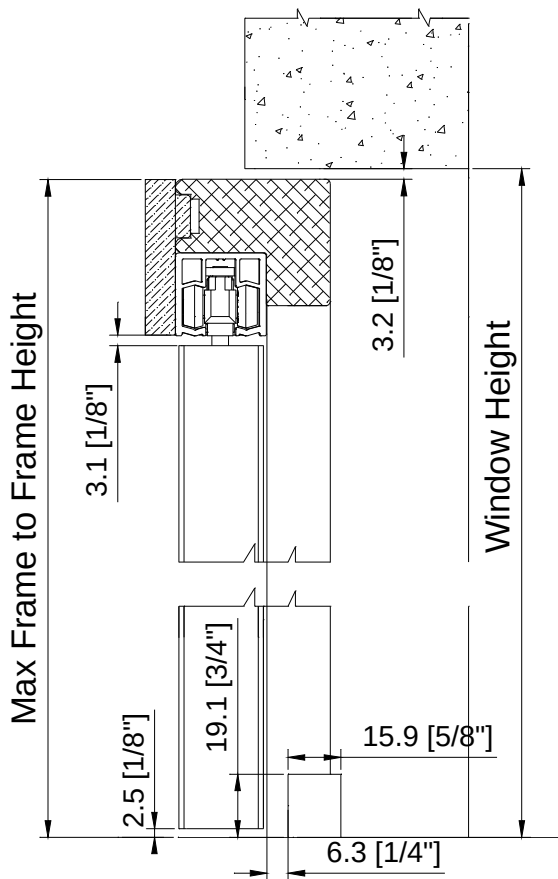


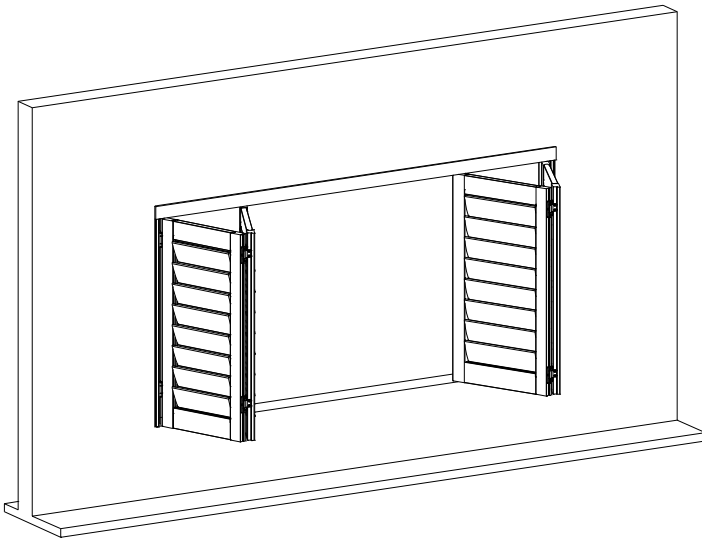


Note: For OM Frame Hinged Bifold shutter not to use as a door, if there is no existing sill at the bottom of the window opening, it is recommended to have the 3" Deco Sill Frame at the bottom.

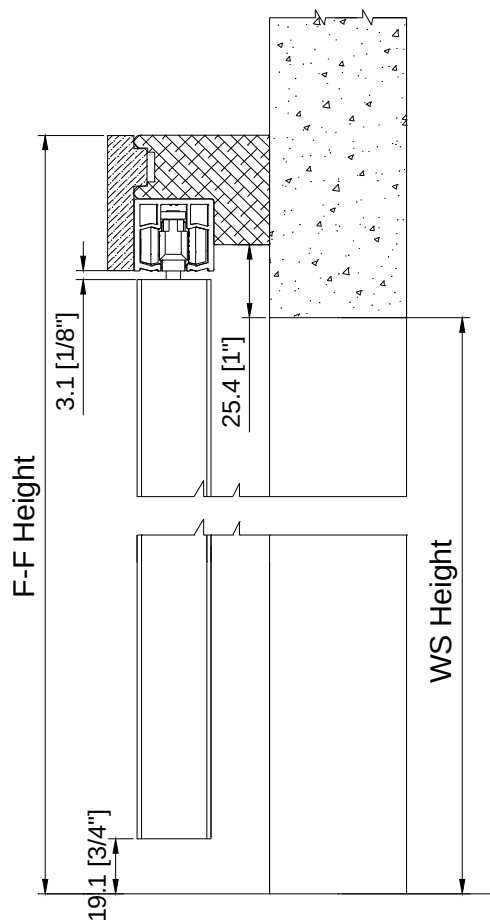
Semi-Inside

Inside Mount





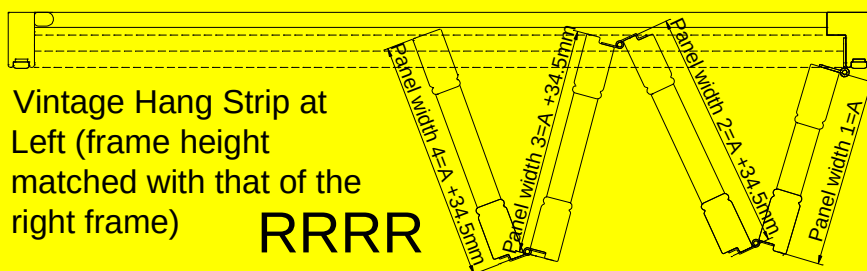
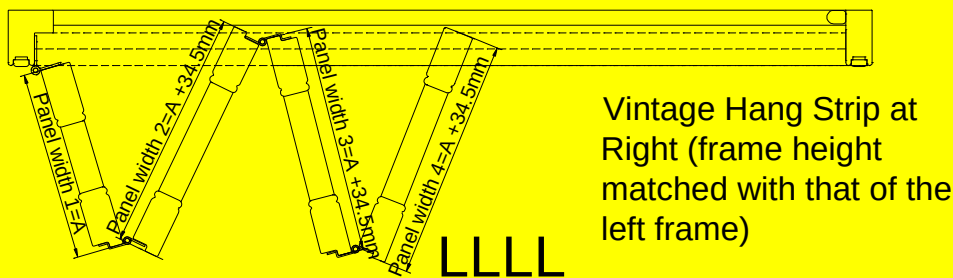
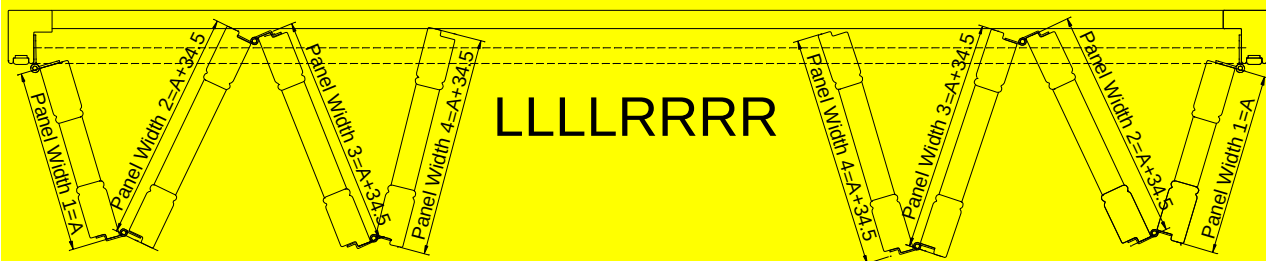
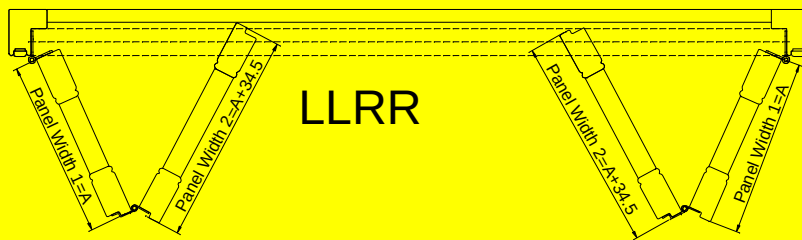
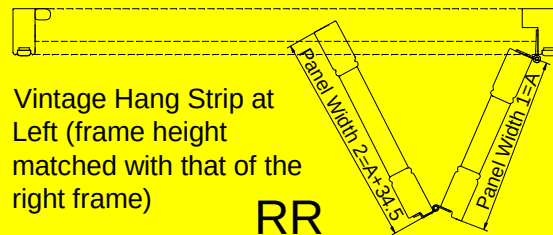
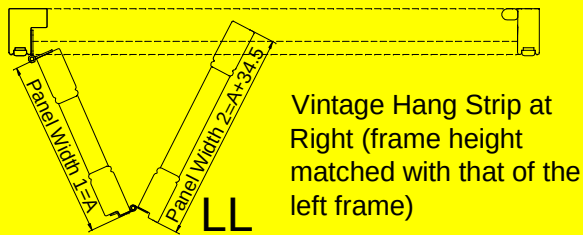
For frame hinged bifold track shutter used as door, the bottom gap will be 19mm by default. Please specify “Used As Door” on the order form.



Panel Configuration (uneven panel width)

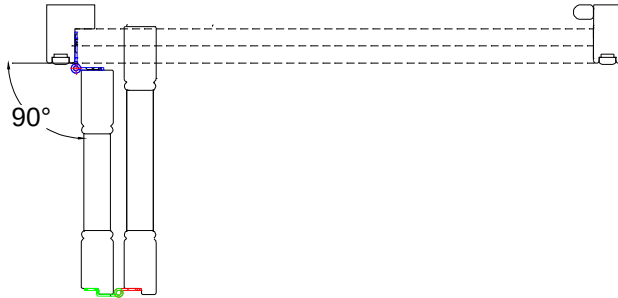
- 1) If the hinge type between panel and frame is 2 3/8" Self Mortise Hinge,
the panel width1 (hinged with Self Mortise Hinge) = A,
the other panels width (panel width2 / panel width3 / panel width4) = $A + 34.5\text{mm}$;
- 2) If the hinge type between panel and frame is Invisible Hinge,
the panel width1 (hinged with Invisible Hinge) = A,
the other panels width (panel width2 / panel width3 / panel width4) = $A + 29.5\text{mm}$;

Take the drawings of the hinge type between panel and frame is 2 3/8" Self Mortise Hinge as examples:



Notes:

1. Only available for 3-Sides + bottom light block or 3-Sides + 3" Deco Sill Frame (4-sides is not available); Panels can be folded and open at 90 degree;



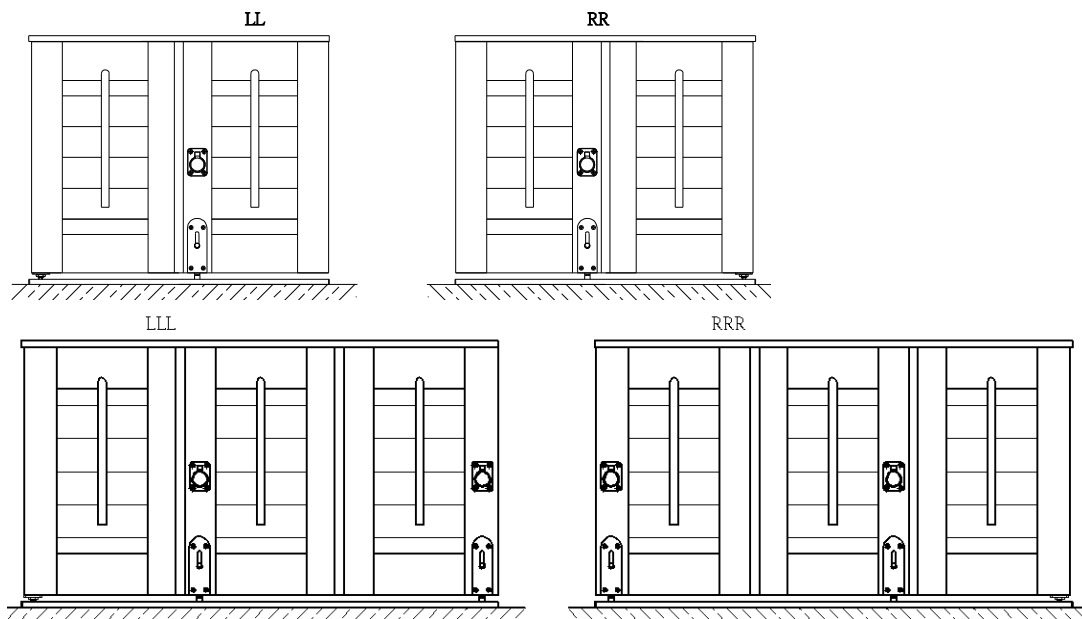
2. Frames available: Only Vintage L Frame

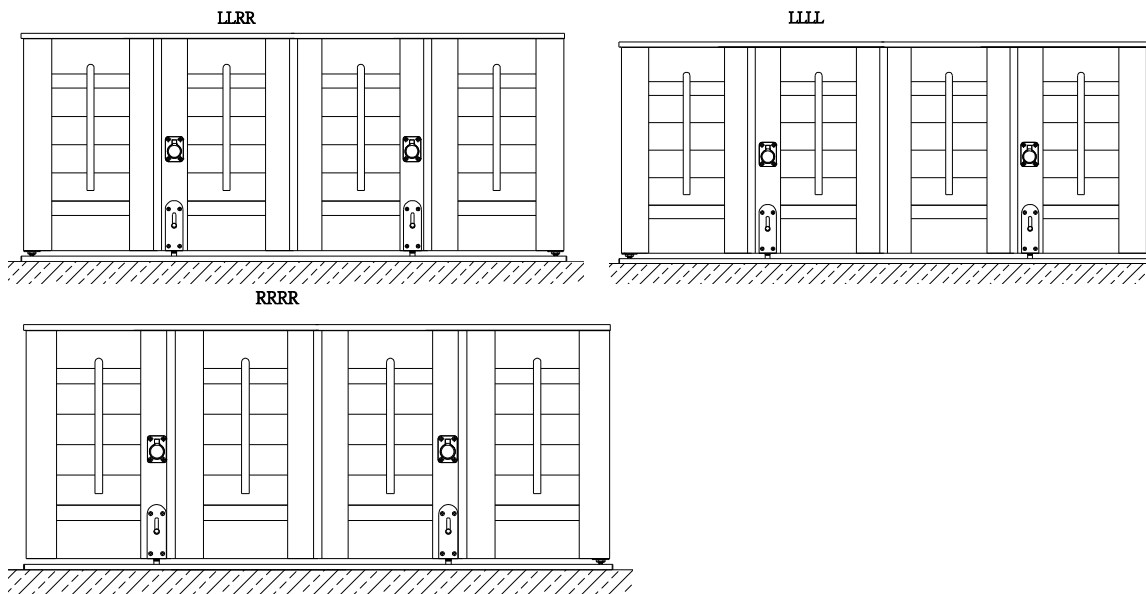
- 1) For top frame, the insert will be replaced by top fascia;
- 2) Top track will be pre-attached to the top frame;
- 3) Frame will be pre-drilled with screw holes same as regular shutters.

Ring Pull: The ring pull will be provided as an option; and for frame hinged bi-fold, the ring pull will be supplied as default; Customer need specify ring pull location on the order form. Ring pull location is measured from the floor to center of the ring pull in height direction.

Ring pull color will match the hinge color.

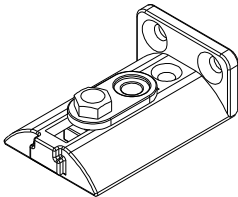
Please refer to below drawing of ring pull position for all shutter type:



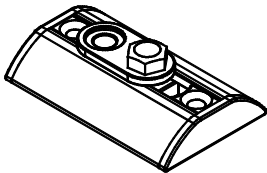


Bottom Pivot Bracket for 90 degree bifold:

For 90° Bifold Track Shutter, there will be 2 types of Bottom Pivot Brackets available.

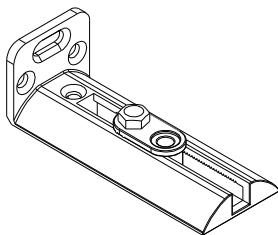


For Inside Mount, Bottom Pivot L Bracket will be supplied as default;

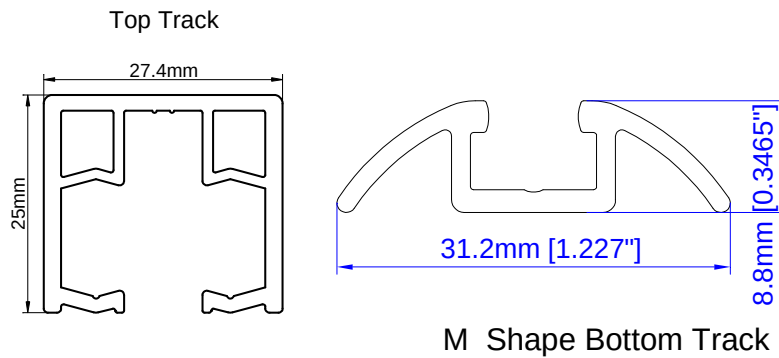


For Semi-Inside or Outside Mount, bottom pivot bracket will be supplied as default.

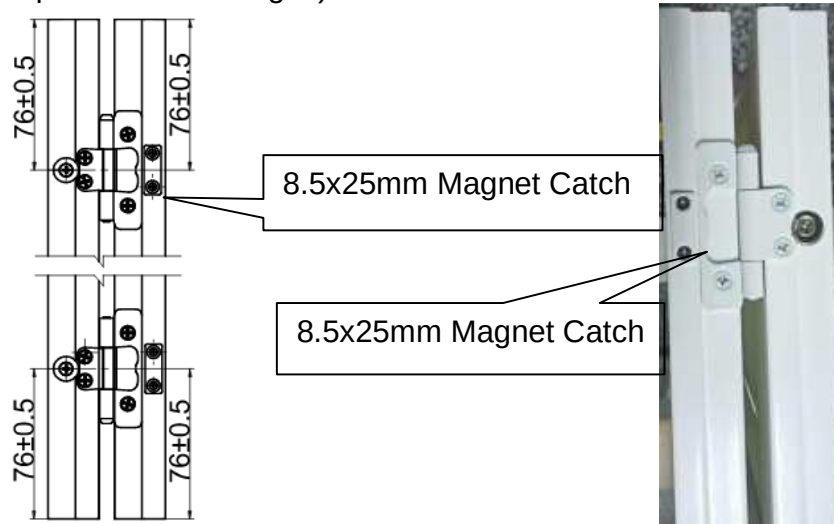
For 180° Bifold Track Shutter, 81mm Bottom Pivot L Bracket will be supplied as default.



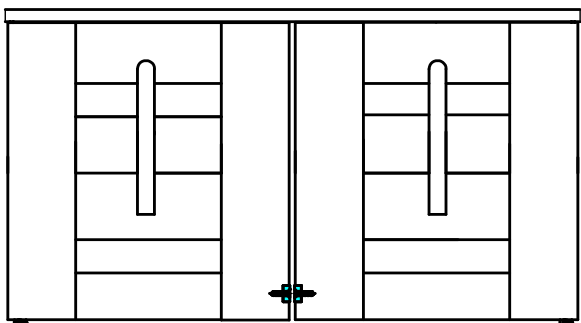
Track Dimension:



Extra 8.5x25mm Magnet Catches and magnets will be supplied for Bifold hinge: (only for the most top and bottom hinges)



Side Magnet: applied to the meeting stiles of LR panels



Extra magnet and Side Magnet:

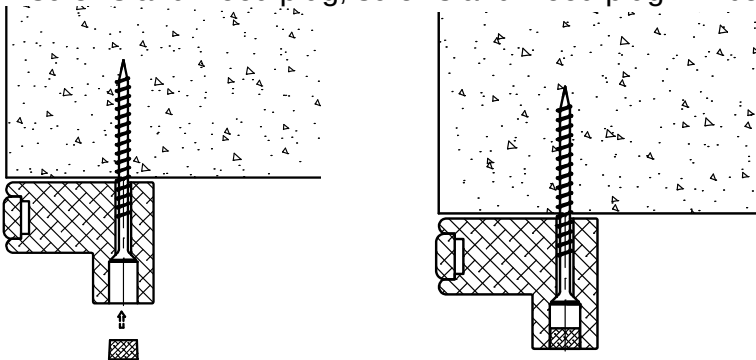


90° Bi Fold Track Shutters Hardware Box

Parts
Wrench
Hinge Pin
Triangle Bracket (based on the length of header)
2 " screws for Triangle bracket
¾ " screws for Triangle bracket
Bottom Pivot Bracket
38.1mm screws for bottom pivot bracket
63.5mm screws for Frame with Predrilled holes

Notes:

1. for all bifold track shutter Outside Mount, the header will be pre-drilled with screw holes in the insert and screws will be supplied by default;
2. for all bifold track shutter Inside/Semi-Inside Mount, the header will be pre-drilled with holes for screws and wood plug, screws and wood plug will be supplied by default



90 Degree Bifold Track & Floating Panels Hanger Location:

LL/RR/LLRR

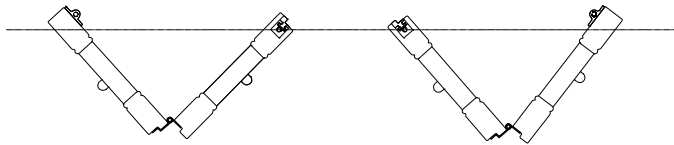


LLLL/RRRR/LLLLRRRR

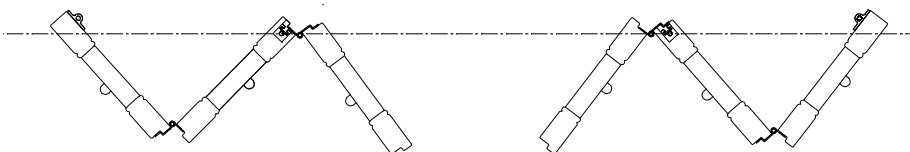


180 Degree Bifold Track Panel Hanger Location:

LL/RR/LLRR

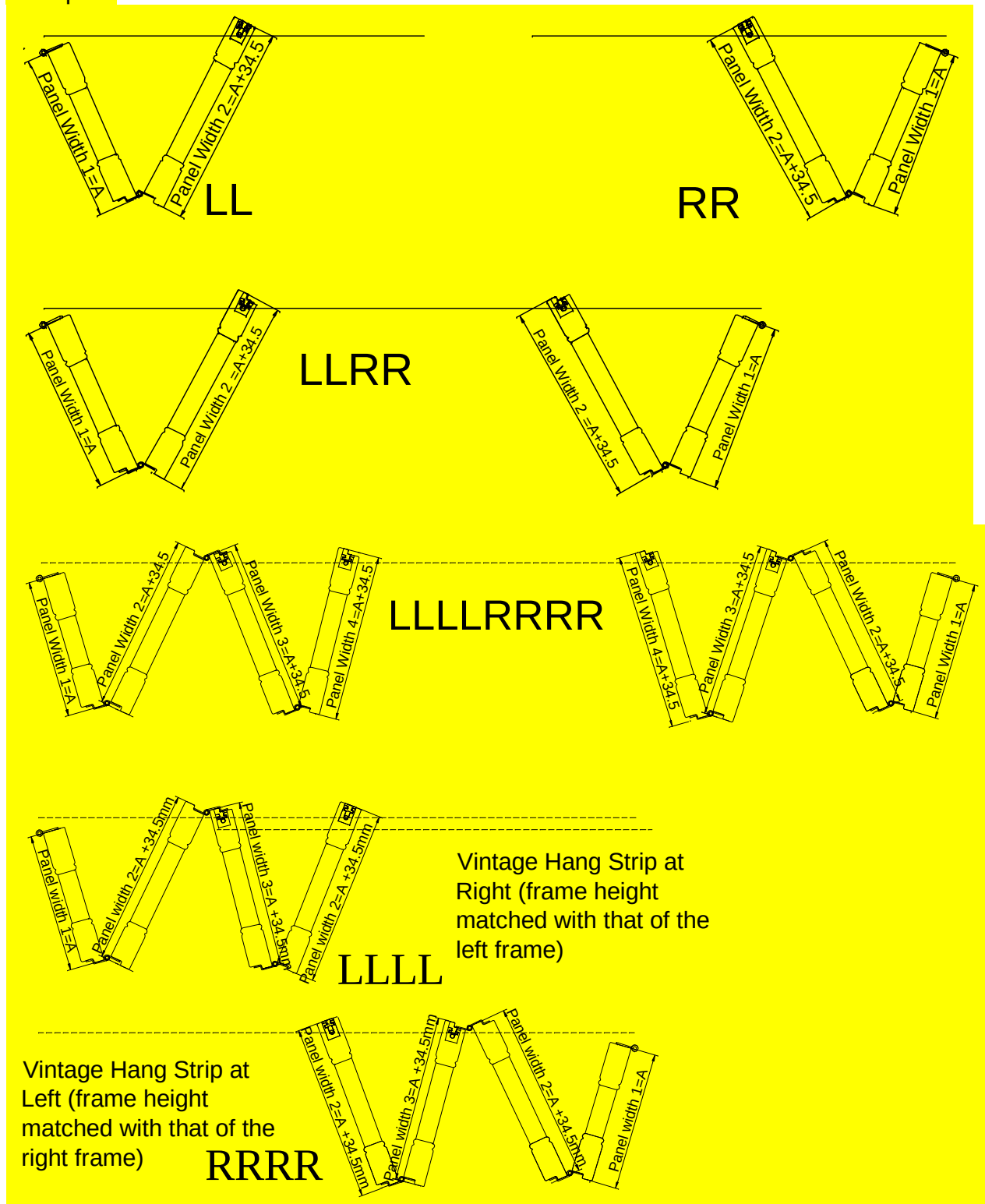


LLL/RRR/LLLRRR



Frame Hinged Bifold Track Panel Hanger Location:

Take the drawings of the hinge type between panel and frame is 2 3/8" Self Mortise Hinge as examples:

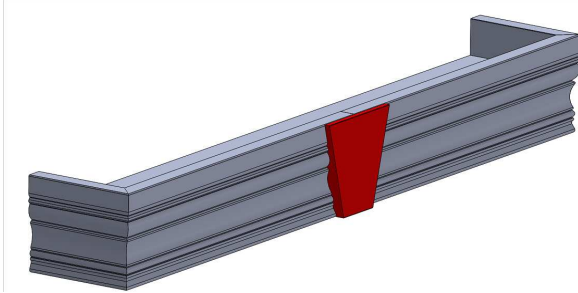


Keystone for Fascia of the Track Shutter

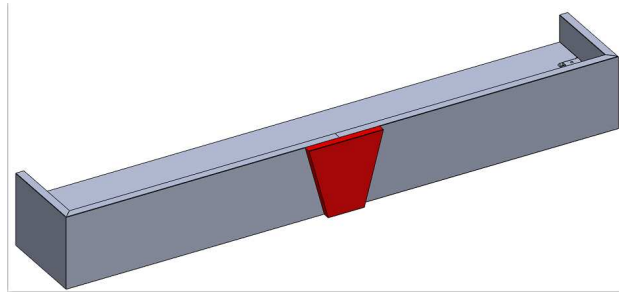
When fascia length exceeds the limitation and needs to split, keystone design fascia will be supplied without surcharge for track shutters.

For fascia length within the split limitation, keystone can also be supplied as an option with surcharge.

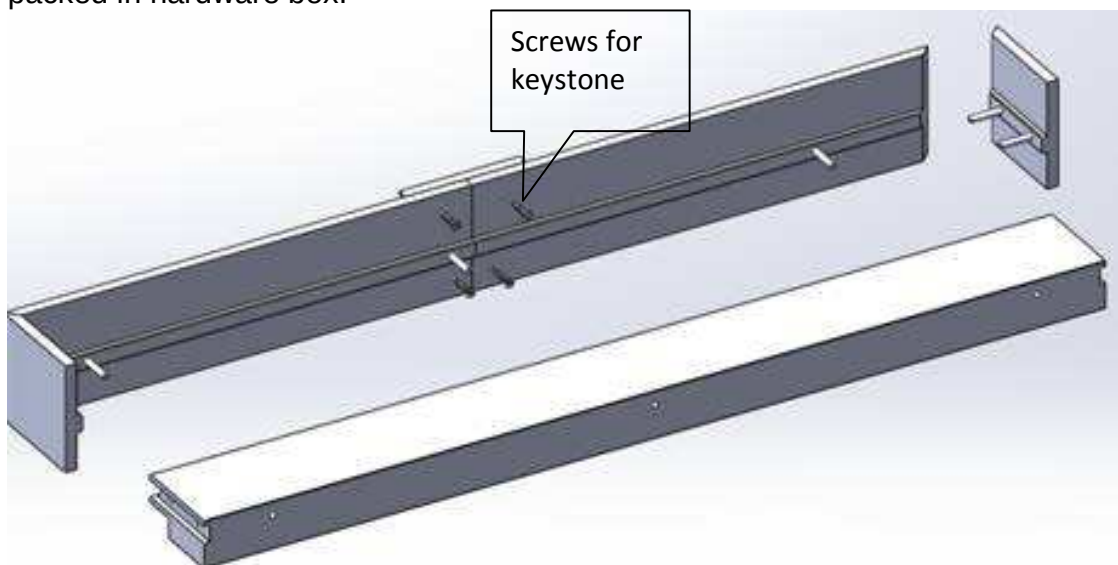
For Deco Fascia:



For Flat Fascia:



The keystone will be installed by screws. We will pre-drill the screw holes and the screws will be packed in hardware box.



Track Shutters—2020 Bypass



Definitions

Closed Bypass - louvers must be in the closed position for panels to bypass each other when sliding.

Open Bypass - louvers can remain open position when panels bypass each other allowing for a clearer view when sliding.

Combi-joined panels - two or more panels combined or hinged together to form a single, wider sliding section, please see the drawings of where hinges will be installed.

Note: the interlocking bottom guide is standard on the closed bypass and optional on the open bypass.

Guidelines and Specs:

<u>Description</u>	<u>Minimum</u>	<u>Maximum</u>		
		1 $\frac{7}{8}$ "	2 $\frac{1}{2}$ " , 3"	3 $\frac{1}{2}$ " ; 4 $\frac{1}{2}$ "
Single-Panel Width (Wood)	6"	30"	36"	42"
Single-Panel Height	10"	132"	132"	132"

Outside Mount

Max frame-to-frame width: window width + 3"

Max frame-to-frame height: window height + 4 $\frac{1}{2}$ "

Semi-Inside Mount

Max frame-to-frame width: window width - $\frac{1}{8}$ "

Max frame-to-frame height: window height + 1 $\frac{1}{8}$ "

Inside Mount

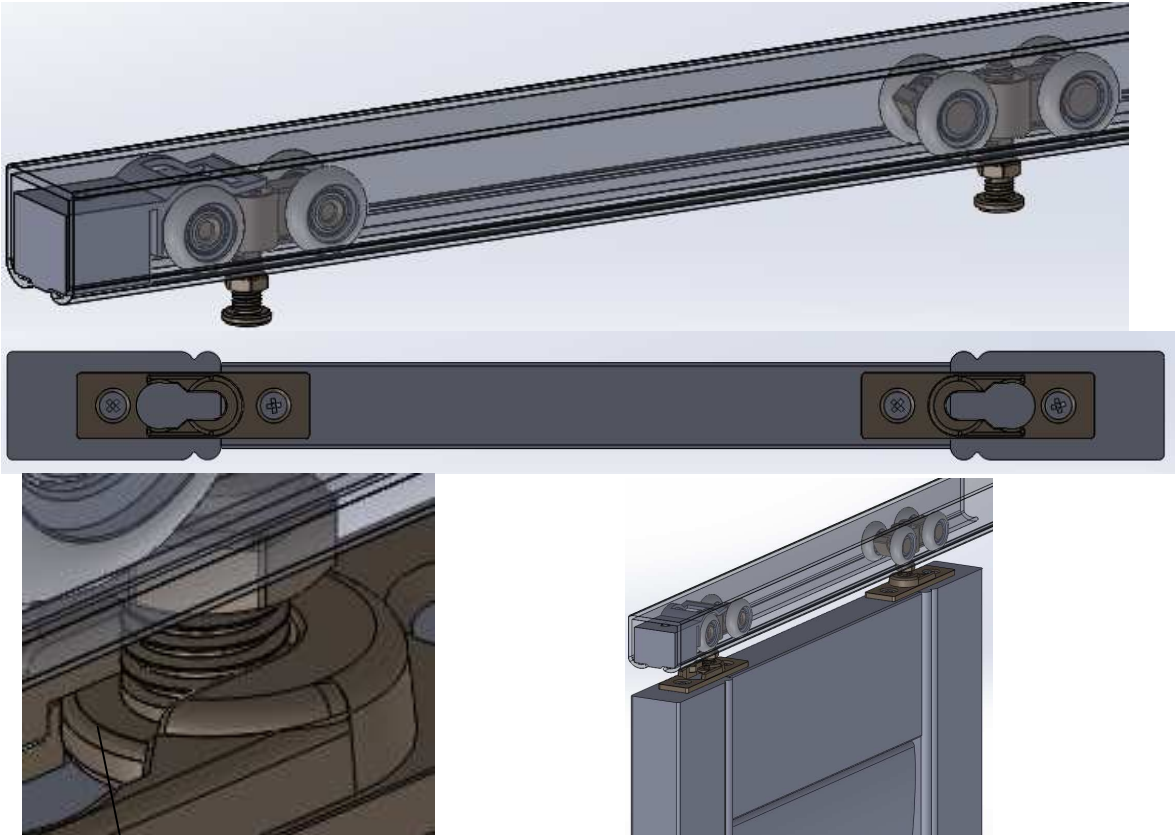
Max frame-to-frame width: window width - $\frac{1}{8}$ "

Max frame-to-frame height: window height - $\frac{1}{8}$ "

Track Shutters—2020 Bypass

Wheel Carriers are in the Track / Wheel Carrier Brackets are on the top of Panel:

Adjustments to panel alignment are made by turning the wheel carrier bolt with the wrench included in the Hardware Box

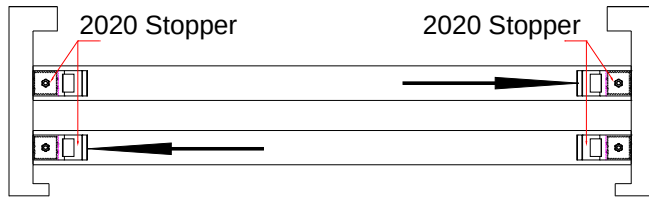


Push wheel carrier into the wheel carrier bracket to seat

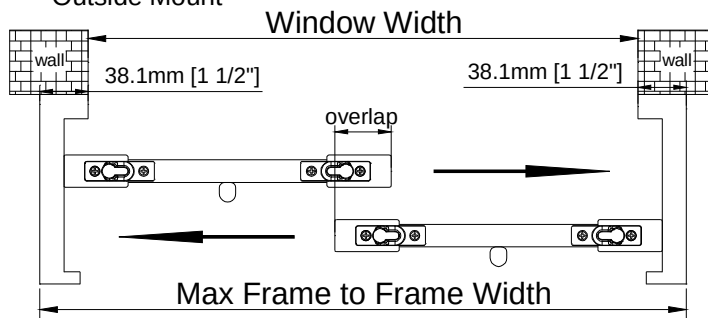
Note: 2020 Stopper will be supplied at both ends of top track by default.

Track Shutters—2020 Bypass

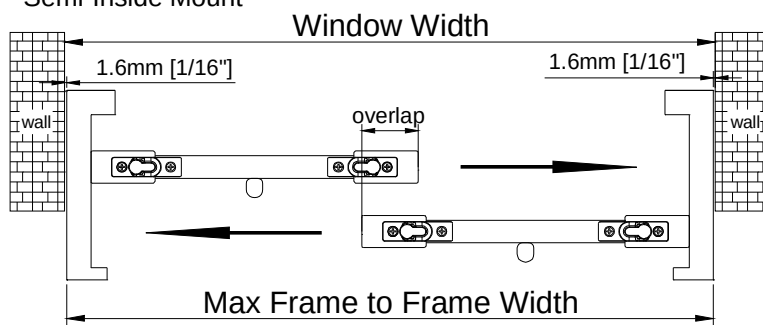
Top View



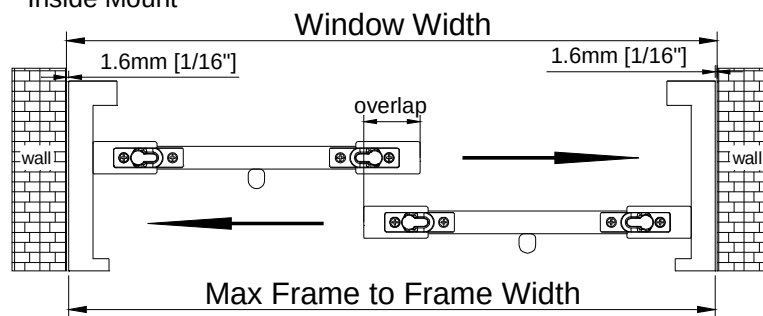
Outside Mount



Semi-Inside Mount

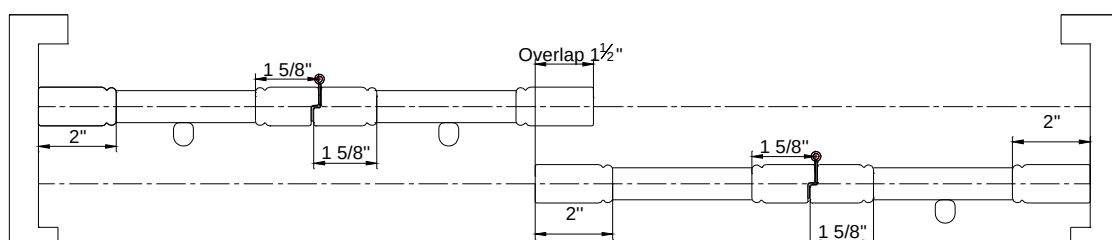


Inside Mount



Top View

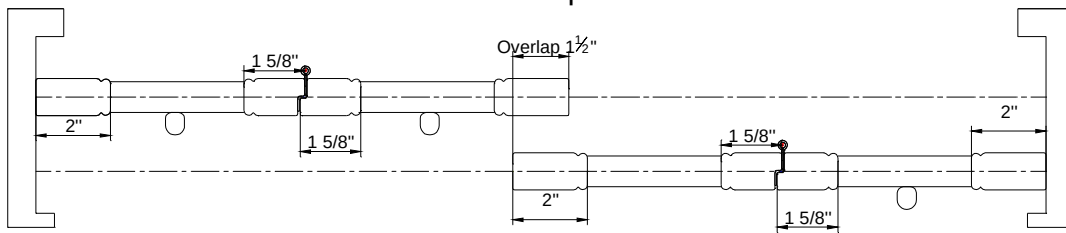
Combi - joined Panels:



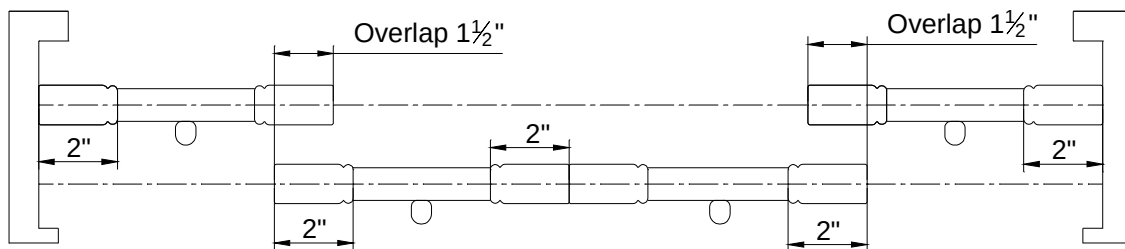
Track Shutters—2020 Bypass

For shutters with 4 or more panels, please select “side open” or “center open”

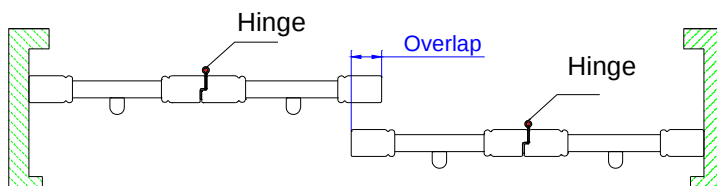
Side Open



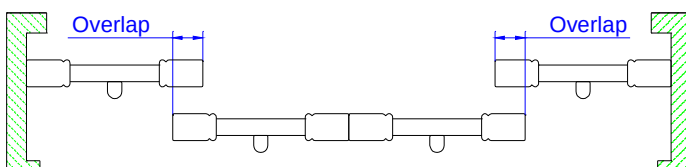
Center Open



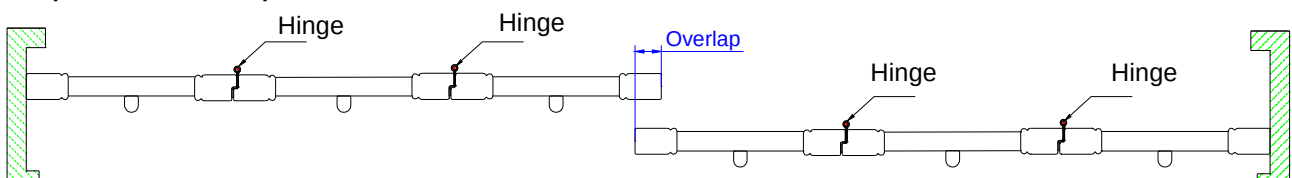
4 panels side open w/Rabbet stile



4 panels center open w/Butt stile



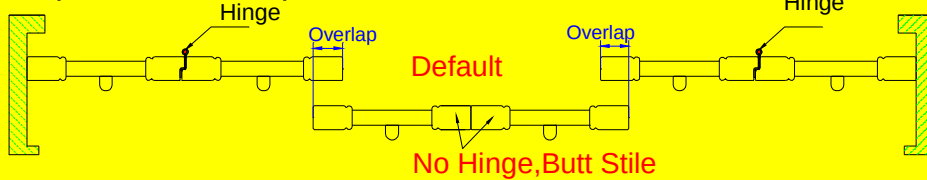
6 panels side open w/Rabbet stile



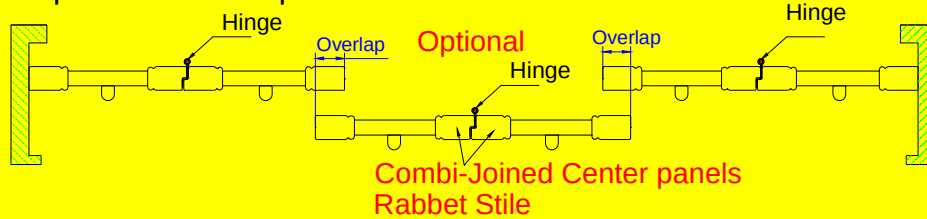
Track Shutters—2020 Bypass

For 6 panels center open configurations, the center panels should split (1/1) by default with the option for them to be combi-joined into a single 2 panel section.

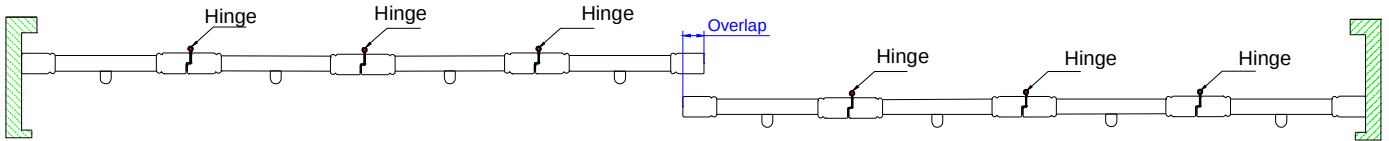
6 panels center open w/Rabbet stile



6 panels center open w/Rabbet stile

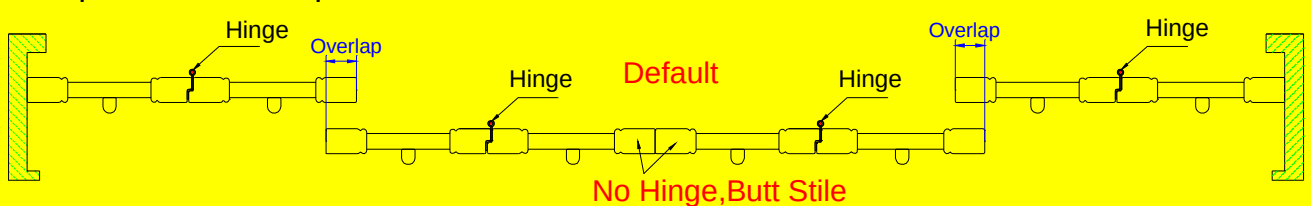


8 panels side open w/Rabbet stile

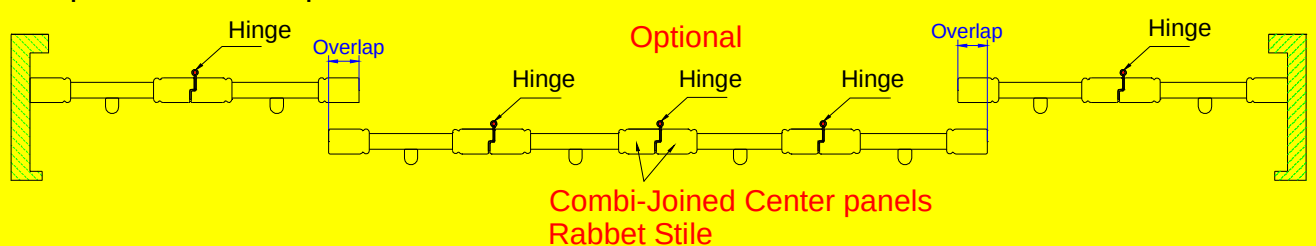


For 8 panels center open configurations, the center panels should split (2/2) by default with the option for them to be combi-joined into a single 4 panel section.

8 panels center open w/Rabbet stile



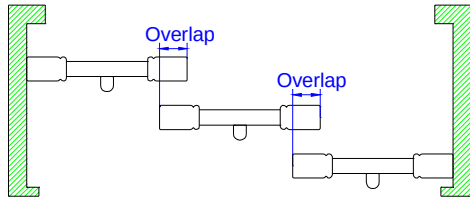
8 panels center open w/Rabbet stile



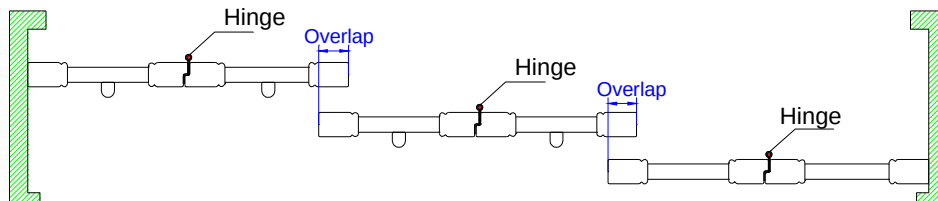
Track Shutters—2020 Bypass

Triple Track

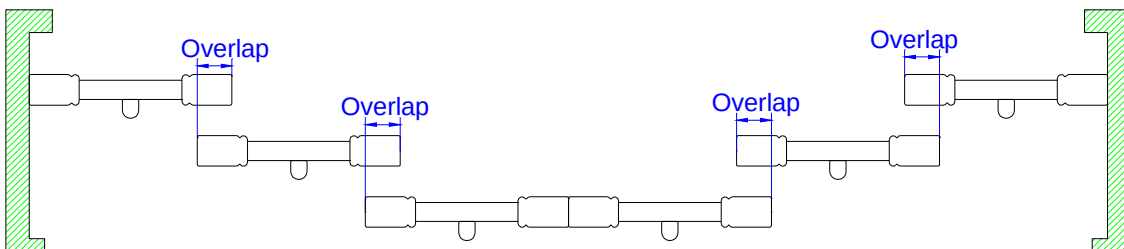
3 panels side open w/Butt stile



6 panels side open w/Rabbit stile



6 panels center open w/Butt stile

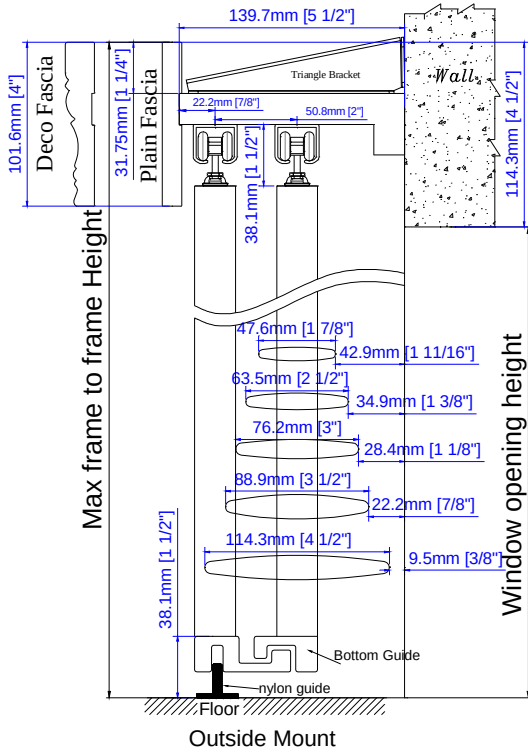


Stile: Only 2", 2 1/4" butt and rabbit stiles are available,
1 5/8" rabbit stile only used for combi-joined panels.
Center stile is not available.

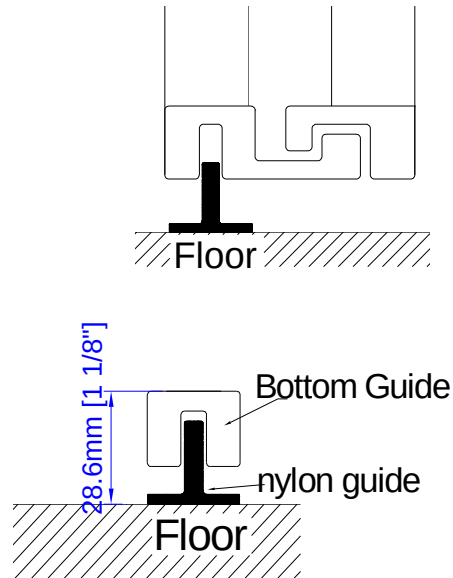
Track Shutters—2020 Bypass

Mounting Options (Closed Bypass)

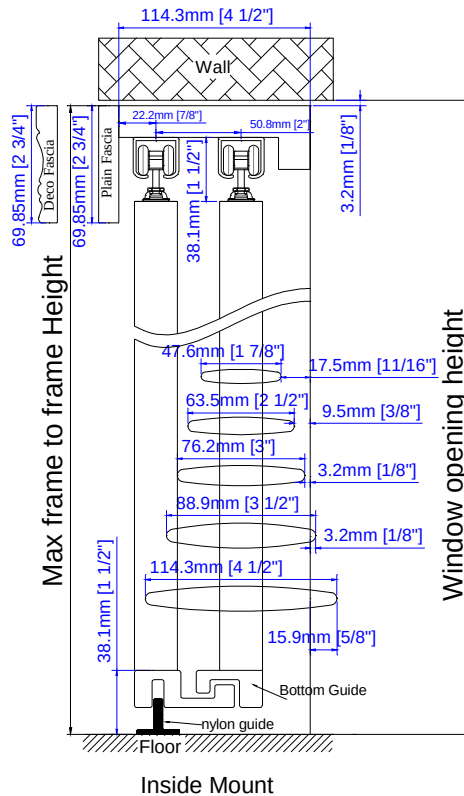
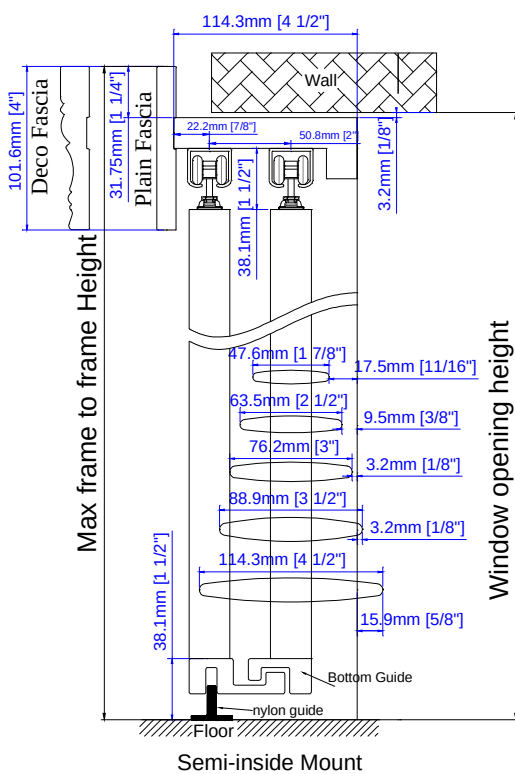
Side View



Bottom Guide

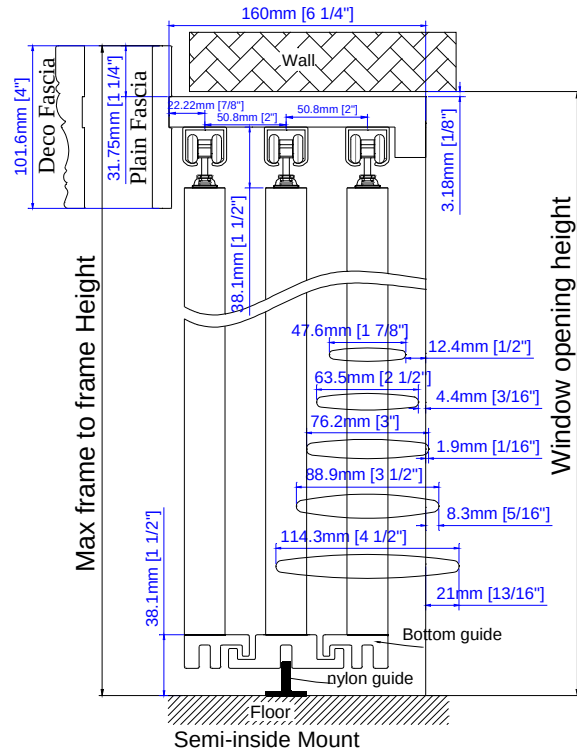
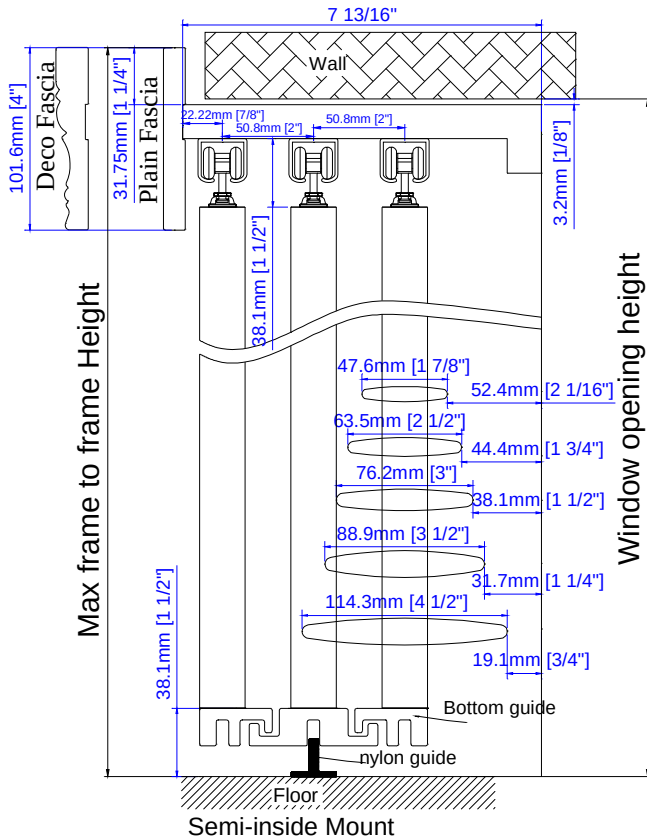
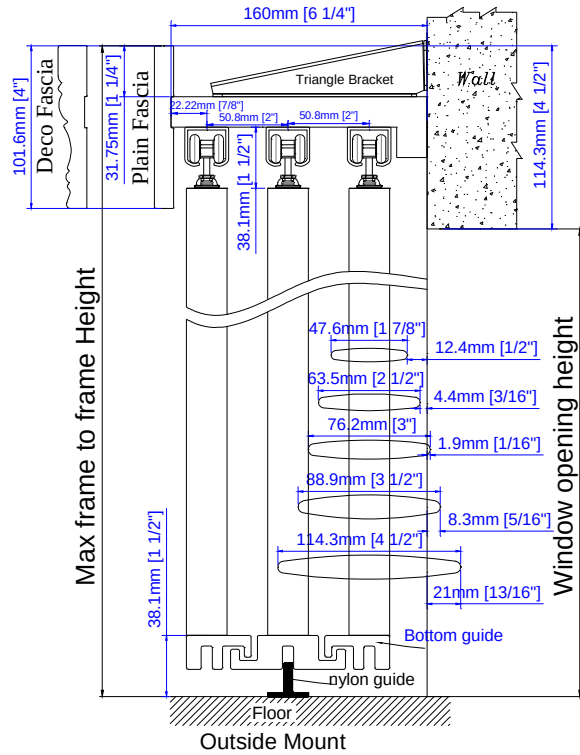
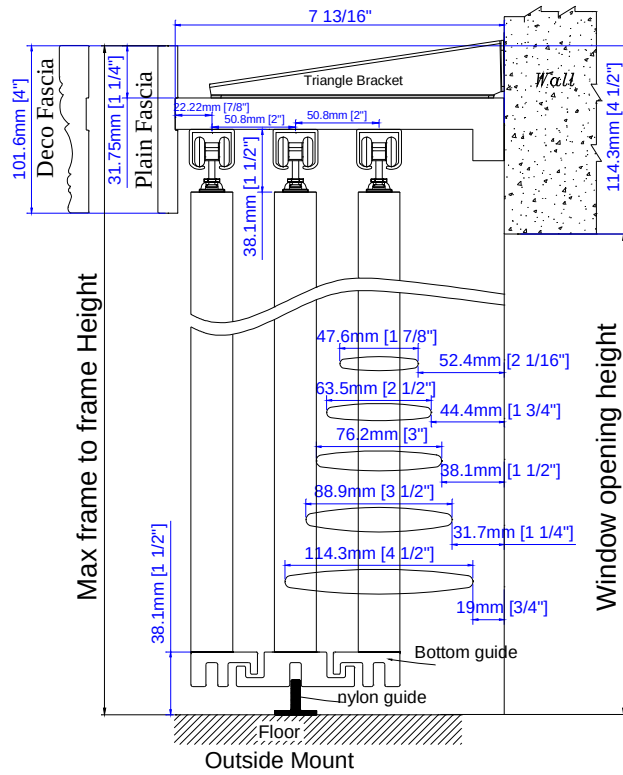


If the closed bypass only uses one track, the U-shaped bottom guide for the open bypass will be used at the bottom of the panel.



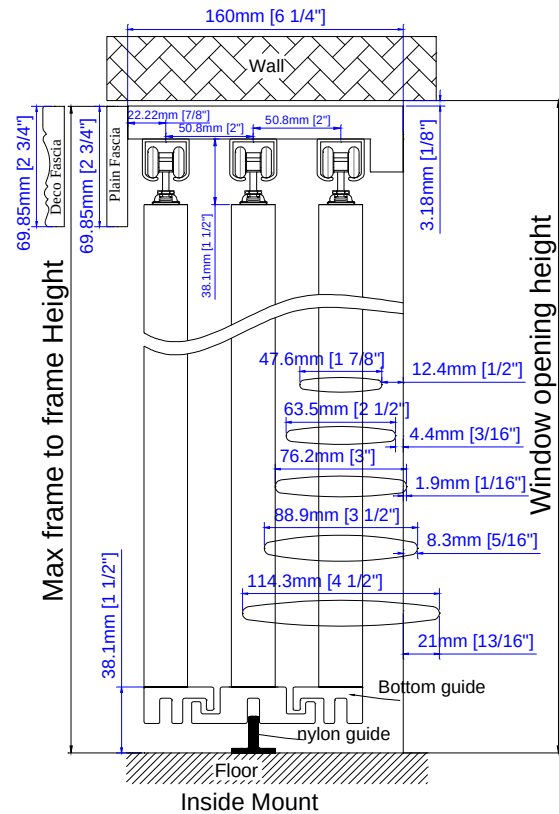
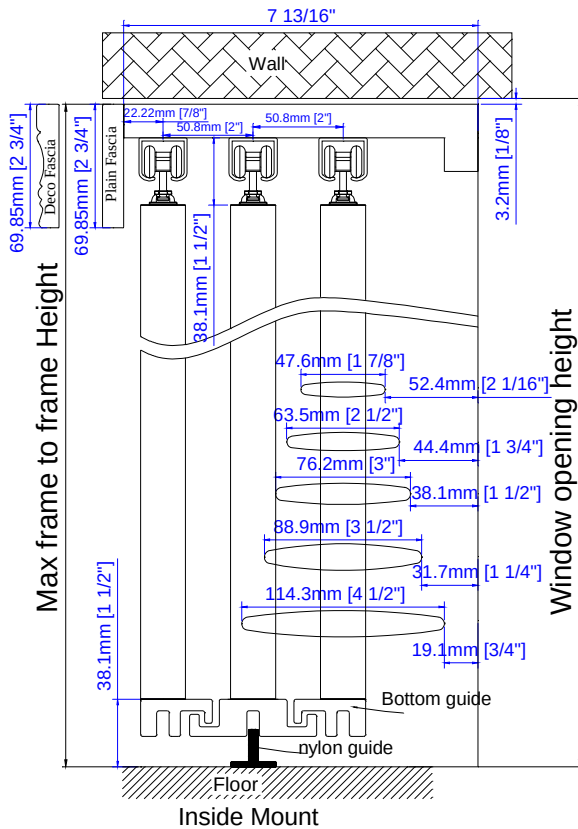
Track Shutters—2020 Bypass

Mounting Options (Closed Bypass with 3 Tracks, Triple tracks):



Track Shutters—2020 Bypass

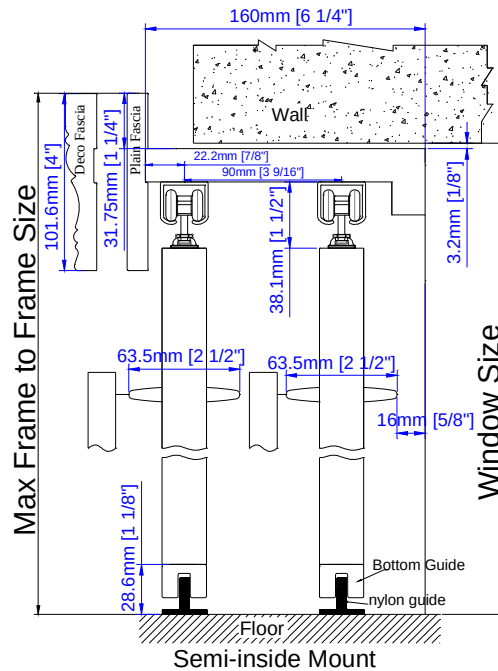
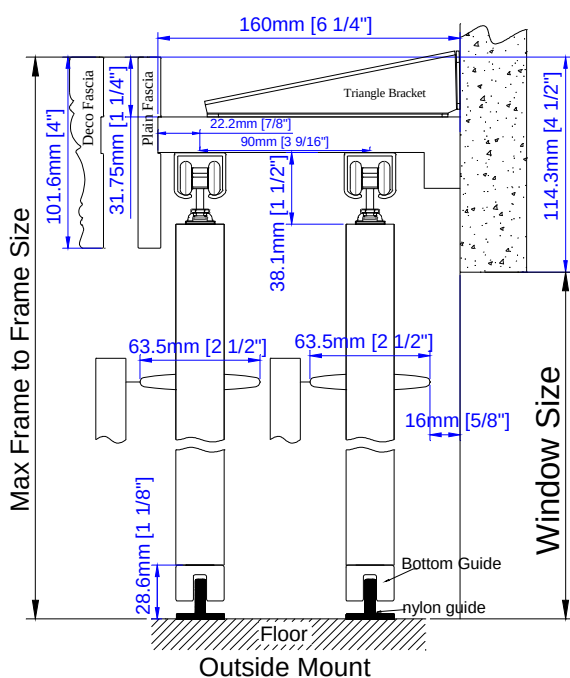
Brightwood
by Norman®



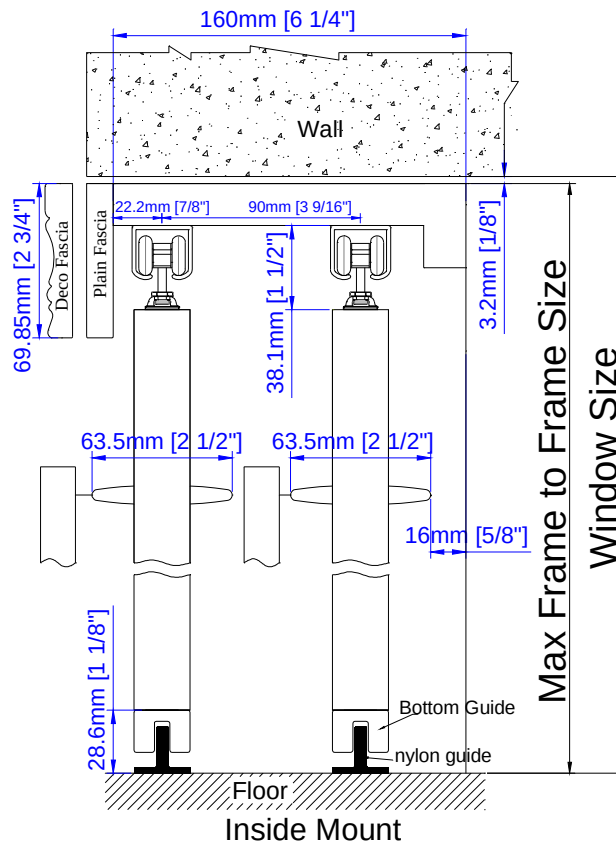
Mounting Options (Opened Bypass)

Only available in 2 1/2", 3" and 3 1/2" louver sizes

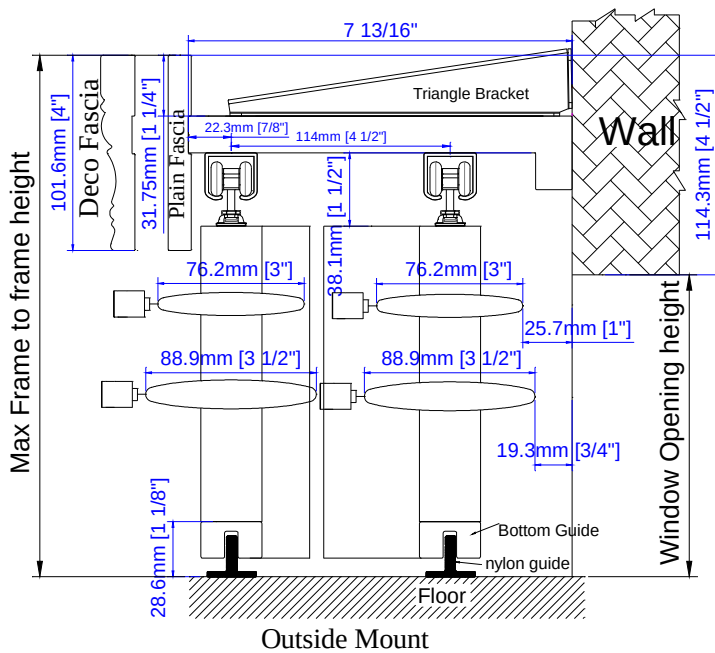
2 1/2" Louver



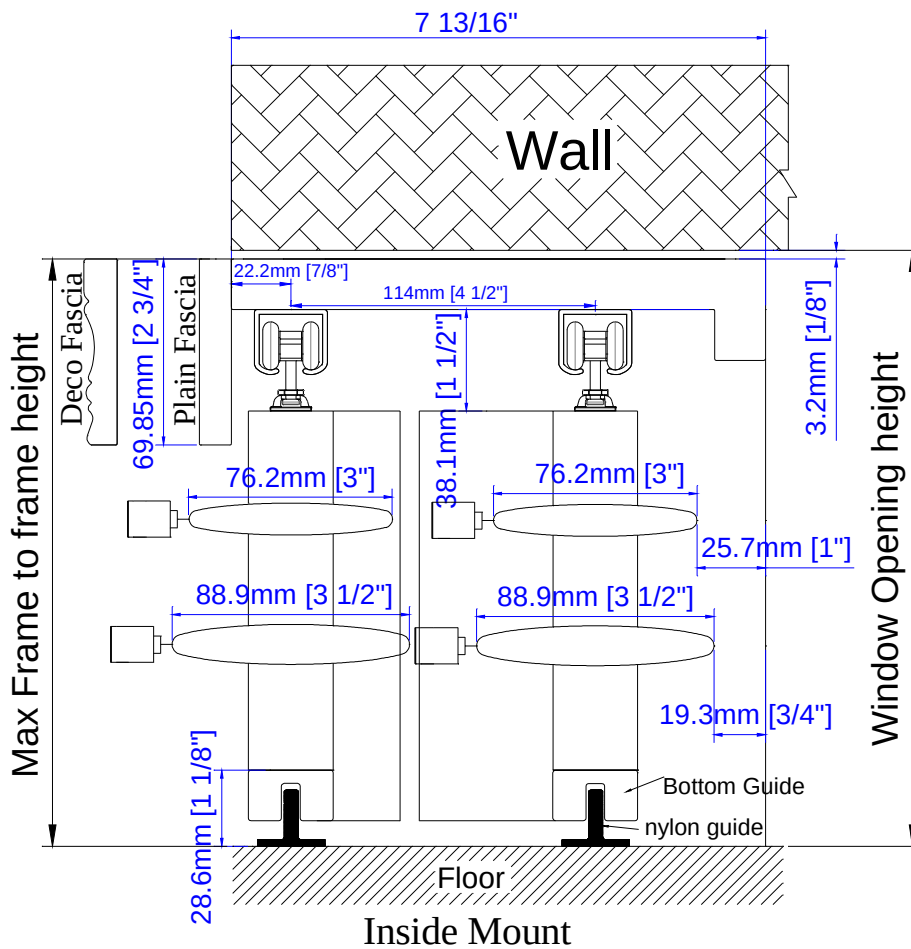
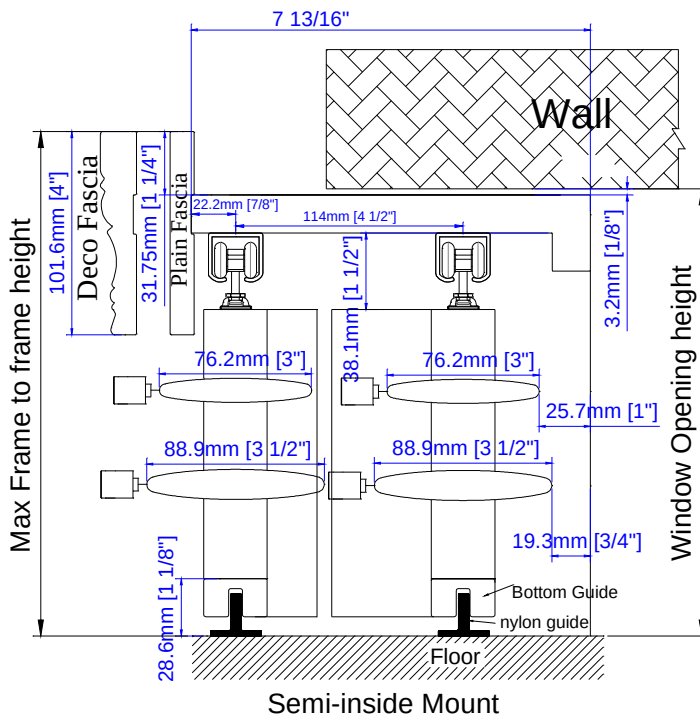
Track Shutters—2020 Bypass



3" / 3 1/2" Louver

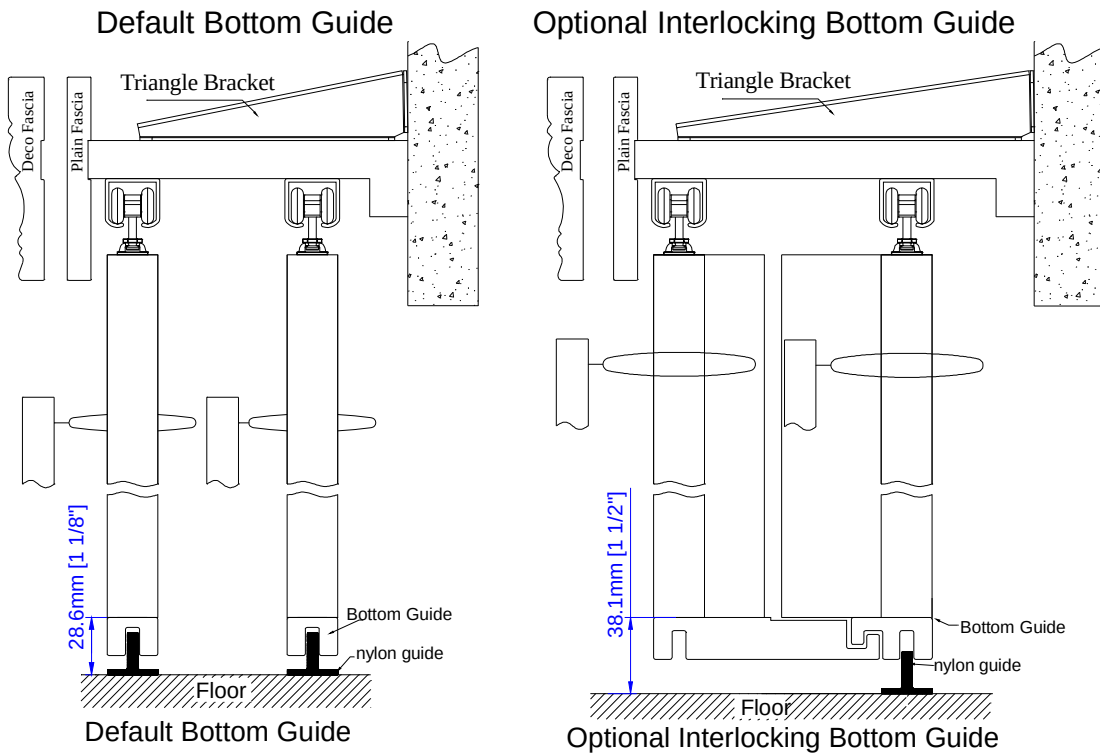


Brightwood™
by Norman®



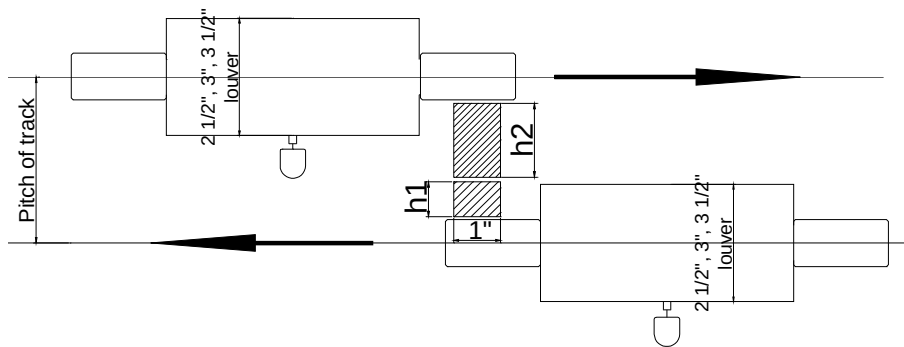
Track Shutters—2020 Bypass

Bottom guide for Open ByPass:

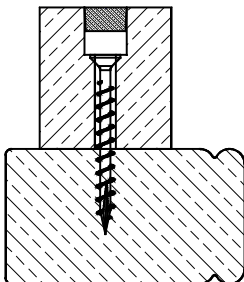


Opened Bypass

Top View



Note: The light block for open bypass will be changed to use screws with wood plugs instead of current dowels.



Track Shutters—2020 Bypass

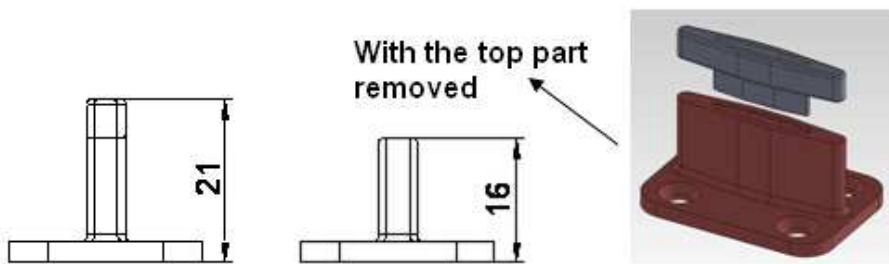
Separation of track

Louver	U shape bottom guide		Interlocking bottom guide	
	Regular Tilt Rod	Invisible tilt	Regular Tilt Rod	Invisible tilt
2.5"	3 9/16"	2 3/4"	3 9/16"	3 9/16"
3" / 3.5"	4 1/2"	3 5/8"	4 1/2"	4 1/2"

Light block thickness for panel

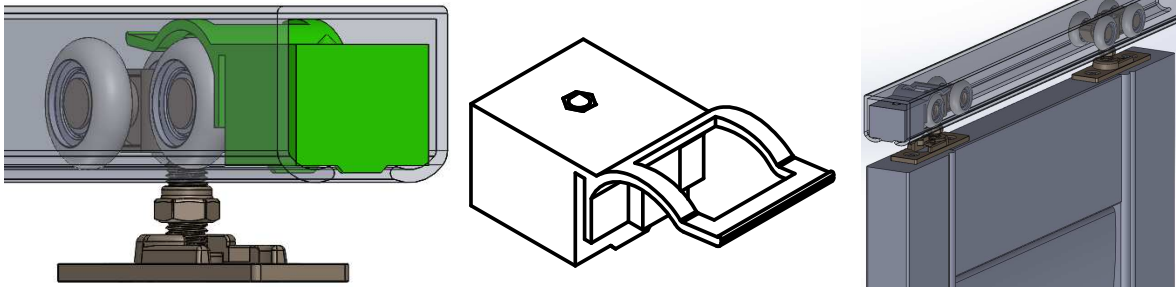
Louver		U shape bottom guide		Interlocking bottom guide	
		Regular Tilt Rod	Invisible tilt	Regular Tilt Rod	Invisible tilt
2.5"	h1	3/4"	3/4"	3/4"	3/4"
	h2	1 9/16"	3/4"	1 9/16"	1 9/16"
3" / 3.5"	h1	1 3/16"	1 3/16"	1 3/16"	1 3/16"
	h2	1 15/16"	1 3/16"	1 15/16"	1 15/16"

Nylon Guide: The nylon guide is composed by 2 parts and will supplied as a whole set by default (bottom part 16mm and top part 5mm), please remove the top part if necessary.



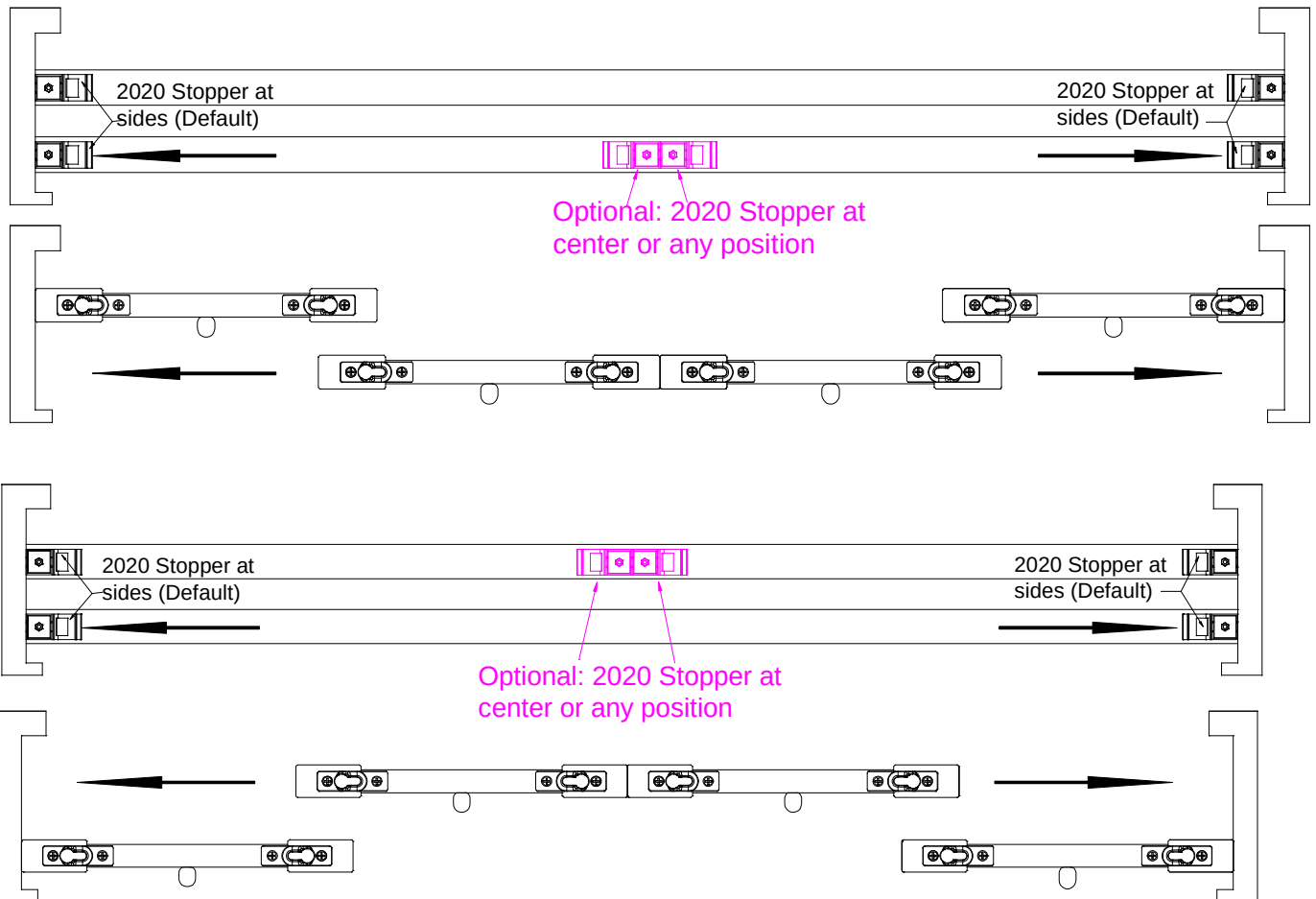
Track Shutters—2020 Bypass

The 2020 Stopper



Note: 2020 Stopper will be supplied at both ends of top track by default.

The 2020 stopper can also be used in the center or any position of the top track as an option. Please specify the actual position when placing an order.



Track Shutters—2020 Bypass



Track Shutters Framing Materials

Framing	Size				Pc(s)
L Shape Header	Header (Closed Bypass)	Header (Opened Bypass)	Supporting Strip		1 pc
(2 in 1)	$\frac{3}{4}" \times 5\frac{1}{2}"^*$ (Default for Outside Mount; Optional for Semi-Inside and Inside Mount)	$\frac{3}{4}" \times 6\frac{1}{4}"^*$ (2½" Louver)	$\frac{3}{4}" \times \frac{3}{4}"$		
Header with Supporting Strip	$\frac{3}{4}" \times 4\frac{1}{2}"^*$ (Default for Semi-Inside and Inside Mount; Optional for Outside Mount)	$\frac{3}{4}" \times 7\frac{13}{16}"^*$ (3", 3½" Louver)			
	Custom * ≤ 8"	Custom * ≤ 8"			
U Shape Side Frame	Side Frame (Closed Bypass)	Side Frame (Opened Bypass)	Supportin g Strip	Side Fascia	2 pcs
(3 in 1)	$\frac{3}{4}" \times 5\frac{7}{8}"^*$ (Default for Outside Mount; Optional for Semi-Inside and Inside Mount)	$\frac{3}{4}" \times 6\frac{5}{8}"^*$ (2½" Louver)	$\frac{3}{4}" \times \frac{3}{4}"$	$\frac{3}{8}" \times 1\frac{1}{4}"$	
Side Frame with Supporting Strip & Side Fascia	$\frac{3}{4}" \times 4\frac{7}{8}"^*$ (Default for Semi-Inside and Inside Mount; Optional for Outside Mount)	$\frac{3}{4}" \times 8\frac{1}{4}"^*$ (3", 3½" Louver)			
	Custom * ≤ 8"	Custom * ≤ 8"			
Top Fascia (Plain/Deco)	4" (Semi-inside and Outside Mount)				1 pc
	$\frac{1}{2}" \times 2\frac{3}{4}"$ (Inside Mount)				
Top Fascia Return	4" (Semi-inside and Outside Mount)				2 pcs
	N/A (Inside Mount)				

Note:

- When the header size is $4\frac{1}{2}"^*$, $5\frac{1}{2}"^*$, $6\frac{1}{4}"^*$ or $7\frac{13}{16}"^*$, the L Shape header and the U Shape Side Frame will be sent integrally. That is:
 1. The header and the supporting strip will be sent integrally as L Shape Header;
 2. The side frame, supporting strip and the side fascia will be sent integrally as U Shape Side Frame.
- For any custom header sizes other than $4\frac{1}{2}"^*$, $5\frac{1}{2}"^*$, $6\frac{1}{4}"^*$ or $7\frac{13}{16}"^*$, the L Shape Header and the U Shape Side Frame will be sent separately. That is:
 1. The header and the supporting strip will be sent separately for L Shape Header;
 2. The side frame, supporting strip and the side fascia will be sent separately for U Shape Side Frame.

- Custom Boxout Width is available for both closed and opened bypass shutters.

Track Shutters—2020 Bypass

- For Outside Mount Bypass, the custom header size can reach to 8", and the total size of custom header size with extension thickness can't be exceeded 8 5/8".
- For Inside Mount and Semi-Inside Mount Bypass, the custom header size can reach to 8", and the extension thickness of header can reach to 2".

Hardware Box

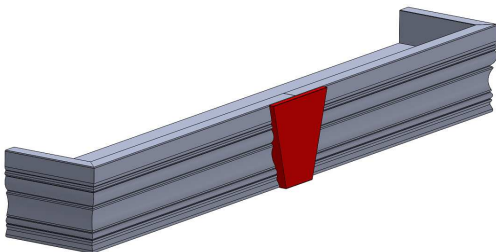
Hardware Box
Part
Wrench
Nylon Guide
3/4" screws for Nylon Guide
Triangle Bracket (based on the length of header)
2" screws for Triangle bracket
3/4" screws for Triangle bracket

Keystone for Fascia of the Track Shutter

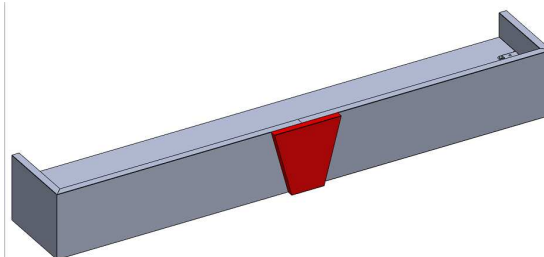
When fascia length exceeds the limitation and needs to split, keystone design fascia will be supplied without surcharge for track shutters.

For fascia length within the split limitation, keystone can also be supplied as an option with surcharge.

For Deco Fascia:

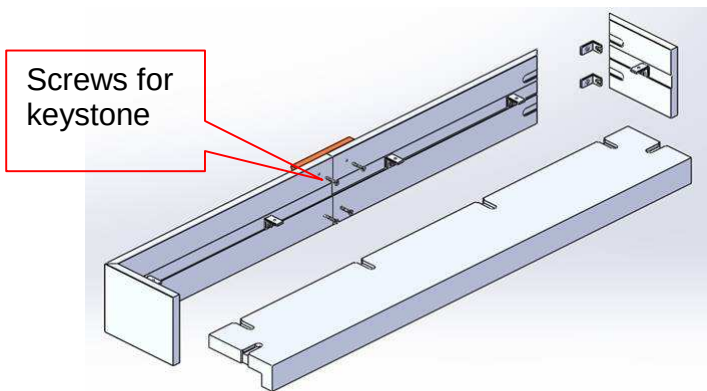


For Plain Fascia:



The keystone will be installed by screws. We will pre-drill the screw holes and the screws will be packed in hardware box.

Track Shutters—2020 Bypass

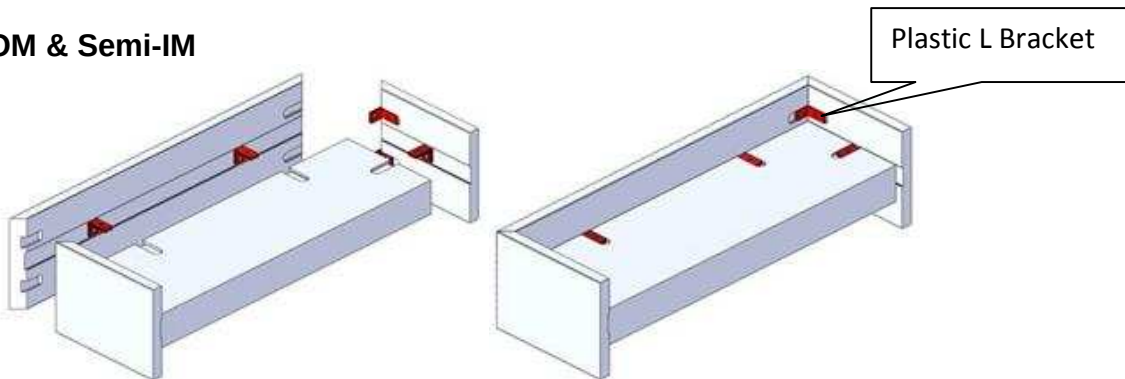


Plastic L Bracket to Connect the Header, Fascia & Fascia Returns

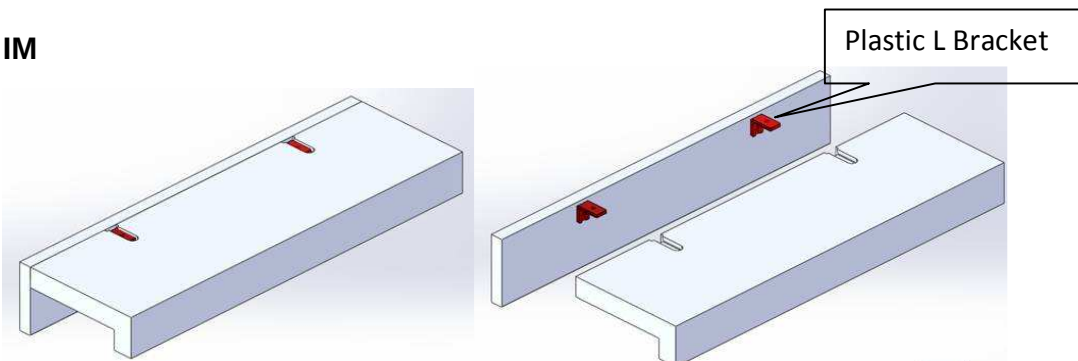


L Bracket will be used to connect the header, fascia and fascia returns for Bypass shutter. This applies for Inside Mount, Semi-Inside Mount & Outside Mount.

OM & Semi-IM



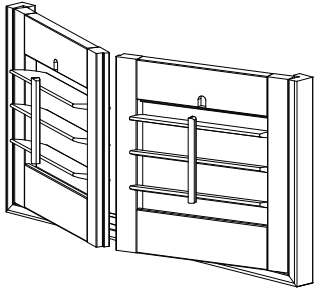
IM



Special Options

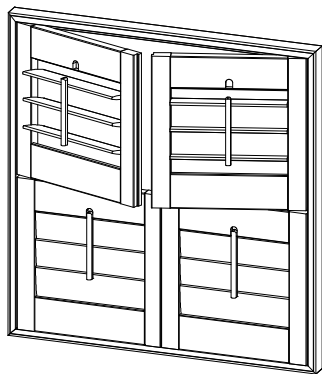
Café* Shutters

- The top of the panel requires a special finish. Please indicate “café shutter” when placing orders.
- The shutter does not come with recessed magnet on top.
- When the height of the framing material (including t-posts) exceeds the panel height, by more than 5", a surcharge will apply.

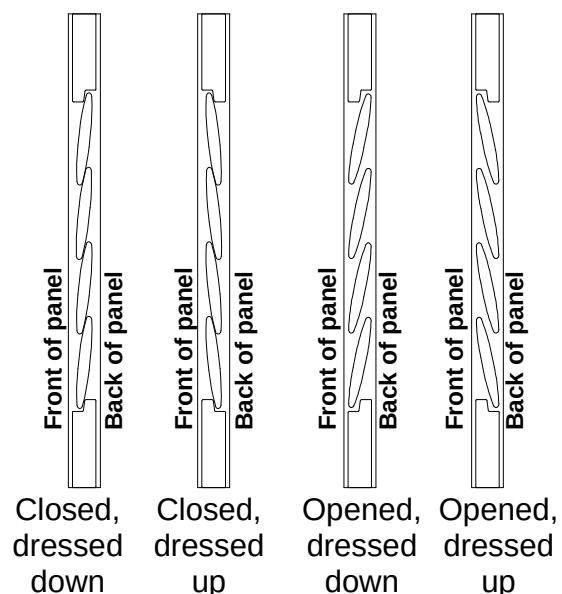
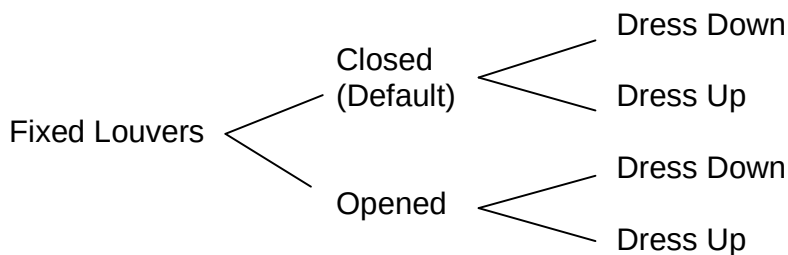


Double-Hung* Shutters

- For total number of panels, please include panels in both rows.
- The default location for the division point is the center unless specified otherwise.
- Horizontal T Post can be added for increased structural integrity.
- The length of Horizontal T post in each section must be less than or equal to 48".
- For Double Hung shutter, whenever an upper or lower panel is larger than 78", we will add one more extra hinge for it, we will also attempt to create larger top/bottom rails for additional strength by reducing the number of louvers by one as long as the rail sizes remain within spec.



Fixed Louvers



Fixed louvers Closed Dress Down and Fixed louvers Opened Dress Down are not available for all specialty shutters except all French Door Type E,F and Type A,B,C&D without fixed louver(s) which did not choose “Arch type” option.

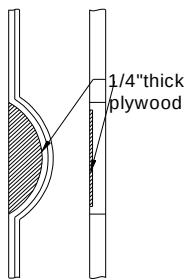
*Some surcharges apply

French Door Contour Collection*

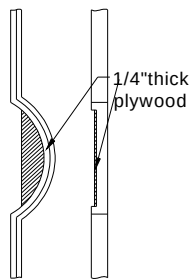
- Flush and offset cutout options are available in both curved and square French door cutout.
 - o Offset cutout is a good option for French Door with molding around the glass.
- All French door cutouts are only available in L frame except Type E and F. The batten is placed behind the Type E/F panel to block light.

Louver size	Batten Width	Batten thickness
1 7/8" & 2 1/2"	3/4"	1"
3" & 3 1/2"	3/4"	1 3/8"
4 1/2"	3/4"	1 7/8"

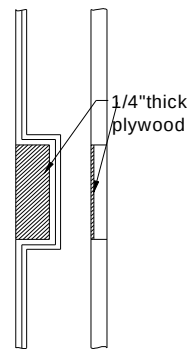
- For measurement details please refer to forms section.
- French Door cutouts with Arch/Quarter Arch at the top are also available except Type E and F.
- Vintage L frame with 1/4" light block and Beaded L frame with 1/4" light block are not available for French door Type A, B, C & D.



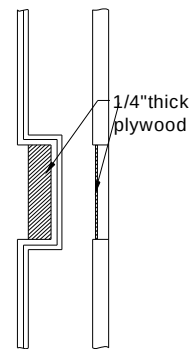
Type A



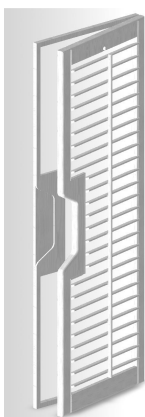
Type B



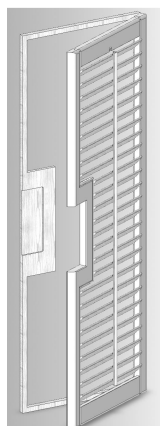
Type C



Type D



Type E



Type F

Auto Tilt[™] (Patented) *

Available in Bifold or Multi-fold panel configuration

Available with Invisible Tilt Only

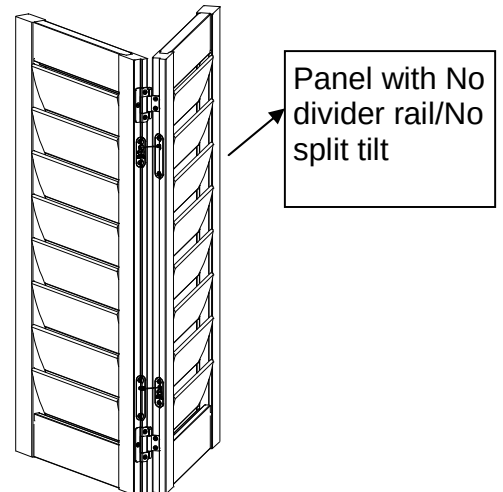
Available in 2" Butt, Astragal and Rabbet Stile

Minimum Louver Qty (Please see below for details)

Ring pulls included. For ring pull location, please refer to the ring pull section.

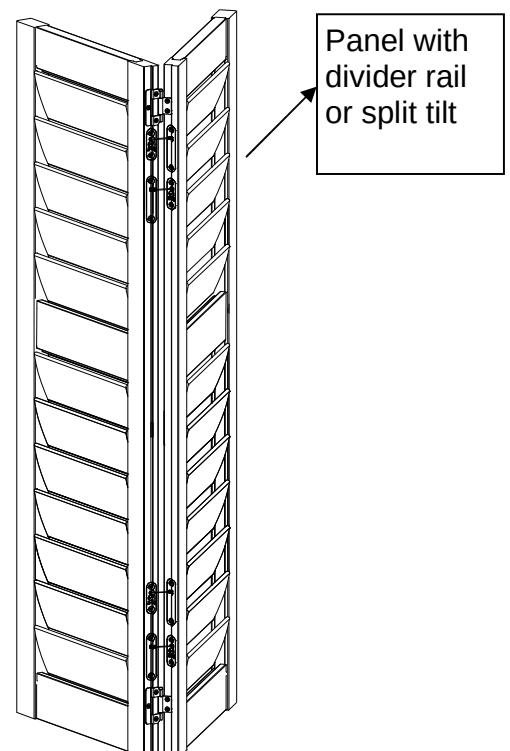
When there is only one louver section (no divider rail/no split tilt)

Louver Size	Rail Size	Minimum Louver qty
2.5"	<4.25"	6
	>=4.25"	4
3"	<3.75"	5
	>=3.75"	4
3.5"	2.75"~6.5"	4
4.5"	2.75"~6.5"	3



When there is more than one louver sections (with divider rail/ split tilt)

Louver Size	Rail Size	Minimum Louver qty
2.5"	<4.25"	5
	>=4.25"	4
3"	2.75"~6.5"	4
3.5"	2.75"~6.5"	3
4.5"	2.75"~6.5"	3



*Some surcharges apply

Custom Colors *

Available in Brightwood shutters.

Option (Color Code)	Sample Size	<u>Confirmation Process</u>
Custom Solid (C01)	A least 4"x4" on a clean, flat surface, such as wood or cardboard (please do not send anything critical as the sample will not be returned)	All orders require customers to confirm an actual swatch which will be returned in the mail before production will commence. An additional shipping and handling fee applies.

Note:

- 1) All custom colors will be finished with the standard gloss used on all Norman products.
- 2) Please be aware that we will only be able to match the tone and not the texture or finish of the sample you submit.
- 3) Only one color revision will be allowed during the color confirmation process.

Disclaimer:

**Sherwin Williams ColorSnap (MWCS),
Sherwin Williams Designer Expressions (MWDE),
Benjamin Moore Classic (MBMC) (M2460009TL) &
Benjamin Moore Preview (MBMP) (M2440564TL)
Benjamin Moore, Off White Collection M2451888TL**

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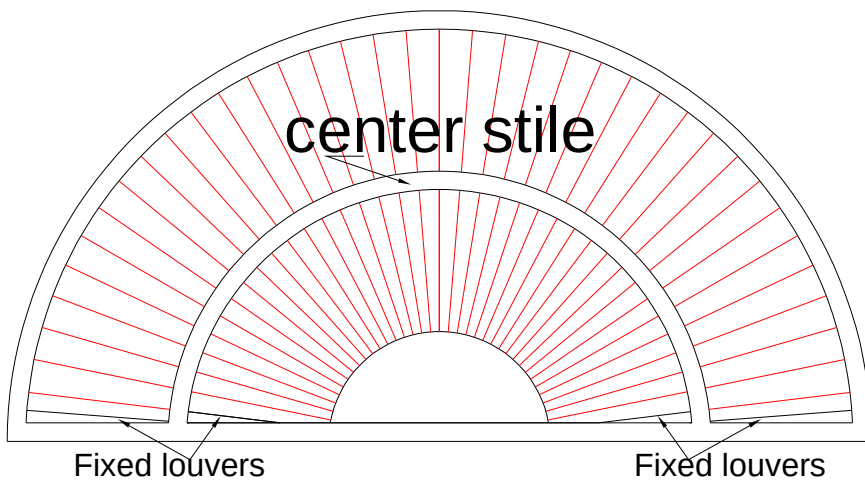
Specialties

Specialty Shutters*

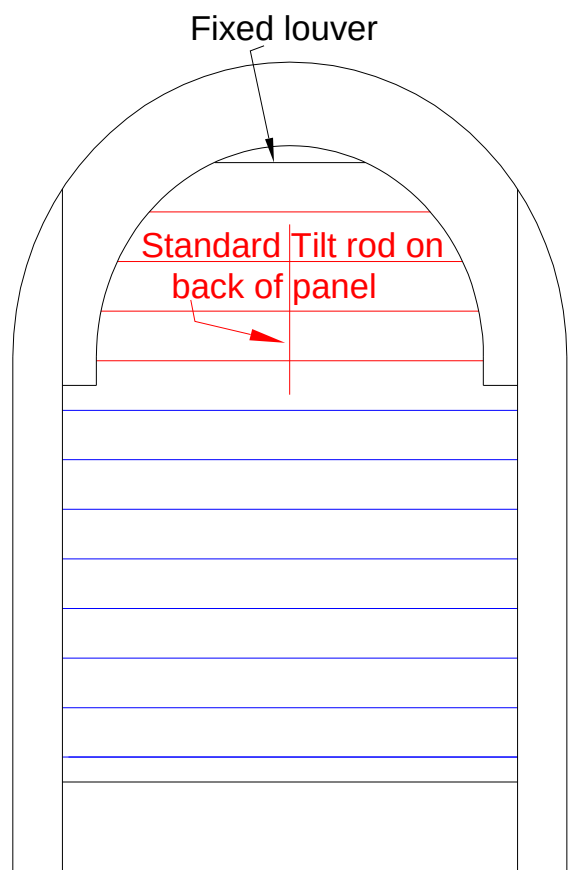
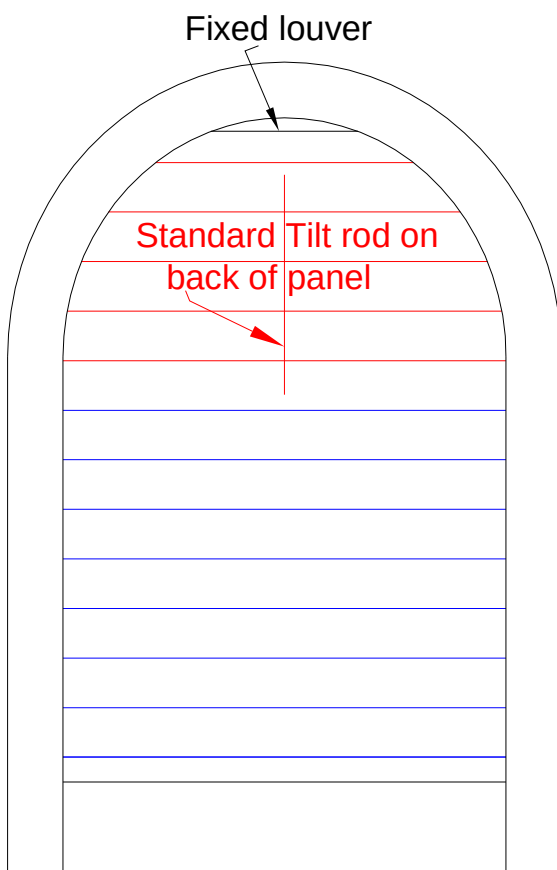
Specialty shutters are made of premium solid wood, finished to match the beauty of all Norman shutters. Templates are required if it is an imperfect specialty. Delivery time is 8 weeks from the order confirmation date for all specialty shutters. Please refer to Specialties and Template Requirement sections for details.

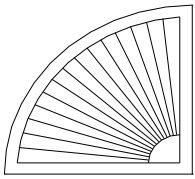
Regarding frame with an insert or solid frame issue:

- 1). For Half Round Sunburst, Quarter Round Sunburst, Eyebrow Sunburst & Elongated Eyebrow Sunburst:
 - All deco frames will be made as solid frames;
 - All Vintage L frames & Vintage hang strip: all top frames (curve shape frame) will be made as a solid frame but the bottom & both sided frames (straight frames) will be made as regular frames with inserts
- 2). For Triangle, Hexagon, Octagon, Peak & Angle top: all frames will be made as regular frames with inserts.
- 3). For Circle, Oval: all frames will be made as solid frames (not with an insert).
- 4). For Sunburst on top with Divider strip: the top frame (curve shape frame) will be made as a solid frame but the bottom & both side frames (straight frames) will be made as regular frames with inserts.
- 5). For Louver Arch with Eyebrow curve shape:
 - All deco frames excluding 2.5" Mission deco frame: all frames will be made as solid frames (not with an insert).
 - All Vintage L frames & Vintage hang strip & 2.5" Mission deco frame: the top frame (curve shape frame) will be made as a solid frame but the bottom & both sided frames (straight frames) will be made as regular frames with inserts;
- 6). For other Specialty Shutters when curve shape is similar as perfect arch (such as Louver Arch except Louver Arch with Eyebrow curve shape in point 5), Horizontal louvers center arch with quarter round side panels(-Standard louvered arch & Continuous arch), Sunburst louvers center arch with quarter round side panels, Arch Top Picture Window with horizontal louvers, Arch Top Picture Window with Sunburst, Standard unit with horizontal arch center etc.):
 - All deco frames will be made as solid frames;
 - All Vintage L frames & Vintage hang strip: all top frames (curve shape frame) will be made as a solid frame but the bottom & both sided frames (straight frames) will be made as regular frames with inserts.
- 7). Hang strip behind the panel is not available for Hexagon and octagon with horizontal louvers.
- 8). All sunburst hubs must be less than or equal to 12".
- 9). If a requested half round sunburst arch will result in louvers that are out of spec, a center stile may be added in order to produce the unit.

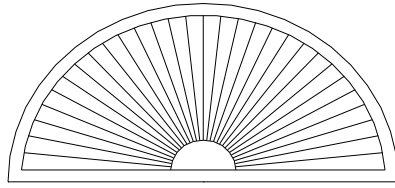


The louvers in the curved section can not be operated by the Invisible Tilt system. They can be independently operated by a separate standard tilt rod installed on back of the panel.

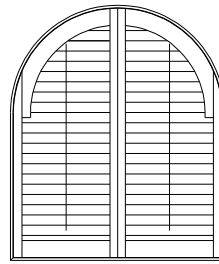




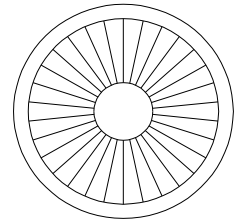
Quarter Round
Sunburst



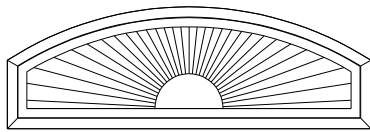
Half Round Sunburst



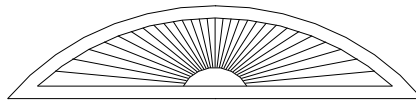
Louvered Arch
w/ Center Tilt Rod



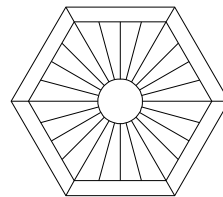
Circle Sunburst



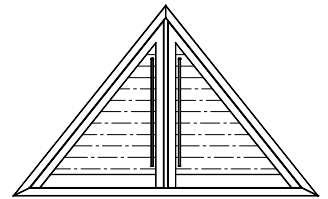
Elongated Eyebrow
Sunburst



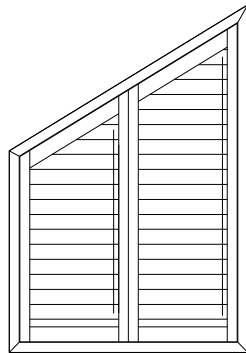
Eyebrow Sunburst



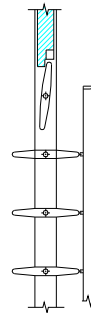
Hexagon Sunburst



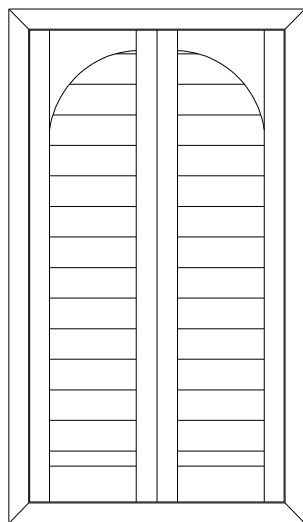
Triangle



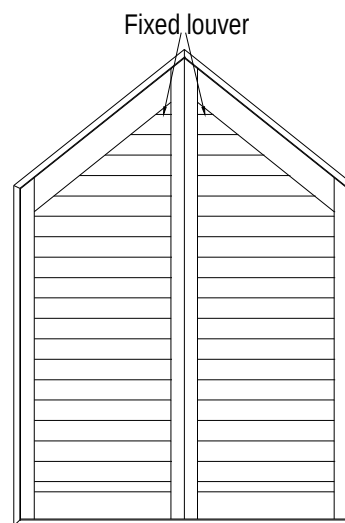
Angle Top



*Top louver of angle top might
be fixed as shown above

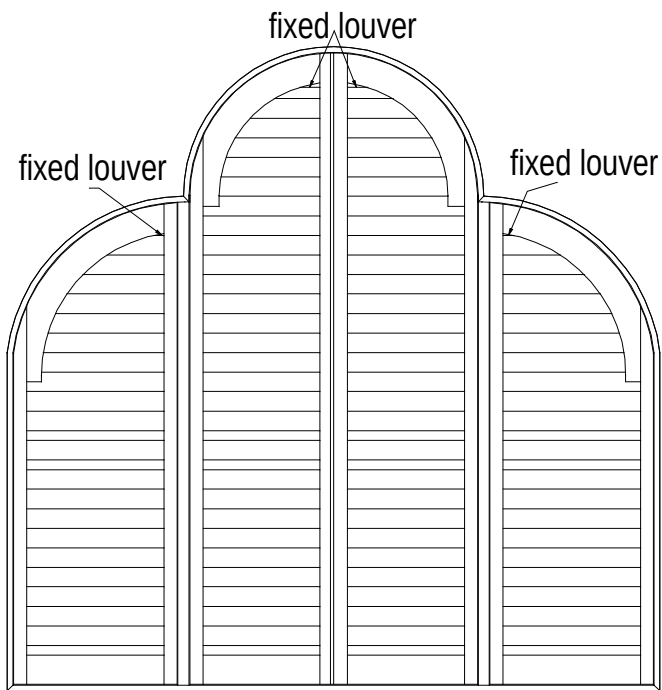


Louver Arch in a Rectangle

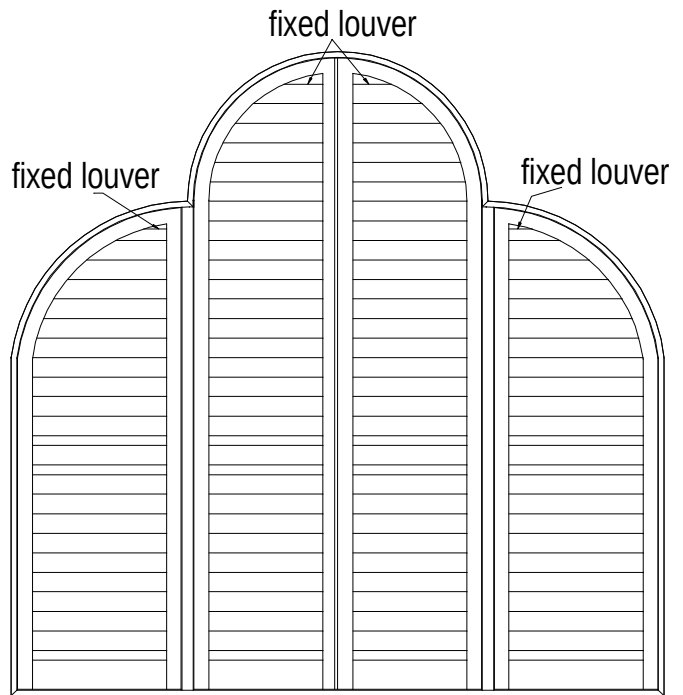


Fixed louver

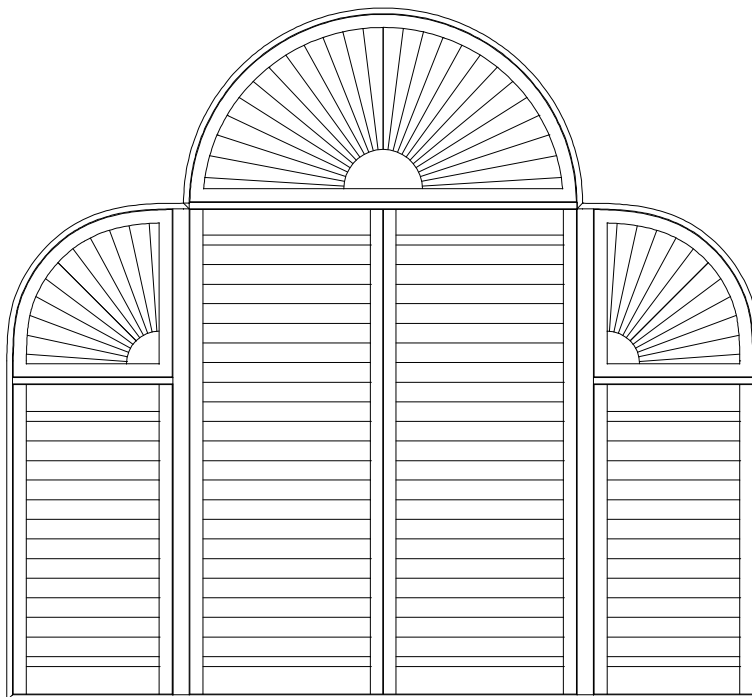
Peak



Horizontal louvers center arch
with quarter round side panels
-Standard louvered arch

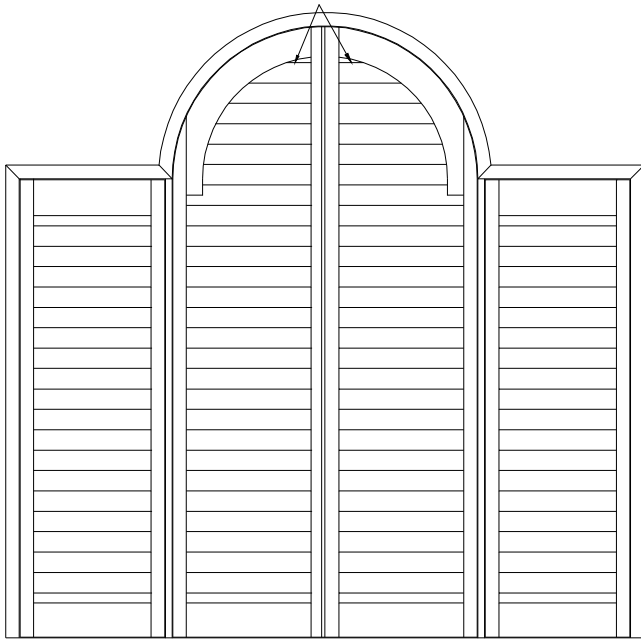


Horizontal louvers center arch
with quarter round side panels
-Continuous arch



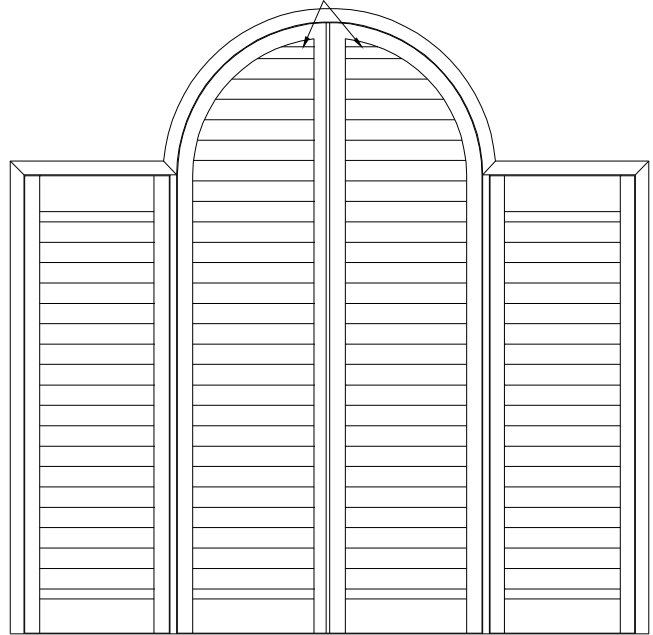
Sunburst louvers center arch with quarter round side panels

Fixed louver

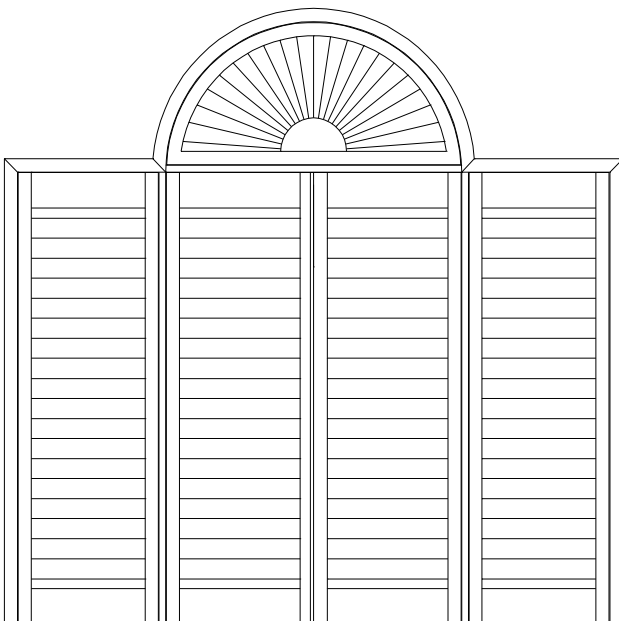


Arch Top Picture Window with
horizontal louvers
-Standard louvered arch

Fixed louver

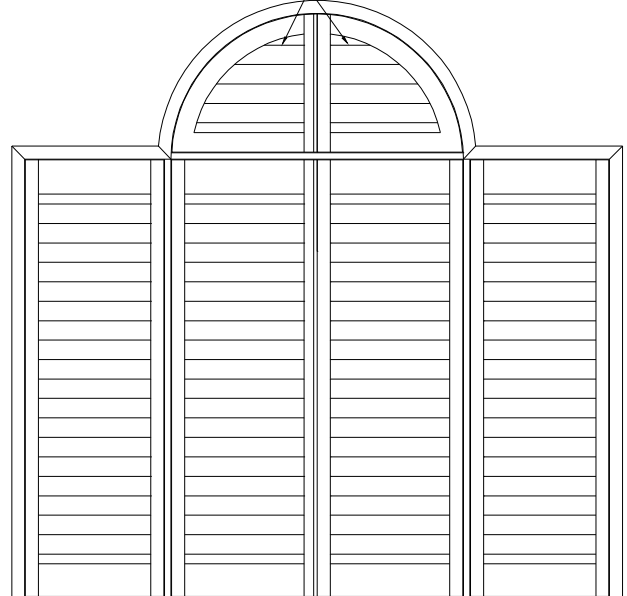


Arch Top Picture Window with
horizontal louvers
-Continuous arch

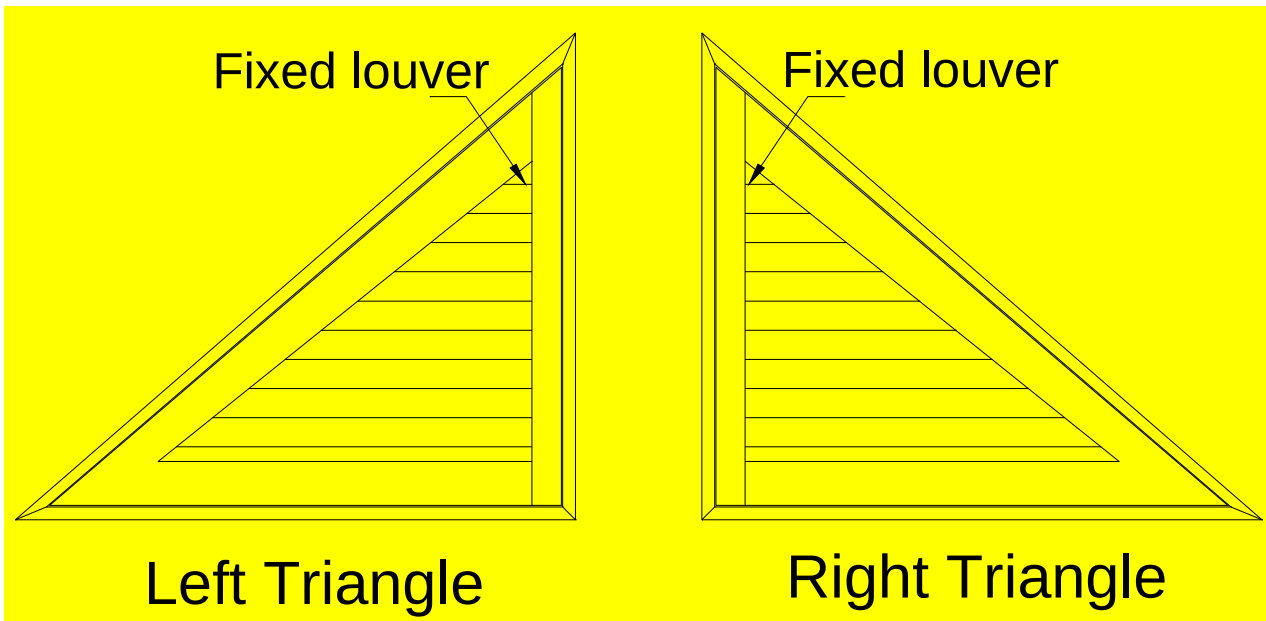


Arch Top Picture Window with Sunburst

fixed louver



Standard unit with horizontal arch center



Pricing

1) By Max Frame-to-Frame Measurement:							
Regular Shutter							
Inside and Outside Mount		Width	x	Height	/ 144	x	\$/sq ft
Track Shutter**							
Inside		Width	x	Height	/ 144	x	\$/sq ft
Semi-Inside and Outside Mount		Width	x	(Height-1½")	/ 144	x	\$/sq ft
2) By Net Panel Measurement:		Width	x	Height	/ 144	x	\$/sq ft
3) By Window Size:							
2" Camber Deco Frame 2" Classic Deco Frame	4-sided	(Width+4")	x	(Height+4")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+4")	x	(Height+4")	/ 144	x	\$/sq ft
	3-sided	(Width+4")	x	(Height+2")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+4")	x	(Height+2")	/ 144	x	\$/sq ft
	2-sided	(Width+4")	x	(Height)	/ 144	x	\$/sq ft
3" Ridge Deco Frame	4-sided	(Width+6")	x	(Height+6")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+6")	x	(Height+6")	/ 144	x	\$/sq ft
	3-sided	(Width+6")	x	(Height+3")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+6")	x	(Height+3")	/ 144	x	\$/sq ft
	2-sided	(Width+6")	x	(Height)	/ 144	x	\$/sq ft
2.5" Mission deco frame	4-sided	(Width+5")	x	(Height+5")	/ 144	x	\$/sq ft
	3-sided	(Width+5")	x	(Height+2.5")	/ 144	x	\$/sq ft
	2-sided	(Width+5")	x	(Height)	/ 144	x	\$/sq ft
3" Crown Z Frame	4-sided	(Width+4 ½")	x	(Height+4½")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+4 ½")	x	(Height+4½")	/ 144	x	\$/sq ft
	3-sided	(Width+4 ½")	x	(Height+2¼")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+4 ½")	x	(Height+2¼")	/ 144	x	\$/sq ft
	2-sided	(Width+4 ½")	x	(Height)	/ 144	x	\$/sq ft

Pricing

2" Bel Air Z Frame 2" Bullnose Z Frame	4-sided	(Width+2 ½")	x	(Height+2½")	/ 144	x	\$/sq ft
	3-sided with Sill Plate	(Width+2 ½")	x	(Height+2½")	/ 144	x	\$/sq ft
	3-sided	(Width+2 ½")	x	(Height+1¼")	/ 144	x	\$/sq ft
	2-sided with Sill Plate	(Width+2 ½")	x	(Height+1¼")	/ 144	x	\$/sq ft
	2-sided	(Width+2 ½")	x	(Height)	/ 144	x	\$/sq ft
1 1/2" Bullnose Z Frame 1 ¼ " Beaded Z Frame/ Bullnose Tilt Out Z Frame	4-sided	(Width+2")	x	(Height+2")	/ 144	x	\$/sq ft
	3-sided with Sill Plate (N/A w/ Tilt out Z)	(Width+2")	x	(Height+2")	/ 144	x	\$/sq ft
	3-sided	(Width+2")	x	(Height+1")	/ 144	x	\$/sq ft
	2-sided with Sill Plate (N/A w/ Tilt out Z)	(Width+2")	x	(Height+1")	/ 144	x	\$/sq ft
	2-sided	(Width+2")	x	(Height)	/ 144	x	\$/sq ft
Vintage Hang Strip	Inside	Width	x	Height	/ 144	x	\$/sq ft
	Outside 4-sided	(Width+3")	x	(Height+3")	/ 144	x	\$/sq ft
	Outside 3-sided	(Width+3")	x	(Height+1½")	/ 144	x	\$/sq ft
	Outside 2-sided	(Width+3")	x	(Height)	/ 144	x	\$/sq ft
Vintage L Frame/ Beaded L Frame	Outside 4-sided	(Width+3")	x	(Height+3")	/ 144	x	\$/sq ft
	Outside 3-sided	(Width+3")	x	(Height+1½")	/ 144	x	\$/sq ft
	Outside 2-sided	(Width+3")	x	(Height)	/ 144	x	\$/sq ft
Track Shutter**							
Inside and Semi-Inside Mount		Width	x	Height	/144	x	\$/sq ft
Outside Mount		(Width+3")	x	(Height+3")	/144	x	\$/sq ft

1 side frame price:

2" Camber Deco Frame	1-sided	(Width+2")	x	(Height)	/ 144	x	\$/sq ft
2" Classic Deco Frame	1-sided	(Width)	x	(Height+2")	/ 144	x	\$/sq ft
3" Ridge Deco Frame	1-sided	(Width+3")	x	(Height)	/ 144	x	\$/sq ft
	1-sided	(Width)	x	(Height+3")	/ 144	x	\$/sq ft
2 1/2" Mission deco frame	1-sided	(Width+2 1/2")	x	(Height)	/ 144	x	\$/sq ft
	1-sided	(Width)	x	(Height+2 1/2")	/ 144	x	\$/sq ft
3" Crown Z Frame	1-sided	(Width+2 1/4")	x	(Height)	/ 144	x	\$/sq ft
	1-sided	(Width)	x	(Height+2 1/4")	/ 144	x	\$/sq ft
2" Bel Air Z Frame	1-sided	(Width+1 1/4")	x	(Height)	/ 144	x	\$/sq ft
2"Bullnose Z Frame	1-sided	(Width)	x	(Height+1 1/4")	/ 144	x	\$/sq ft
1 1/2"Bullnose Z Frame	1-sided	(Width+1")	x	(Height)	/ 144	x	\$/sq ft
1 1/4" Beaded Z Frame	1-sided	(Width)	x	(Height+1")	/ 144	x	\$/sq ft
Bullnose Tilt Out Z Frame							
Vintage Hang Strip	Outside 1-sided	(Width+1 1/2")	x	(Height)	/ 144	x	\$/sq ft
	Outside 1-sided	(Width)	x	(Height+1 1/2")	/ 144	x	\$/sq ft
Vintage L Frame	Outside 1-sided	(Width+1 1/2")	x	(Height)	/ 144	x	\$/sq ft
Beaded L Frame							
Beaded L with 1/4" Light Block							
Beaded L with 1/2" Buildout	Outside 1-sided	(Width)	x	(Height+1 1/2")	/ 144	x	\$/sq ft
Beaded L with 1" Buildout							
Vintage L with 1/4" Light Block							
Vintage L with 1/2" Buildout							
Vintage L with 1" Buildout							